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United States Patent Application Publication

20250270636

Kind Code

A1

Publication Date

August 28, 2025

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MULTI-FLUOROPHORE SINGLE NUCLEOTIDE COMPLEXES FOR SEQUENCING

Abstract

The present disclosure relates in some aspects to methods, systems, and kits for sequencing a template nucleic acid molecule using nucleotide-fluorophore complexes. In some embodiments, each nucleotide-fluorophore complex comprises: (i) a single nucleotide attached to a core; and (ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide. In some embodiments, the methods provided herein achieve improved signal intensity and sensitivity.

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Family ID: 1000008477570

Appl. No.: 19/060538

Filed: February 21, 2025

Related U.S. Application Data

us-provisional-application US 63557233 20240223

Publication Classification

Int. Cl.: C12Q1/6869 (20180101); C12Q1/6841 (20180101)

U.S. Cl.:

CPC C12Q1/6869 (20130101); C12Q1/6841 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application No. 63/557,233, filed Feb. 23, 2024, entitled “MULTI-FLUOROPHORE SINGLE NUCLEOTIDE COMPLEXES FOR SEQUENCING,” which is herein incorporated by reference in its entirety for all purposes.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

[0002] The content of the electronic sequence listing (202412022100seqlist.xml; Size: 10,114 bytes; and Date of Creation: Feb. 10, 2025) is herein incorporated by reference in its entirety.

FIELD

[0003] The present disclosure relates in some aspects to methods for sequencing nucleic acid molecules, including methods for in situ sequencing and analysis of target nucleic acids in a biological sample.

BACKGROUND

[0004] Nucleic acid sequencing is a versatile tool that helps scientists advance the understanding of biology and has wide-ranging applications in various fields, such as medical diagnostics, biotechnology, forensic biology, and virology. Currently, there are several sequencing methods available, including Maxam-Gilbert sequencing, Sanger (chain-termination) sequencing, and next-generation sequencing (NGS) techniques. Despite advances in nucleic acid sequencing, many challenges remain unaddressed.

SUMMARY

[0005] NGS sequencing-by-synthesis (SBS) is based on incorporation of a fluorescent, reversibly terminated nucleotide into an extended priming strand, where the incorporated nucleotide is complementary to a nucleotide in the template nucleic acid molecule that is being probed. In such methods, each nucleotide is labelled with a single fluorophore which can limit sensitivity and signal intensity. One approach to improved SBS is to increase the number of fluorophores associated with a single nucleotide (e.g., 3 fluorophores per nucleotide). In some embodiments, provided herein are methods for sequencing a template nucleic acid molecule using nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex comprises: i) a single nucleotide attached to a core; and ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide. In some embodiments, the methods provided herein achieve improved signal intensity, signal-to-noise ratios, and sensitivity. Also provided herein are systems and kits for sequencing a template nucleic acid molecule using nucleotide-fluorophore complexes.

[0006] In some aspects, provided herein is a method for sequencing a template nucleic acid molecule, comprising: a) contacting a priming strand bound to the template nucleic acid molecule with: A) a polymerase; and B) a first composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex comprises: i) a single nucleotide attached to a core; and ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide; b) incorporating one of the nucleotide-fluorophore complexes of the first composition of nucleotide-fluorophore complexes into the priming strand to form an extended priming strand; and c) detecting the incorporated nucleotide-fluorophore complex to identify a complementary nucleotide in the template nucleic acid molecule.

[0007] In some aspects, provided herein is a method for sequencing a template nucleic acid molecule, comprising: (a) contacting a priming strand bound to the template nucleic acid molecule with: (A) a polymerase; and (B) a composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex comprises: (i) a single nucleotide attached to a core; and (ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore

complex comprises no more than one nucleotide; (b) incorporating one of the nucleotide-fluorophore complexes of the composition of nucleotide-fluorophore complexes into the priming strand to form an extended priming strand; and (c) detecting the incorporated nucleotide-fluorophore complex.

[0008] In some of any of the embodiments herein, the ratio of the fluorophores to the nucleotide in each of the nucleotide-fluorophore complexes of the first composition is at least 3:1.

[0009] In some of any of the embodiments herein, the method further comprises removing the fluorophores from the incorporated nucleotide-fluorophore complex.

[0010] In some of any of the embodiments herein, the method further comprises, after the detecting step, cleaving the core from the nucleotide of the incorporated nucleotide-fluorophore complex, leaving an incorporated nucleotide.

[0011] In some of any of the embodiments herein, the core of each of the nucleotide-fluorophore complexes comprises an avidin or streptavidin core, and wherein the nucleotide in each complex is bound to the core via a biotinylated linker. In any of the embodiments herein, at least a subset of the fluorophores in each nucleotide-fluorophore complex are bound to the core via biotin, optionally wherein at least a subset of the fluorophores in each nucleotide-fluorophore complex are bound to the core via a biotinylated linker. In any of the embodiments herein, the biotinylated linker is a biotinylated linear polyethylene glycol (PEG) linker. In any of the embodiments herein, the core of each of the nucleotide-fluorophore complexes comprises a branched polymer core, and wherein the nucleotide and the plurality of fluorophores in each nucleotide-fluorophore complex are attached to the branched polymer core.

[0012] In some of any of the embodiments herein, the nucleotide comprises a reversible terminator moiety, optionally wherein the method further comprises cleaving the reversible terminator moiety after detecting the incorporated nucleotide-fluorophore complex.

[0013] In some of any of the embodiments herein, the method further comprises: (d) contacting a priming strand bound to the template nucleic acid molecule with: (A) a polymerase; and (B) an additional composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex of the additional composition comprises: (i) a single nucleotide attached to a core; and (ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide; (c) incorporating one of the nucleotide-fluorophore complexes of the additional composition of nucleotide-fluorophore complexes into the extended priming strand; and (f) detecting the incorporated nucleotide-fluorophore complex of the additional composition of nucleotide-fluorophore complexes to identify a complementary nucleotide at a subsequent position in the template nucleic acid molecule.

[0014] In some of any of the embodiments herein, the method further comprises: (d) contacting a priming strand bound to the template nucleic acid molecule with: (A) an additional polymerase; and (B) an additional composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex of the additional composition comprises: (i) a single nucleotide attached to a core; and (ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide; (e) incorporating one of the nucleotide-fluorophore complexes of the additional composition of nucleotide-fluorophore complexes into the extended priming strand; and (f) detecting the incorporated nucleotide-fluorophore complex of the additional composition of nucleotide-fluorophore complexes.

[0015] In any of the embodiments herein, the method comprises repeating steps (d) through (f) for at least one additional cycle using at least one additional composition comprising nucleotide-fluorophore complexes. In any of the embodiments herein, the method comprises repeating steps (d) through (f) for at least one additional cycle using at least one additional composition comprising nucleotide-fluorophore complexes to identify at least one additional complementary nucleotide in the template nucleic acid molecule. In any of the embodiments herein, the at least one additional cycle comprises at least 2, 5, 10, 20, 30, 40, or 50 additional cycles. In any of the embodiments

herein, the at least one additional cycle comprises at least 2 additional cycles.

[0016] In some of any of the embodiments herein, the nucleotide-fluorophore complexes of the first and/or additional composition(s) comprise four sets of nucleotide-fluorophore complexes, each set comprising a same nucleobase that differs from the other sets, and wherein, for each set, the plurality of fluorophores is a plurality of a same fluorophore that differs from the other sets. In some of any of the embodiments herein, the nucleotide-fluorophore complexes of the composition comprises four sets of nucleotide-fluorophore complexes, each set comprising a same nucleobase that differs from the other sets, and wherein, for each set, the plurality of fluorophores is a plurality of a same fluorophore that differs from the other sets.

[0017] In some of any of the embodiments herein, the nucleotide-fluorophore complexes of the first and/or additional composition(s) comprise three sets of nucleotide-fluorophore complexes, each set comprising a same nucleobase that differs from the other sets, and wherein, for each set, the plurality of fluorophores is a plurality of a same fluorophore that differs from the other sets, and wherein the first and/or additional composition(s) further comprise a set of unlabeled nucleotides comprising a same nucleobase that differs from the nucleobase of the three sets of nucleotide-fluorophore complexes.

[0018] In some of any of the embodiments herein, the nucleobase of the nucleotide in each of the nucleotide-fluorophore complexes is selected from the group consisting of A, T, U, C, and G. In any of the embodiments herein, the nucleobase of the nucleotide in each of the nucleotide-fluorophore complexes is selected from the group consisting of A, T, C, and G.

[0019] In some of any of the embodiments herein, the template nucleic acid molecule comprises a DNA molecule. In any of the embodiments herein, the template nucleic acid molecule comprises an RNA molecule.

[0020] In some of any of the embodiments herein, the template nucleic acid molecule comprises a target analyte nucleic acid molecule. In any of the embodiments herein, the template nucleic acid molecule comprises a barcode sequence associated with a target analyte. In any of the embodiments herein, the method further comprises hybridizing a circularizable probe or probe set to the target analyte or to a labeling agent bound to the target analyte and ligating the circularizable probe or probe set to form a circularized probe, wherein the method further comprises performing rolling circle amplification of the circularized probe to generate the template nucleic acid molecule. In any of the embodiments herein, the circularizable probe or probe set is a padlock probe. In any of the embodiments herein, the target analyte nucleic acid molecule comprises an mRNA molecule.

[0021] In some of any of the embodiments herein, the template nucleic acid molecule to be sequenced is attached to a solid support. In any of the embodiments herein, the solid support comprises a sequencing flow cell.

[0022] In some of any of the embodiments herein, the template nucleic acid molecule is sequenced in situ in a cell sample or tissue sample. In any of the embodiments herein, the cell sample comprises a layer of cells deposited on a surface.

[0023] In some of any of the embodiments herein, the method further comprises, after step (c), identifying a complementary nucleotide in the template nucleic acid based on the detecting.

[0024] In some aspects, provided herein is a kit for sequencing a template nucleic acid molecule comprising: a composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex comprises: i) a single nucleotide attached to a core; and ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide. In some of any of the embodiments herein, the composition of nucleotide-fluorophore complexes comprises at least three sets of nucleotide-fluorophore complexes, wherein each set comprises a same nucleobase that differs from the other sets, and wherein, for each set, the plurality of fluorophores is a plurality of same fluorophores that differ from the other sets. In any of the embodiments herein, the composition comprises four sets of nucleotide-fluorophore complexes. In any of the embodiments herein, the composition further

comprises unlabeled nucleotides, wherein the unlabeled nucleotides each comprise a same nucleobase that is different from the nucleobase of the nucleotide-fluorophore complexes in the sets. In some of any of the embodiments herein, the core of each of the nucleotide-fluorophore complexes comprises an avidin or streptavidin core, and wherein the nucleotide in each complex is bound to the core via a biotinylated linker. In any of the embodiments herein, at least a subset of the fluorophores in each nucleotide-fluorophore complex are bound to the core via biotin, optionally wherein at least a subset of the fluorophores in each nucleotide-fluorophore complex are bound to the core via a biotinylated linker. In any of the embodiments herein, the biotinylated linker is a biotinylated linear polyethylene glycol (PEG) linker. In any of the embodiments herein, of the nucleotide-fluorophore complexes comprises a branched polymer core, and wherein the nucleotide and the plurality of fluorophores in each nucleotide-fluorophore complex are attached to the branched polymer core.

[0025] In some aspects, provided herein is a composition comprising: a composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex comprises: i) a single nucleotide attached to a core; and ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide. In any of the embodiments herein, the nucleotide comprises a reversible terminator moiety. In some of any of the embodiments herein, the composition of nucleotide-fluorophore complexes comprises at least three sets of nucleotide-fluorophore complexes, wherein each set comprises a same nucleobase that differs from the other sets, and wherein, for each set, the plurality of fluorophores is a plurality of same fluorophores that differ from the other sets. In any of the embodiments herein, the composition comprises four sets of nucleotide-fluorophore complexes. In any of the embodiments herein, the composition further comprises unlabeled nucleotides, wherein the unlabeled nucleotides each comprise a nucleobase that is different from the nucleobase of the nucleotide-fluorophore complexes in the sets. In some of any of the embodiments herein, the core of each of the nucleotide-fluorophore complexes comprises an avidin or streptavidin core, and wherein the nucleotide in each complex is bound to the core via a biotinylated linker. In any of the embodiments herein, at least a subset of the fluorophores in each nucleotide-fluorophore complex are bound to the core via biotin, optionally wherein at least a subset of the fluorophores in each nucleotide-fluorophore complex are bound to the core via a biotinylated linker. In any of the embodiments herein, the biotinylated linker is a biotinylated linear polyethylene glycol (PEG) linker. In some of any of the embodiments herein, the core of each of the nucleotide-fluorophore complexes comprises a branched polymer core, and wherein the nucleotide and the plurality of fluorophores in each nucleotide-fluorophore complex are attached to the branched polymer core.

[0026] In some embodiments, provided herein is a system comprising: one or more processors; and a memory communicatively coupled to the one or more processors and configured to store instructions that, when executed by the one or more processors, cause the system to perform the method of any of the preceding embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The drawings illustrate certain features and advantages of this disclosure. These embodiments are not intended to limit the scope of the appended claims in any manner.

[0028] FIG. 1 provides a non-limiting example of a process flowchart for sequencing a template nucleic acid molecule in accordance with one implementation of the methods described herein.

[0029] FIG. 2 depicts a system for performing a sequencing assay, in accordance with some implementations of the methods described herein.

[0030] FIG. 3 depicts a computer system or computer network, in accordance with some instances

of the systems described herein.

[0031] FIG. 4 provides a schematic illustration of a method for sequencing a template nucleic acid molecule in accordance with one implementation of the methods described herein.

DETAILED DESCRIPTION

[0032] All publications, comprising patent documents, scientific articles and databases, referred to in this application are incorporated by reference in their entirety for all purposes to the same extent as if each individual publication were individually incorporated by reference. If a definition set forth herein is contrary to or otherwise inconsistent with a definition set forth in the patents, applications, published applications and other publications that are herein incorporated by reference, the definition set forth herein prevails over the definition that is incorporated herein by reference.

[0033] The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described.

I. Overview

[0034] Methods for sequencing nucleic acid molecules described herein are based on incorporating a nucleotide-fluorophore complex into the 3' end of a priming strand to form an extending priming strand. The nucleotide-fluorophore complex comprises: (i) a single nucleotide attached to a core; and (ii) a plurality of fluorophores attached to the core; wherein the nucleotide-fluorophore complex comprises no more than one nucleotide. The disclosed methods, compositions, kits, and systems for performing sequencing of a template nucleic acid molecule are applicable to both in situ sequencing and flow cell sequencing. In such embodiments, the plurality of fluorophores associated with each nucleotide-fluorophore complex improves signal intensity, signal-to-noise ratios, and sensitivity.

[0035] Additional aspects of the methods, compositions, kits, and systems disclosed herein are described in the sections below.

II. Compositions

[0036] In some embodiments, provided herein is a composition for sequencing a template nucleic acid molecule comprising nucleotide-fluorophore complexes. Each nucleotide-fluorophore complex comprises: (i) a single nucleotide attached to a core; and (ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide. Contacting the composition comprising nucleotide-fluorophore complexes and a polymerase with a primed template nucleic acid results in the incorporation of one of the nucleotide-fluorophore complexes of the composition into the priming strand to form an extended priming strand. The plurality of fluorophores attached to the core allows for a brighter fluorescence signal when detecting the incorporated nucleotide-fluorophore complex to identify a complementary nucleotide in the template nucleic acid molecule.

Nucleotide-Fluorophore Complexes

[0037] In some embodiments, a composition comprises 1, 2, 3, 4 or more sets of nucleotide-fluorophore complexes, wherein each set comprises a same nucleobase that differs from the other sets. Each nucleotide-fluorophore complex of the composition comprises a single nucleotide attached to the core. In some embodiments, a first set of nucleotide-fluorophore complexes comprises a nucleobase selected from the group consisting of ATP, ADP, AMP, dATP, dADP, and dAMP. In some embodiments, a second set of nucleotide-fluorophore complexes comprises a nucleobase selected from the group consisting of TTP, TDP, TMP, dTTP, dTDP, dTMP, UTP, UDP, UMP, dUTP, dUDP, and dUMP. In some embodiments, a third set of nucleotide-fluorophore complexes comprises a nucleobase selected from the group consisting of CTP, CDP, CMP, dCTP, dCDP, and dCMP. In some embodiments, a fourth set of nucleotide-fluorophore complexes comprises a nucleobase selected from the group consisting of GTP, GDP, GMP, dGTP, dGDP, and dGMP. In some embodiments, each set of nucleotide-fluorophore complexes comprises a nucleobase selected from the group consisting of ATP, ADP, AMP, dATP, dADP, dAMP, TTP, TDP,

TMP, dTTP, dTDP, UTP, UDP, UMP, dUTP, dUDP, CTP, CDP, CMP, dCTP, dCDP, dCMP, GTP, GDP, GMP, dGTP, dGDP, and dGMP, wherein each set comprises a same nucleobase that differs from the other sets.

[0038] In some embodiments, the composition comprises 1, 2, 3, 4 or more sets of nucleotide-fluorophore complexes, wherein each set comprises a same nucleobase that differs from the other sets, and wherein, for each set, the plurality of fluorophores is a plurality of a same fluorophore that differ from the other sets. In some embodiments, the composition comprises four sets of nucleotide-fluorophore complexes. In some embodiments, a first set of nucleotide-fluorophore complexes comprises a plurality of fluorophores, wherein the plurality of fluorophores is a plurality of a first type of fluorophore. In some embodiments, a second set of nucleotide-fluorophore complexes comprises a plurality of fluorophores, wherein the plurality of fluorophores is a plurality of a second type of fluorophore. In some embodiments, a third set of nucleotide-fluorophore complexes comprises a plurality of fluorophores, wherein the plurality of fluorophores is a plurality of a third type of fluorophore. In some embodiments, a fourth set of nucleotide-fluorophore complexes comprises a plurality of fluorophores, wherein the plurality of fluorophores is a plurality of a fourth type of fluorophore. Examples of suitable fluorophores are described elsewhere herein.

[0039] In some embodiments, the composition further comprises unlabeled nucleotides, wherein the unlabeled nucleotides each comprise a nucleobase that is different from the nucleobase of the nucleotide-fluorophore complexes in the sets. In such embodiments, detecting an incorporated unlabeled nucleotide comprises detecting an absence of a signal. In some embodiments, the composition comprises three sets of nucleotide-fluorophore complexes and one set of unlabeled nucleotides. In some embodiments, a first set of nucleotide-fluorophore complexes comprises a nucleobase selected from the group consisting of ATP, ADP, AMP, dATP, dADP, and dAMP and a plurality of fluorophores, wherein the plurality of fluorophores is a plurality of a first type of fluorophore. In some embodiments, a second set of nucleotide-fluorophore complexes comprises a nucleobase selected from the group consisting of TTP, TDP, TMP, dTTP, dTDP, dTMP, UTP, UDP, UMP, dUTP, dUDP, and dUMP and a plurality of fluorophores, wherein the plurality of fluorophores is a plurality of a second type of fluorophore. In some embodiments, a third set of nucleotide-fluorophore complexes comprises a nucleobase selected from the group consisting of CTP, CDP, CMP, dCTP, dCDP, and dCMP and a plurality of fluorophores, wherein the plurality of fluorophores is a plurality of a third type of fluorophore. In some embodiments, a set of unlabeled nucleotides comprises a nucleobase selected from the group consisting of GTP, GDP, GMP, dGTP, dGDP, and dGMP. In some embodiments, the first, second, and third set of nucleotide-fluorophore complexes and the set of unlabeled nucleotides comprise a single nucleobase selected from the group consisting of ATP, ADP, AMP, dATP, dADP, dAMP, TTP, TDP, TMP, dTTP, dTDP, dTMP, UTP, UDP, UMP, dUTP, dUDP, dUMP, CTP, CDP, CMP, dCTP, dCDP, dCMP, GTP, GDP, GMP, dGTP, dGDP, and dGMP, wherein each set comprises a same nucleobase that differs from the other sets.

A Single Nucleotide of a Nucleotide-Fluorophore Complex

[0040] In some embodiments, a nucleotide-fluorophore complex as described herein comprises a single nucleotide attached to a core. In some embodiments, each nucleotide-fluorophore complex comprises no more than one nucleotide.

[0041] In some embodiments, the number of nucleotides attached to each core represents an averaged number for a composition of nucleotide-fluorophore complexes or a mixture comprising nucleotide-fluorophore complexes. In some embodiments, the average number of nucleotides attached to each core in a composition of nucleotide-fluorophore complexes or a mixture comprising nucleotide-fluorophore complexes is between about 0.5 to about 2. In some embodiments, the average number of nucleotides attached to each core in a composition of nucleotide-fluorophore complexes or a mixture comprising nucleotide-fluorophore complexes is less than about 2. In some embodiments, the average number of nucleotides attached to each core

in a composition of nucleotide-fluorophore complexes or a mixture comprising nucleotide-fluorophore complexes is about 1. In some embodiments, the majority of the nucleotide-fluorophore complexes in a composition or mixture comprises 1 nucleotide. In some embodiments, at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, or at least about 95% of the nucleotide-fluorophore complexes in a composition or mixture comprise 1 nucleotide. When preparing the nucleotide-fluorophore complexes, the reaction conditions (e.g., relative concentrations of free nucleotide to core) can be optimized to achieve a desired average number of nucleotides per core (e.g., 1 nucleotide per core).

[0042] In some embodiments, a nucleotide is any of a variety of naturally-occurring nucleotides and/or functional analogs thereof (e.g., nucleotide analogs capable of hybridizing to a nucleic acid sequence in a sequence-specific/correctly base-paired manner) to probe a template nucleic acid sequence. Naturally-occurring nucleotides include deoxyribonucleotides (found in DNA) that comprise a deoxyribose sugar moiety, and ribonucleotides (found in RNA) that comprise a ribose sugar moiety. Naturally-occurring deoxyribonucleotides comprise a nucleobase (or “base”) selected from the group consisting of adenine (A), thymine (T), cytosine (C), or guanine (G). Naturally-occurring ribonucleotides comprise a nucleobase selected from the group consisting of uracil (U), adenine (A), cytosine (C), or guanine (G). In some embodiments, the nucleotides are terminated (e.g., reversibly terminated). In some embodiments, the nucleotides are conjugated to a detectable label, e.g., a fluorophore. In some embodiments, the nucleotides are conjugated to other moieties, e.g., reactive functional groups. In some embodiments, the nucleotides are attached to the core through a linker, e.g., a biotinylated linker.

[0043] In some embodiments, the single nucleotide attached to each core of the nucleotide-fluorophore complex comprises any suitable reversible terminator moiety. In some embodiments, the nucleotide is a 3'-O-blocked reversibly terminated nucleotide. In some embodiments, 3'-O-blocked reversibly terminated nucleotide is, e.g., a 3'-O-azidomethyl deoxynucleotide triphosphate (3'-O-azidomethyl dNTP), a 3'-O-allyl deoxynucleotide triphosphate (3'-O-allyl-dNTP), 3'-O-acetate deoxynucleotide triphosphate (3'-O-acetate dNTP), or a 3'-O-amino deoxynucleotide triphosphate (3'-O—NH₂ dNTP). In some embodiments, 3' reversibly terminated nucleotide may be a 3'-unblocked reversibly terminated nucleotide.

Cores

[0044] In some embodiments, a nucleotide-fluorophore complex as described herein comprises a core, wherein a single nucleotide and a plurality of fluorophores are attached to the core.

[0045] In some embodiments, a core of a nucleotide-fluorophore complex is an avidin core. In some embodiments, the avidin core is selected from the group consisting of streptavidin, tamavidin, traptavidin, xenavidin, bradavidin, AVR2, AVR4, and homologs thereof. In some embodiments, the core is a streptavidin core. In some embodiments, the core comprises at least 1, at least 2, at least 3, or at least 4 biotin binding sites. In some embodiments, the core comprises 4 biotin binding sites.

[0046] In some embodiments, the single nucleotide and at least a subset of the plurality of fluorophores are attached to the core through affinity binding (e.g., binding of biotin by avidin or streptavidin). In some embodiments, the single nucleotide of a nucleotide-fluorophore complex is bound to the core via a biotinylated linker (e.g., a biotinylated linear polyethylene glycol (PEG) linker, wherein biotin is attached to one end of the linear PEG linker and the nucleotide is attached to the other end of the linear PEG linker). In some embodiments, at least a subset of the fluorophores in each nucleotide-fluorophore complex are bound to the core via biotin (e.g., a fluorophore is functionalized with a biotin molecule, wherein the avidin or streptavidin core binds to the biotin molecule). In some embodiments, at least a subset of the fluorophores in each nucleotide-fluorophore complex are bound to the core via a biotinylated linker (e.g., a biotinylated linear polyethylene glycol (PEG) linker, wherein biotin is attached to one end of the linear PEG linker and the fluorescent dye is attached to the other end of the linear PEG linker). In some

embodiments, the ratio of the fluorophores to the nucleotide attached to each core is about 3:1.

[0047] In some instances, binding of biotin by avidin or streptavidin is performed under any suitable conditions. In some instances, the binding of biotin occurs in water, aqueous buffer, or cell culture media. In some instances, the binding of biotin is performed at physiological pH (e.g., about 7.4).

[0048] In some embodiments, the core of a nucleotide-fluorophore complex is a branched polymer; a dendrimer; a cross-linked polymer core such as an agarose, polyacrylamide, acrylate, methacrylate, cyanoacrylate, methyl methacrylate core; a glass core; a ceramic core; a metal core; a quantum dot; a liposome; an emulsion core, or any other suitable core (e.g., nanoparticle cores, microparticle cores, or the like).

[0049] In some embodiments, the core of a nucleotide-fluorophore complex is a branched polymer core (e.g., a polymer comprising a plurality of branches). In some embodiments, branched polymer core comprises a configuration selected from the group consisting of stellate (“starburst”), aggregated stellate (“helter skelter”), bottle brush, and dendrimer. In some embodiments, the branched polymer core radiates from a central attachment point or central moiety, or incorporates multiple branch points, such as, for example, about 2, about 3, about 4, about 5, about 6, about 7, about 8, about 9, about 10, or more branch points. In some embodiments, each subunit of a polymer constitutes a separate branch point. In some embodiments, a branched polymer core comprises an even number of branches. In some embodiments, a branched polymer core comprises an odd number of branches.

[0050] In some embodiments, the branched polymer comprises polyethylene glycol (PEG), polypropylene glycol, polyvinyl alcohol polylactic acid polyglycolic acid, polyglycine, polyvinyl acetate, a dextran, or other such polymers, or copolymers incorporating any two or more of the foregoing. In some embodiments, the branched polymer core comprises a copolymer, wherein the copolymer comprises at least a subset of monomer units that are fluorescent. In some embodiments, the branched polymer core comprises a copolymer, wherein the copolymer comprises at least a subset of monomer units that each comprise a functional group. Examples of the functional group include but are limited to hydroxyl, ester, amine, carbonate, acetal, aldehyde, aldehyde hydrate, alkenyl, acrylate, methacrylate, acrylamide, active sulfone, hydrazide, thiol, alkanoic acid, acid halide, isocyanate, isothiocyanate, maleimide, vinylsulfone, dithiopyridine, vinylpyridine, iodoacetamide, epoxide, glyoxal, dione, mesylate, tosylate, and tresylate.

[0051] In some embodiments, the length and size of a branch of a branched polymer core differs based on the type of polymer. In some embodiments, a branch of a branched polymer core has a length of between about 1 and about 1,000 nm, between about 1 and about 100 nm, between about 1 and about 200 nm, between about 1 and about 300 nm, between about 1 and about 400 nm, between about 1 and about 500 nm, between about 1 and about 600 nm, between about 1 and about 700 nm, between about 1 and about 800 nm, or between about 1 and about 900 nm, or more, or having a length falling within or between any of the values disclosed herein.

[0052] In some embodiments, the branched polymer core comprises a size corresponding to an apparent molecular weight of about 500 Da, about 1 kDa, about 2 kDa, about 3 kDa, about 4 kDa, about 5 kDa, about 10 kDa, about 15 kDa, about 20 kDa, about 30 kDa, about 50 kDa, about 80 kDa, about 100 kDa, or any value within a range defined by any two of the foregoing. In some embodiments, the apparent molecular weight of a polymer is calculated from the known molecular weight of a representative number of subunits, as determined by size exclusion chromatography, as determined by mass spectrometry, or as determined by any other suitable method.

[0053] In some embodiments, the branches of a branched polymer core has a size corresponding to an apparent molecular weight of about 50 Da, about 100 Da, about 500 Da, about 1 kDa, about 2 kDa, about 3 kDa, about 4 kDa, about 5 kDa, about 10 kDa, about 15 kDa, about 20 kDa, about 30 kDa, about 50 kDa, about 80 kDa, about 100 kDa, or any value within a range defined by any two of the foregoing. In some embodiments, the apparent molecular weight of a polymer is calculated

from the known molecular weight of a representative number of subunits, as determined by size exclusion chromatography, as determined by mass spectrometry, or as determined by any other suitable method. The polymer can have multiple branches. In some embodiments, the number of branches in the polymer is about 2, about 3, about 4, about 5, about 6, about 7, about 8, about 12, about 16, about 24, about 32, about 64, about 128 or more, or a number falling within a range defined by any two of these values.

[0054] In some embodiments, the nucleotide of a nucleotide-fluorophore complex is attached to the branched polymer core. In some embodiments, the nucleotide is covalently attached to the branched polymer core. In some embodiments, the nucleotide is attached to the branched polymer core through 5' end of the nucleotide. In some embodiments, the nucleotide is attached to one end of a branch of the branched polymer core. In some embodiments, the nucleotide is attached to a location within a branch of the branched polymer core. In some embodiments, the nucleotide of the nucleotide-fluorophore complex is sterically accessible to one or more proteins, one or more enzymes, and a priming strand bound to a template nucleic acid molecule.

[0055] In some embodiments, the plurality of fluorophores of a nucleotide-fluorophore complex is attached to the branched polymer core. In some embodiments, the plurality of fluorophores is covalently attached to the branched polymer core. In some embodiments, a fluorophore of the plurality is attached to one end of a branch of the branched polymer core. In some embodiments, a fluorophore of the plurality is attached to a location within a branch of the branched polymer core.

[0056] In some embodiments, the branched polymer core comprises a first functional group and the nucleotide comprises a second functional group, wherein a coupling reaction between the first functional group and the second functional group covalently attaches the nucleotide to the branched polymer core. In some embodiments, the branched polymer core comprises plurality of a third functional group and the fluorophores of the plurality each comprise a fourth functional group, wherein a coupling reaction between a third functional group of the plurality and a fourth functional group of the plurality covalently attaches a fluorophore to the branched polymer core. In some embodiments, the coupling reaction between a first functional group and a second functional group and the coupling reaction between a third functional group and a fourth functional group are the same. In some embodiments, the coupling reaction between a first functional group and a second functional group and the couple reaction between a third functional group and a fourth functional group are the different. Any suitable coupling reaction can be used (e.g., a click chemistry reaction).

[0057] In some embodiments, the branched polymer core does not comprise a photo emitting or photo absorbing monomer unit.

Fluorophores of a Nucleotide-Fluorophore Complex

[0058] In some embodiments, a nucleotide-fluorophore complex as described herein comprises a plurality of fluorophores attached to the core.

[0059] In some embodiments, the plurality of fluorophores attached to the core of each nucleotide-fluorophore complex comprises at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, or at least 10 fluorophore molecules. In some embodiments, the plurality of fluorophores attached to the core of each nucleotide-fluorophore complex comprises at least 3 fluorophore molecules.

[0060] In some embodiments, a set of nucleotide-fluorophore complexes comprises a degree of labelling (e.g., a molar ratio in the form of label/protein, sometimes abbreviated as DoL). In some embodiments, the DoL is high enough to produce a strong fluorescence signal without substantial self-quenching of the plurality of fluorophores. In embodiments, the DoL is about 1 to about 10, about 3 to about 8, about 3.5 to about 7, or about 1.5 to about 4.

[0061] In some embodiments, the ratio of the fluorophores to the single nucleotide in each of the nucleotide-fluorophore complexes is at least about 2:1, at least about 3:1, at least about 4:1, at least about 5:1, at least about 6:1, at least about 7:1, at least about 8:1, or at least about 10:1. In some

embodiments, the ratio of the fluorophores to the single nucleotide in each of the nucleotide-fluorophore complexes is at least about 3:1. In some embodiments, the ratio of the fluorophores to the single nucleotide in each of the nucleotide-fluorophore complexes is about 3:1. In some embodiments, the ratio of the fluorophores to the single nucleotide in each of the nucleotide-fluorophore complexes represents an averaged ratio for a composition of nucleotide-fluorophore complexes or a mixture comprising nucleotide-fluorophore complexes. In some embodiments, a majority of the nucleotide-fluorophore complexes in a composition or mixture comprises a ratio of 3:1 fluorophores to nucleotide. In some embodiments, at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, or at least about 95% of the nucleotide-fluorophore complexes in a composition or mixture comprise a ratio of 3:1 fluorophores to nucleotide. When preparing the nucleotide-fluorophore complexes, the reaction conditions (e.g., relative concentrations of free nucleotide and fluorophore to core) can be optimized to achieve a desired ratio of the fluorophores to the single nucleotide in each of the nucleotide-fluorophore complexes (e.g., 3:1).

[0062] It is contemplated that increasing the ratio of the fluorophores to the single nucleotide in a nucleotide-fluorophore complex as described herein will increase the signal intensity associated with the nucleotide-fluorophore complex incorporated into the priming strand.

[0063] In some embodiments, a fluorophore of the plurality in a nucleotide-fluorophore complex is attached directly to the core. In some embodiments, a fluorophore of the plurality in a nucleotide-fluorophore complex is covalently or noncovalently attached to the core.

[0064] In some embodiments, a fluorophore of the plurality in a nucleotide-fluorophore complex is attached indirectly to the core (e.g., through a linker).

[0065] In some embodiments, the core is an avidin or streptavidin core, wherein at least a subset of the fluorophores in the nucleotide-fluorophore complex is attached to the core via biotin (e.g., a fluorophore is functionalized with a biotin molecule, wherein the avidin or streptavidin core binds to the biotin molecule). In some embodiments, at least a subset of the fluorophores in the nucleotide-fluorophore complex is attached to the core via a biotinylated linker (e.g., a biotinylated linear polyethylene glycol (PEG) linker, wherein biotin is attached to one end of the linear PEG linker and the fluorescent dye is attached to the other end of the linear PEG linker). In some embodiments, at least a subset of the fluorophores in each nucleotide-fluorophore complex are attached directly to the avidin or streptavidin (e.g., conjugation of amino acid-reactive fluorescent dyes (e.g., N-hydroxy-succinimidyl-ester fluorescein (NHS-Fluorescein) to avidin or streptavidin). Any suitable amino acid-reactive fluorescent dye can be used to label the avidin or streptavidin core. Other suitable coupling reactions can be used for conjugating fluorophores to the avidin or streptavidin core (e.g., click-chemistry).

[0066] In some embodiments, the core is an avidin or streptavidin core, wherein all of the fluorophores in each nucleotide-fluorophore complex are attached to the core via biotin. In some embodiments, all of the fluorophores in each nucleotide-fluorophore complex are attached to the core via a biotinylated linker (e.g., a biotinylated linear polyethylene glycol (PEG) linker).

[0067] In some embodiments, the core is an avidin or streptavidin core, wherein all of the fluorophores in each nucleotide-fluorophore complex are attached covalently to the avidin or streptavidin core (e.g., conjugation of amino acid-reactive fluorescent dyes to avidin or streptavidin).

[0068] In some embodiments, the core is a branched polymer core, wherein the plurality of fluorophores in each nucleotide-fluorophore complex are attached to the branched polymer core. In some embodiments, a fluorophore of the plurality in each nucleotide-fluorophore complex is covalently attached to the core the branched polymer core.

[0069] In some embodiments, a plurality of fluorophores in each nucleotide-fluorophore complex is selected from a group that includes: 7-AAD (7-Aminoactinomycin D), Acridine Orange (+DNA), Acridine Orange (+RNA), Alexa Fluor® 350, Alexa Fluor® 430, Alexa Fluor® 488,

Alexa Fluor® 532, Alexa Fluor® 546, Alexa Fluor® 555, Alexa Fluor® 568, Alexa Fluor® 594, Alexa Fluor® 633, Alexa Fluor® 647, Alexa Fluor® 660, Alexa Fluor® 680, Alexa Fluor® 700, Alexa Fluor® 750, Allophycocyanin (APC), AMCA/AMCA-X, 7-Aminoactinomycin D (7-AAD), 7-Amino-4-methylcoumarin, 6-Aminoquinoline, Aniline Blue, ANS, APC-Cy7, ATTO-TAG™ CBQCA, ATTO-TAG™ FQ, Auramine O-Feulgen, BCECF (high pH), BFP (Blue Fluorescent Protein), BFP/GFP FRET, BOBO™-1/BO-PRO™-1, BOBO™-3/BO-PRO™-3, BODIPY® FL, BODIPY® TMR, BODIPY® TR-X, BODIPY® 530/550, BODIPY® 558/568, BODIPY® 564/570, BODIPY® 581/591, BODIPY® 630/650-X, BODIPY® 650-665-X, BTC, Calcein, Calcein Blue, Calcium Crimson™, Calcium Green-1™, Calcium Orange™, Calcofluor® White, 5-Carboxyfluorescein (5-FAM), 5-Carboxynaphthofluorescein, 6-Carboxyrhodamine 6G, 5-Carboxytetramethylrhodamine (5-TAMRA), Carboxy-X-rhodamine (5-ROX), Cascade Blue®, Cascade Yellow™, CCF2 (GeneBLAzer™), CFP (Cyan Fluorescent Protein), CFP/YFP FRET, Chromomycin A3, CI-NERF (low pH), CPM, 6-CR 6G, CTC Formazan, Cy2®, Cy3®, Cy3.5®, Cy5®, Cy5.5®, Cy7®, Cychrome (PE-Cy5), Dansylamine, Dansyl cadaverine, Dansylchloride, DAPI, Dapoxyl, DCFH, DHR, DIA (4-Di-16-ASP), DiD (DilC18 (5)), DIDS, Dil (DilC18 (3)), DiO (DiOC18 (3)), DiR (DilC18 (7)), Di-4 ANEPPS, Di-8 ANEPPS, DM-NERF (4.5-6.5 pH), DsRed (Red Fluorescent Protein), EBFP, ECFP, EGFP, ELF®-97 alcohol, Eosin, Erythrosin, Ethidium bromide, Ethidium homodimer-1 (EthD-1), Europium (III) Chloride, 5-FAM (5-Carboxyfluorescein), Fast Blue, Fluorescein-dT phosphoramidite, FITC, Fluo-3, Fluo-4, FluorX®, Fluoro-Gold™ (high pH), Fluoro-Gold™ (low pH), Fluoro-Jade, FM® 1-43, Fura-2 (high calcium), Fura-2/BCECF, Fura Red™ (high calcium), Fura Red™/Fluo-3, GeneBLAzer™ (CCF2), GFP Red Shifted (rsGFP), GFP Wild Type, GFP/BFP FRET, GFP/DsRed FRET, Hoechst 33342 & 33258, 7-Hydroxy-4-methylcoumarin (pH 9), 1,5 IAEDANS, Indo-1 (high calcium), Indo-1 (low calcium), Indodicarbocyanine, Indotricarbocyanine, JC-1, 6-JOE, JOJO™-1/JO-PRO™-1, LDS 751 (+DNA), LDS 751 (+RNA), LOLO™-1/LO-PRO™-1, Lucifer Yellow, LysoSensor™ Blue (pH 5), LysoSensor™ Green (pH 5), LysoSensor™ Yellow/Blue (pH 4.2), LysoTracker® Green, LysoTracker® Red, LysoTracker® Yellow, Mag-Fura-2, Mag-Indo-1, Magnesium Green™, Marina Blue®, 4-Methylumbelliferone, Mithramycin, MitoTracker® Green, MitoTracker® Orange, MitoTracker® Red, NBD (amine), Nile Red, Oregon Green® 488, Oregon Green® 500, Oregon Green® 514, Pacific Blue, PBF1, PE (R-phycoerythrin), PE-Cy5, PE-Cy7, PE-Texas Red, PerCP (Peridinin chlorophyll protein), PerCP-Cy5.5 (TruRed), PharRed (APC-Cy7), C-phycoerythrin, R-phycoerythrin (PE), PI (Propidium Iodide), PKH26, PKH67, POPO™-1/PO-PRO™-1, POPO™-3/PO-PRO™-3, Propidium Iodide (PI), PyMPO, Pyrene, Pyronin Y, Quantam Red (PE-Cy5), Quinacrine Mustard, R670 (PE-Cy5), Red 613 (PE-Texas Red), Red Fluorescent Protein (DsRed), Resorufin, RH 414, Rhod-2, Rhodamine B, Rhodamine Green™, Rhodamine Red™, Rhodamine Phalloidin, Rhodamine 110, Rhodamine 123, 5-ROX (carboxy-X-rhodamine), S65A, S65C, S65L, S65T, SBF1, SITS, SNAFL®-1 (high pH), SNAFL®-2, SNARF®-1 (high pH), SNARF®-1 (low pH), Sodium Green™, SpectrumAqua®, SpectrumGreen® #1, SpectrumGreen® #2, SpectrumOrange®, SpectrumRed®, SYTOR 11, SYTOR 13, SYTOR 17, SYTOR 45, SYTOX® Blue, SYTOX® Green, SYTOX® Orange, 5-TAMRA (5-Carboxytetramethylrhodamine), Tetramethylrhodamine (TRITC), Texas Red®/Texas Red®-X, Texas Red®-X (NHS Ester), Thiadiazolecarbocyanine, Thiazole Orange, TOTO®-1/TO-PRO®-1, TOTO®-3/TO-PRO®-3, TO-PRO®-5, Tri-color (PE-Cy5), TRITC (Tetramethylrhodamine), TruRed (PerCP-Cy5.5), WW 781, X-Rhodamine (XRITC), Y66F, Y66H, Y66W, YFP (Yellow Fluorescent Protein), YOYO®-1/YO-PRO®-1, YOYO®-3/YO-PRO®-3, 6-FAM (Fluorescein), 6-FAM (NHS Ester), 6-FAM (Azide), HEX, TAMRA (NHS Ester), Yakima Yellow, MAX, TET, TEX615, ATTO 488, ATTO 532, ATTO 542, ATTO 550, ATTO 565, ATTO Rho101, ATTO 590, ATTO 633, ATTO 647N, TYE 563, TYE 665, TYE 705, 5' IRDye® 700, 5' IRDye® 800, 5' IRDye® 800CW (NHS Ester), WellRED D4 Dye, WellRED D3 Dye, WellRED D2 Dye, Lightcycler® 640 (NHS Ester), and Dy 750 (NHS Ester).

Linkers

[0070] In some embodiments, the single nucleotide of each nucleotide-fluorophore complex is attached to the core through a linker. In some embodiments, the nucleotide is attached to the linker through 5' end of the nucleotide. In some embodiments, the nucleotide is sterically accessible to one or more proteins, one or more enzymes, and a priming strand bound to a template nucleic acid molecule.

[0071] In some embodiments, a fluorophore of the plurality of fluorophores is attached to the core through a linker. In some embodiments, a single fluorophore is attached to the core through a single linker. In some embodiments, a plurality of fluorophores (e.g., 2, 3, 4, or more fluorophores) are attached to the core through a single linker.

[0072] In some embodiments, a linker attaching the nucleotide to the core and a linker attaching a fluorophore of the plurality to the core are of the same type. In some embodiments, a linker attaching the nucleotide to the core and a linker attaching a fluorophore of the plurality to the core are different.

[0073] In some embodiments, a linker comprises a polymer chain. In some embodiments, the polymer chain comprises at least about 2, at least about 5, at least about 8, at least about 10, at least about 12, at least about 15, at least about 20, at least about 25, at least about 50, at least about 75, or at least about 100 monomer units. In some embodiments, the polymer chain is polyethylene glycol, polypropylene glycol, polyvinyl acetate, polylactic acid, or polyglycolic acid. In some embodiments, the polymer chain is a linear or branched molecule. In some embodiments, the linker comprises a polyethylene glycol (PEG) linker. In some embodiments, the PEG linker comprises at least about 2, at least about 5, at least about 8, at least about 10, at least about 12, at least about 15, at least about 20, at least about 25, at least about 50, at least about 75, or at least about 100 ethylene glycol units. In some embodiments, the PEG linker comprises about 2 to about 24 ethylene glycol units.

[0074] In some embodiments, a linker is a biotinylated linker and the core is avidin or streptavidin core. In some embodiments, the biotinylated linker is a biotinylated linear polyethylene glycol (PEG) linker (e.g., wherein biotin is attached to one end of the linear PEG linker and the fluorescent dye or nucleotide is attached to the other end of the linear PEG linker).

[0075] In some embodiments, the linker comprises a cleavable linker. In some embodiments, the cleavable linker comprises a photocleavable linker, a Pd-cleavable linker, a phosphine-cleavable linker, or a disulfide linker. In some embodiments, cleavage of the linker releases the core from the nucleotide incorporated into the priming strand. In some embodiments, cleavage of the linker releases at least a subset of fluorophores of the plurality from the core. In some embodiments, the nucleotide and at least a subset of fluorophores of the plurality are each attached to the core through a cleavable linker. In some embodiments, the nucleotide and at least a subset of fluorophores of the plurality are each attached to the core through the same type of cleavable linker. In some embodiments, the nucleotide and at least a subset of fluorophores of the plurality are cleaved from the core at the same time. In some embodiments, the nucleotide and at least a subset of fluorophores of the plurality are attached to the core through different cleavable linkers. In some embodiments, the nucleotide and at least a subset of fluorophores of the plurality are cleaved from the core sequentially (in either order). In some embodiments, the linker does not comprise a photo emitting or photo absorbing group.

[0076] In some embodiments, a linker comprises a photocleavable linker. Any suitable photocleavable linker can be used (see, e.g., Seo et al. (2005), PNAS 102 (17): 5926-5931, incorporated by reference herein in its entirety). In some embodiments, the photocleavable linker comprises a nitrobenzyl group. For instance, a photocleavable nitrobenzyl linker can be cleaved using laser irradiation (355 nm, 10 seconds, 1.5 W/cm²).

[0077] In some embodiments, a linker comprises a Pd-cleavable linker. Any suitable Pd-cleavable linker can be used (see, e.g., Ju et al. (2006), PNAS 103 (52): 19635-19640, incorporated by

reference herein in its entirety). In some embodiments, the Pd-cleavable linker comprises an allyl group. For instance, a Pd-cleavable allyl linker can be cleaved using incubation with a Na.sub.2PdCl.sub.4/P(PhSO.sub.3Na).sub.3 mixture (30 seconds at 70° C.).

[0078] In some embodiments, a linker comprises a phosphine-cleavable linker. Any suitable phosphine-cleavable linker can be used (see, e.g., Guo et al. (2008), PNAS 105 (27): 9145-9150, incorporated by reference herein in its entirety). In some embodiments, the phosphine-cleavable linker comprises an azide group. For instance, a phosphine-cleavable azide linker can be cleaved using incubation with a Tris(2-carboxyethyl) phosphine (TCEP) mixture (15 minutes at 65° C.).

[0079] In some embodiments, a linker comprises a disulfide bond. For instance, the disulfide bond can be cleaved using incubation with a reducing agent, such as beta-mercaptoethanol, TCEP, or dithiothreitol (DTT).

III. Nucleic Acid Sequencing Using Nucleotide-Fluorophore Complexes

[0080] The sequencing methods described herein are useful for multi-cycle sequencing approaches where nucleotides of nucleotide strand are “interrogated” by binding to a complementary nucleotide. For example, the sequencing methods described herein are applicable to both in situ sequencing applications (e.g., in situ sequencing of endogenous nucleic acid sequences and/or target-specific barcode sequences associated with target analytes of interest that are distributed within a cell or tissue sample) and to more conventional “sequencing in a flow cell” applications (e.g., sequencing of endogenous nucleic acid sequences extracted from a cell or tissue sample). The in situ and flow cell sequencing approaches differ in terms of the sample preparation steps required, as described elsewhere herein, but can share common features in terms of the cyclic series of steps performed to identify nucleotides base-by-base in a template nucleic acid sequence (e.g., a target analyte sequence and/or an associated target-specific barcode sequence).

[0081] In some embodiments, the sequencing methods described herein comprise contacting a template nucleic acid molecule with a sequencing primer designed to hybridize to a portion of the template nucleic acid molecule. In some embodiments, the sequencing primer comprises a free 3'-hydroxyl group at its 3' terminus, and a primer extension reaction is performed to incorporate a nucleotide of a nucleotide-fluorophore complex into the priming strand. In some embodiments, an unlabeled nucleotide is incorporated into the priming strand.

[0082] In some embodiments, the sequencing methods described herein comprises performing a cyclic series of base-by-base sequencing reactions, where each sequencing cycle comprises: (a) contacting the priming strand bound to the template nucleic acid molecule with a polymerase and a composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex comprises: i) a single nucleotide attached to a core; and ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide; (b) incorporating one of the nucleotide-fluorophore complexes of the first composition of nucleotide-fluorophore complexes into the priming strand to form an extended priming strand; and (c) detecting the incorporated nucleotide-fluorophore complex. In some embodiments, the method further comprises identifying a complementary nucleotide in the template nucleic acid molecule based on the detecting step. In some embodiments, detecting the presence of the nucleotide-fluorophore complex comprises detecting a fluorescence signal associated with a fluorescently-labeled nucleotide-fluorophore complex. In some embodiments, identifying a complementary nucleotide in the template nucleic acid molecule comprises detecting an absence of a signal (e.g., an unlabeled nucleotide of the composition is incorporated into the priming strand to form an extended priming strand).

Template Nucleic Acid Molecules

[0083] In some embodiments, the template nucleic acid molecule includes a target analyte nucleic acid molecule (e.g., a DNA molecule, an RNA molecule, or an mRNA molecule). In some embodiments, the template nucleic acid includes a reporter oligonucleotide, such as a barcode.

[0084] In some embodiments, the template nucleic acid molecule is a DNA molecule. Examples of

DNA template nucleic acid molecules include DNA molecules such as single-stranded DNA (ssDNA), double-stranded DNA (dsDNA), genomic DNA, methylated DNA, specific methylated DNA sequences, fragmented DNA, mitochondrial DNA, in situ synthesized PCR products, and RNA/DNA hybrids. In some embodiments, the DNA molecule is copied from another nucleic acid molecule (e.g., DNA or RNA such as mRNA).

[0085] In some embodiments, the template nucleic acid molecule is an RNA molecule. Examples of RNA template nucleic acid molecules include RNA molecules such as various types of coding and non-coding RNA. Examples of the different types of RNA molecules include messenger RNA (mRNA), including a nascent RNA, a pre-mRNA, a primary-transcript RNA, and a processed RNA, such as a capped mRNA (e.g., with a 5' 7-methyl guanosine cap), a polyadenylated mRNA (poly-A tail at 3' end), and a spliced mRNA in which one or more introns have been removed. Also included in the analytes disclosed herein are non-capped mRNA, a non-polyadenylated mRNA, and a non-spliced mRNA. In some embodiments, the RNA template nucleic acid molecule is copied from another nucleic acid molecule (e.g., DNA or RNA such as viral RNA) present in a tissue sample. Examples of a non-coding RNAs (ncRNA) that is not translated into a protein include transfer RNAs (tRNAs) and ribosomal RNAs (rRNAs), as well as small non-coding RNAs such as microRNA (miRNA), small interfering RNA (siRNA), Piwi-interacting RNA (piRNA), small nucleolar RNA (snoRNA), small nuclear RNA (snRNA), extracellular RNA (exRNA), small Cajal body-specific RNAs (scaRNAs), and the long ncRNAs such as Xist and HOTAIR. In some embodiments, the RNA is small (e.g., less than 200 nucleic acid bases in length) or large (e.g., RNA greater than 200 nucleic acid bases in length). Examples of small RNAs include 5.8S ribosomal RNA (rRNA), 5S rRNA, tRNA, miRNA, siRNA, snoRNAs, piRNA, tRNA-derived small RNA (tsRNA), and small rDNA-derived RNA (srRNA). In some embodiments, the RNA is double-stranded RNA or single-stranded RNA. In some embodiments, the RNA is circular RNA. In some embodiments, the RNA is a bacterial rRNA (e.g., 16s rRNA or 23s rRNA).

[0086] In some embodiments, the template nucleic acid comprises a nucleic acid analyte derived from a biological sample and/or a reporter oligonucleotide or nucleic acid marker associated with an analyte from a biological sample. Such analytes can be or derived from any biological sample. In some embodiments, the template nucleic acid comprises a nucleic acid analyte and/or a reporter oligonucleotide or nucleic acid marker associated with an analyte present in a biological sample, and the template nucleic acid molecule is sequenced at a location in the biological sample. Methods and compositions disclosed herein may be used for analyzing a biological sample, which may be obtained from a subject using any of a variety of techniques including, but not limited to, biopsy, surgery, and laser capture microscopy (LCM), and generally includes cells and/or other biological material from the subject. In addition to the subjects described above, in some embodiments, a biological sample is obtained from a prokaryote such as a bacterium, an archa, a virus, or a viroid. In some embodiments, a biological sample is obtained from non-mammalian organisms (e.g., a plant, an insect, an arachnid, a nematode, a fungus, or an amphibian). In some embodiments, a biological sample is obtained from a eukaryote, such as a tissue sample, a patient derived organoid (PDO) or patient derived xenograft (PDX). In some embodiments, a biological sample from an organism comprises one or more other organisms or components therefrom. For example, a mammalian tissue section may comprise a prion, a viroid, a virus, a bacterium, a fungus, or components from other organisms, in addition to mammalian cells and non-cellular tissue components. In some embodiments, subjects from which biological samples are obtained are healthy or asymptomatic individuals, individuals that have or are suspected of having a disease (e.g., a patient with a disease such as cancer) or a pre-disposition to a disease, and/or individuals in need of therapy or suspected of needing therapy.

[0087] In some embodiments, a template nucleic acid includes a reporter oligonucleotide or marker associated with the presence of an analyte (e.g., an endogenous analyte) in a sample. Such analytes may include nucleic acid analytes and/or non-nucleic acid analytes. Examples of non-nucleic acid

analytes include, but are not limited to, lipids, carbohydrates, peptides, proteins, glycoproteins (N-linked or O-linked), lipoproteins, phosphoproteins, specific phosphorylated or acetylated variants of proteins, amidation variants of proteins, hydroxylation variants of proteins, methylation variants of proteins, ubiquitylation variants of proteins, sulfation variants of proteins, viral coat proteins, extracellular and intracellular proteins, antibodies, and antigen binding fragments. In some embodiments, the analyte is inside a cell or on a cell surface, such as a transmembrane analyte or one that is attached to the cell membrane. In some embodiments, the analyte is an organelle (e.g., nuclei or mitochondria). In some embodiments, the analyte is an extracellular analyte, such as a secreted analyte. Examples of analytes include, but are not limited to, a receptor, an antigen, a surface protein, a transmembrane protein, a cluster of differentiation protein, a protein channel, a protein pump, a carrier protein, a phospholipid, a glycoprotein, a glycolipid, a cell-cell interaction protein complex, an antigen-presenting complex, a major histocompatibility complex, an engineered T-cell receptor, a T-cell receptor, a B-cell receptor, a chimeric antigen receptor, an extracellular matrix protein, a posttranslational modification (e.g., phosphorylation, glycosylation, ubiquitination, nitrosylation, methylation, acetylation or lipidation) state of a cell surface protein, a gap junction, and an adherens junction.

[0088] In some embodiments, a template nucleic acid molecule is a denatured nucleic acid, wherein the resulting denatured nucleic acid is single-stranded. The nucleic acid may be denatured, for example, optionally using formamide, heat, or both formamide and heat. In some embodiments, the nucleic acid is not denatured for use in a method disclosed herein.

[0089] In some embodiments, methods and compositions disclosed herein are used to analyze any number of template nucleic acid molecules (e.g., nucleic acid analytes and/or analyte-associated barcode sequences) or fragments thereof. For example, in some embodiments, the number of analytes that are analyzed is at least about 2, at least about 3, at least about 4, at least about 5, at least about 6, at least about 7, at least about 8, at least about 9, at least about 10, at least about 11, at least about 12, at least about 13, at least about 14, at least about 15, at least about 20, at least about 25, at least about 30, at least about 40, at least about 50, at least about 100, at least about 1,000, at least about 10,000, at least about 100,000 or more different analytes present in a region of a sample (e.g., a cell sample or tissue sample) or tethered within individual features on a substrate (e.g., a flow cell surface).

Nucleotide-Fluorophore Complex

[0090] In some embodiments, each cycle of a cyclic series of base-by-base sequencing reactions performed as part of the disclosed methods for in situ or flow cell sequencing comprises contacting priming strands bound to template nucleic acid molecules with a composition comprising nucleotide-fluorophore complexes. Each nucleotide-fluorophore complex of the composition comprises: i) a single nucleotide attached to a core; and ii) a plurality of fluorophores attached to the polymer core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide. In some embodiments, each cycle of a cyclic series of base-by-base sequencing reactions comprises contacting the priming strand bound to a template nucleic acid molecule with a composition comprising a plurality of sets of nucleotide-fluorophore complexes (e.g., 2, 3, or 4 sets of nucleotide-fluorophore complexes). In some embodiments, each set of the of nucleotide-fluorophore complexes comprises a same nucleobase that differs from the other sets.

[0091] In some embodiments, nucleotide-fluorophore complexes of a composition contacted with the primed template nucleic acid molecule(s) in each cycle of a multicycle sequencing process comprise A, T, U, C, and/or G. In some embodiments, nucleotide-fluorophore complexes of a composition contacted with the primed template nucleic acid molecule(s) in each cycle of a multicycle sequencing process comprise A, T, C, and/or G.

[0092] In some embodiments, the composition comprising nucleotide-fluorophore complexes contacted with the primed template nucleic acid molecule(s) is the same in each cycle of a multicycle sequencing process (e.g., the composition comprises the same set of nucleotide-

fluorophore complexes in each cycle, wherein each set of nucleotide-fluorophore complexes comprises the same selection of A, T, U, C, and/or G).

[0093] In some embodiments, the composition comprising nucleotide-fluorophore complexes contacted with the primed template nucleic acid molecule(s) is different between at least 2 cycles of a multicycle sequencing process (e.g., the composition comprises different sets of nucleotide-fluorophore complexes in different cycles, wherein the sets of nucleotide-fluorophore complexes comprise a different selection of A, T, U, C, and/or G, and/or wherein the sets of nucleotide-fluorophore complexes comprise a different selection fluorophores).

[0094] In some embodiments, a composition comprises 4 sets of nucleotide-fluorophore complexes, wherein each set comprises a same nucleobase (e.g., selected from A, T, U, C, and/or G) that differs from the other sets, and wherein, for each set, the plurality of fluorophores is a plurality of a same fluorophore that differs from the other sets.

[0095] In some embodiments, a composition comprises a set of unlabeled nucleotides (e.g., not conjugated to a fluorophore) in addition to the nucleotide-fluorophore complexes. In some embodiments, a set of unlabeled nucleotides is used to implement different detection schemes (e.g., two color, three color, or four color detection schemes). In some embodiments, a composition comprises three sets of nucleotide-fluorophore complexes, each set comprising a same nucleobase (e.g., selected from A, T, U, C, and/or G) that differs from the other sets, and wherein, for each set, the plurality of fluorophores is a plurality of a same fluorophore that differs from the other sets, and wherein the composition further comprises a set of unlabeled nucleotides comprising a same nucleobase that differs from the nucleobase of the three sets of nucleotide-fluorophore complexes. In some embodiments, a set of unlabeled nucleotides is used to implement the readout of target-specific barcode designs used to minimize optical crowding when performing in situ sequencing (see, e.g., PCT International Patent Application Publication Nos. WO 2022/060889 and WO 2023/220300, and U.S. Patent Publication Nos. US20220084629A1 and US20240084378A1, each of which is herein incorporated by reference in its entirety).

Polymerases

[0096] Examples of polymerases that are used for performing the disclosed methods include, but are not limited to, DNA polymerases (e.g., Taq DNA polymerase), RNA polymerases, and/or reverse transcriptases.

[0097] In some embodiments, the polymerase of the polymerase conjugate is a DNA polymerase. Examples of DNA polymerases include Taq polymerase, 9°N-7 DNA polymerase (or variants thereof, for example, D141A/E143A/A485L), phi29 (φ29) polymerase, Klenow fragment, *Bacillus stearothermophilus* DNA polymerase (BST), T4 DNA polymerase, T7 DNA polymerase, and DNA polymerase I. In some embodiments, DNA polymerases that have been engineered or mutated to have desirable characteristics are employed. In some embodiments, the polymerase is phi29 DNA polymerase. In some embodiments, the polymerase of the polymerase conjugate is a DNA polymerase and the template nucleic acid molecule includes DNA. In some embodiments, the polymerase of the polymerase conjugate is a DNA polymerase and the nucleotide molecules include deoxyribonucleotides.

[0098] In some embodiments, the DNA polymerase is Taq polymerase or a functional variant thereof. Taq polymerase is a heat stable polymerase from *Thermus aquaticus*. An example Taq polymerase sequence is:

TABLE-US-00001 (SEQ ID NO: 1)

GMLPLFEPKGRVLLVDGHHLAYRTFHALKGLTTSRGEPVQAVYGFAKSL
KALKEDGDAVIVVFDKAPSRHEAYGGYKAGRPTPEDFPRQLALIKEL
VDLLGLARLEVPGYEADDVLASLAKKAEKEGYEVRILTADKDLYQLLSDR
IHVLHPEGYLITPAWLWEKYGLRPDQWADYRALTGDESDNLPGVKGIGEK
TARKLLEWGSLEALLKNLDRPKPAIREKILAHMDDLKLSWDLAKVRTDL
PLEVDFAKRREPDRERLXAFLERLEFGSLLHEFGLLSPKXLXEAPWPPP ERAFVP.

[0099] In some embodiments, the DNA polymerase is phi29 DNA polymerase or a functional variant thereof. The DNA polymerase of phi29 (a phage of *Bacillus subtilis*) has high processivity and fidelity. An example phi29 DNA polymerase sequence is:

TABLE-US-00002 (SEQ ID NO: 2)

MKHMPRKMYSCDFETTTKVEDCRVWAYGYMNIEDHSEYKIGNSLDEFMAW
VLKVQADLYFHNLFKDFGAFIINWLERNGFKWSADGLPNTYNTIISRMGQW
YMIDICLGKYGKRKIHTVIYDSLKKLPFPVKKIAKDFKLTVLKGDIDYHK
ERPVGKITPEEYAYIKNDIQIIAEALLIQFKQGLDRMTAGSDSLKGFKD
IITTKKFKKVFP TSLGLDKEVRYAYRGGFTWLNDRFKEKEIGEGMVFDV
NSLYPAQMYSRLLPYGEPVFEFGKYVWDEDYPLHIQHIRCEFELKEGYIP
TIQIKRSRFYKGNEYLKSSGGEIADLWLSNVDLELMKEHYDLYNVEYISG
LKFKATTGLFKDFIDKWTYIKTTSEGAIKQLAKLMLNSLYGKFASNP DVT
GKVPY LKENGALGFRLGEEETKDPVYTPMGVFITAWARYTTITAAQACYD
RIIYCDTDSIHLTGTEIPDVIKDIVDPKKLGYWAHESTFKRAKYLRQKTY
IQDIYMKEVDGKLVEGSPDDYTDIKFSVKCAGMTDKIKKEVTFENFKVGF
SRKMKPKPVQVPGGVVLVDDTFTIK.

[0100] In some embodiments, the DNA polymerase is a 9°N-7 DNA polymerase or a functional variant thereof (e.g., D141A/E143A/A485L). 9°N-7 is a strain of *Thermococcus* sp. An example of a 9°N-7 DNA polymerase sequence is:

TABLE-US-00003 (SEQ ID NO: 3)

MILDTDYITENGKPVIRVFKKENGEFKIEYDRTFEPYFYALLKDDSAIED
VKKVTAKRHGTVVKVKRAEKVQKKFLGRPIEVWKLYFNHPQDVP AIRDRI
RAHPAVVDIYEYDIPFAKRYLIDKGLIPMEGDEELTMLAFDIETLYHEGE
EFGTGPILMISYADGSEARVITWKKIDLPYVDVVSTEKEMIKRFLRVVRE
KDPDVLITYNGDNFDFAYLKKRCEELGIKFTLGRDGSEP KIQRMGDRFAV
EVKGRIHFDLYPVIRRTINLPTYTLEAVYEAVFGKPKEKVYAE EIAQAW
SGEGLERVARYSMEDAKVTYELGREFFPMEAQLSRLIGQSLWDVSRSTG
NLVEWFLLRKAYKRNELAPNKPDERELARRRGGYAGGYVKEPERGLWDNI
VYLDFRSLYPSIIITHNVSPDTLNREGCKEYDVAPEVGHKFCKDFPGFIP
SLLGDLLEERQKIKRKM KATVDPLEKKLLDYRQRAIKILANSFYGYGYA
KARWYCKECAESVTAWGREYIEMVIRELEEKFGFKVLYADTDGLHATIPG
ADAETVKKKAKEFLKYINPKLPGLLELEYEGFYVRGFFVTKKKYAVIDEE
GKITTRGLEIVRRDWSEIAKETQARVLEAILKHGDVEEAVRIVKEVTEKL
SKYEVPPEKLV IHEQITRDLRDYKATGPHVAVAKRLAARGVKIRPGTVIS
YIVLKGSGRIGDRAIPADEFDPTKHRYDAEYYIENQVLP AVERILKAFGY
RKEDLRYQKTKQVGLGAWLKVKGKK.

[0101] In some embodiments, the DNA polymerase is DNA polymerase I or a functional fragment thereof (e.g., a Klenow fragment). Klenow fragment is an exonuclease deficient fragment of DNA polymerase I. An example of DNA polymerase I sequence is:

TABLE-US-00004 (SEQ ID NO: 4)

MVQIPQNPLILVDGSSYLYRAYHAFPLTNSAGEPTGAMYGV LNMRLSLI
MQYKPTHAAVVFDAKGKTRDELFEHYKSHRPPMPDDLRAQIEPLHAMVK
AMGLPLLAVSGVEADDVIGTLAREAEKAGRPVLISTGDKDMAQLVTPNIT
LINTMTNTILGPEEVVNKYGVPELIIIDFLALMGDSSDNIPGVPGVGEKT
AQALLQGLGGLDTLYAEPEKIAGLSFRGAKTMAAKLEQNKEVAYLSYQLA
TIKTDVELELTCEQLEVQQPAAEELLGLFKKYEFKRWTADVEAGKWLQAK
GAKPAAKPQETSVADEAPEVTATVISYDNYVTILDEETLKAWIAKLEKAP
VFAFDTETDSL DNISANLVGLSFAIEPGVAAYIPVAHDYLDAPDQISRER
ALELLKPLLEDEKALKVGQNLKYDRGILANYGIELRGIAFDTMLESYILN
SVAGRHDMDSLAERWLKHKTTTFEEIAGKGKNQLTFNQIALEEAGRYAAE

DADVTLQLHKLKQKHKGPLNVFENIEMPLVPVLSRIERNNGVKIDPK
VLHNHSEELTLRLAELEKKAHEIAGEEFNLSSTKQLQTLFEKQGKPLK
KTPGGAPSTSEEVLLEALDYPLPKVILEYRGLAKLKSTYTDKLPLMINP
KTGRVHTSYHQAVTATGRLSSTDPNLQNIQVRNEEGRRIRQAFIAPEDYV
IVSADYSQIELRIMAHLSRDKGLLTAFAGKDIHRATAAEVFGPLPLETVT
SEQRRSAKAINFGLIYGMSAFGLARQLNIPRKEAQKYMDLYFERYPGVLE
YMERTRAQAKEQGYVETLDGRRLLYLPDIKSSNGARRAAAERAAINAPMQG
TAADIKRAMIAVDAWLQAEQPRVRMIMQVHDELVEVHKDDVDAVAKQI
HQLMENCTRLDVPLLVEVGSGENWDQAH.

In some embodiments, a Klenow fragment includes positions 324-928 with respect to SEQ ID NO: 4.

[0102] In some embodiments, the polymerase of the polymerase conjugate is a reverse transcriptase. Reverse transcriptases typically have RNA-dependent DNA polymerase activity and DNA-dependent DNA polymerase activity. Examples of reverse transcriptases include Moloney murine leukemia virus (MMLV) reverse transcriptase, HIV-1 reverse transcriptase, and avian myeloblastosis virus (AMV) reverse transcriptase. In some embodiments, the reverse transcriptase lacks (e.g., is mutated to lack) ribonuclease activity. In some embodiments, ribonuclease activity degrades template particularly during longer incubation times such as when reverse transcribing longer cDNAs. In some embodiments, the polymerase of the polymerase conjugate is a reverse transcriptase and the template nucleic acid molecule is an RNA molecule. In some embodiments, the polymerase of the polymerase conjugate is a reverse transcriptase and the nucleotide molecules include deoxyribonucleotide molecules.

[0103] In some embodiments, the reverse transcriptase is an MMLV reverse transcriptase or a functional variant thereof. An example of an MMLV reverse transcriptase sequence is:

TABLE-US-00005 (SEQ ID NO: 5)

AFPLERPDWDYTTQAGRNHLVHYRQLLLAGLQNAGRSPTNLAKVKGITQG
PNESPSAFLERLKEAYRRYTPYDPEDPGQETNVSMSFIWQSAPDIGRKLK
RLEDLKSKTLGDLVREAEKIFNKRETPEEREERIRRETEEKEERRRTVDE
QKEKERDRRRHREMSKLLATVVIGQEQRQGERKRPQLDKDQCAYCKEK
GHWAKDCPKKPRGPRGPRPQTSLLTLGDXGGQGQDPPPEPRITLKVGGQP
VTFLVDTGAQHSVLTQNPGLSDKSAWVQGATGGKRYRWTTDRKVHLATG
KVTHSFLHVPDCPYPLLGRDLLTKLKAQIHFEQSGAQVVGPMGQPLQVLT
LNIEDEYRLHETSKEPDVSLGFTWLSDFPQAWAESGGMGLAVRQAPLIIP
LKATSTPVSQKQYPMSEARLGKIPHIQRLLDQGILVPCQSPWNTPLLVP
KKPGTNDYRPVQDLREVNRVEDIHPTVPNPYNLLSGLPPSHQWYTVLDL
KDAFFCLRLHPTSQPLFAFEWRDPEMGISGQLTWTRLPGGFKNSPTLFDE
ALHRDLADFR. Residues 431-560 of SEQ ID NO: 5 provide reverse transcriptase activity.

[0104] In some embodiments, the reverse transcriptase is an HIV-1 reverse transcriptase or a functional variant thereof. An example of an HIV-1 reverse transcriptase sequence is:

TABLE-US-00006 (SEQ ID NO: 6)

PISPIEPVPVKLKPGMDGPKVKQWPLTEEKIKALVEICTEMEKEGKISKI
GPENPYNTPVFAIKKKDSTRWRKLVDFRELNRKTQDFWEVQLGIPHPAGL
KKKRSVTVLDVGDAYFSVPLDKEFRKYTAFTIPSINNETPGIRYQYNVLP
QGWKGSPIFQSSMTKILEPFRKQNPDIVIYQYMDDLYVGSDDLEIGQHRT
KIEELRQHLLKWGFTTPDKKHQKEPPFLWMGYEHHPDKWTVQPIVLPEKD SWTVNDIQK.

[0105] In some embodiments, the polymerase is selected from Taq polymerase, 9°N-7 DNA polymerase or a functional variant thereof (e.g., D141A/E143A/A485L), and a Klenow fragment of DNA polymerase I. In some embodiments, the polymerase is not labeled with a detectable label.

Workflows for Performing Sequencing

[0106] The disclosed sequencing methods may be applied to both in situ sequencing and flow cell

sequencing applications, where the sequencing reactions are substituted for the stepwise nucleotide incorporation reactions used to probe a template nucleic acid sequence in, e.g., a conventional sequencing-by-synthesis (SBS) method.

[0107] In the case of in situ sequencing, in some embodiments, the disclosed methods comprise performing all or a subset of the following steps (in addition to a cyclic series of base-by-base sequencing reactions): [0108] (i) preparing the biological sample (e.g., by fixing, sectioning, embedding, and/or clearing a cell or tissue sample, as described elsewhere herein). [0109] (ii) contacting target analytes (e.g., target nucleic acid analytes and/or protein analytes) within the prepared sample with target-specific probes, as described elsewhere herein. In some embodiments, the target-specific probes comprise, e.g., target-specific linear and/or circularizable nucleic acid probes (e.g., padlock probes) designed to hybridize directly or indirectly to specific target nucleic acid analytes. In some embodiments, the target-specific linear and/or circularizable nucleic acid probes comprise primer binding sites and/or target-specific barcode (or identifier) sequences. In some embodiments, the target-specific probes comprise, e.g., target-specific antibodies designed to bind to specific target protein analytes, where the antibodies are conjugated to nucleic acid sequences. In some embodiments, the conjugated nucleic acid sequences comprise primer binding sites and/or target-specific barcode (or identifier) sequences. [0110] (iii) optionally performing a reverse transcription reaction (e.g., if the probed target nucleic acid analytes comprise RNA molecules) to create cDNA copies of RNA target molecules. [0111] (iv) optionally amplifying the probed target analyte molecules and/or their associated target-specific barcode sequences (e.g., using rolling circle amplification (RCA) in the case that target-specific circularizable probes were used to probe target analyte molecules and/or associated barcode sequences). [0112] (v) contacting the optionally amplified target nucleic acid analytes and/or associated target-specific barcode sequences with sequencing primers designed to hybridize directly or indirectly to the target nucleic acid analytes and/or their associated target-specific barcode sequences. In some embodiments, the sequencing primers comprise free 3'-hydroxyl groups at their 3' termini, and a primer extension reaction is performed to incorporate a single nucleotide of a nucleotide-fluorophore complex at 3' termini of a bound primer (i.e., 3' termini of the priming strands).

[0113] In the case of flow cell sequencing, in some embodiments, the disclosed methods comprise performing all or a subset of the following steps (in addition to a cyclic series of base-by-base sequencing reactions): [0114] (i) extraction and purification of nucleic acid molecules (e.g., endogenous nucleic acid sequences) from a biological sample, as described elsewhere herein. [0115] (ii) preparation of a sequencing library comprising template nucleic acid molecules (e.g., the endogenous nucleic acid sequences or fragments thereof) that have been end-repaired and ligated to adapter sequences, as described elsewhere herein. [0116] (iii) optionally performing nucleic acid amplification of all or a portion of the sequencing library, as described elsewhere herein. [0117] (iv) immobilizing the template nucleic acid molecules (e.g., denatured, single-stranded template nucleic acid molecules) from the sequencing library on an inner surface of a flow cell using capture probes (e.g., complementary adapter sequences) that have been tethered to the flow cell surface. [0118] (v) performing clonal amplification of the immobilized template nucleic acid molecules to create clusters comprising, e.g., thousands or tens of thousands of copies of the template nucleic acid molecule immobilized at each of a plurality of locations on the flow cells surface. [0119] (vi) contacting the template nucleic acid molecules in each clonally-amplified cluster with sequencing primers designed to hybridize to, e.g., the adapter sequences ligated to the template nucleic acid molecules. In some embodiments, the sequencing primers comprise free 3'-hydroxyl groups at their 3' termini, and a primer extension reaction is performed to incorporate a single nucleotide of a nucleotide-fluorophore complex at 3' termini of a bound primer (i.e., 3' termini of the priming strands).

[0120] In the case of either in situ sequencing or flow cell sequencing, in some embodiments, the disclosed methods comprise: performing a cyclic series of base-by-base sequencing reactions,

where each sequencing cycle comprises: [0121] (a) contacting each priming strand bound to a template nucleic acid molecule with a polymerase and a first composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex comprises: i) a single nucleotide attached to a core; and ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide; [0122] (b) incorporating one of the nucleotide-fluorophore complexes of the first composition of nucleotide-fluorophore complexes into the priming strand to form an extended priming strand; and [0123] (c) detecting the incorporated nucleotide-fluorophore complex. In some embodiments, the method further comprises identifying a complementary nucleotide in the template nucleic acid molecule based on the detecting step. In some embodiments, detecting the presence of the incorporated nucleotide-fluorophore complex comprises detecting a fluorescence signal associated with a nucleotide-fluorophore complex. In some embodiments, identifying a complementary nucleotide in the template nucleic acid molecule comprises detecting an absence of a signal (e.g., an unlabeled nucleotide of the composition is incorporated into the priming strand to form an extended priming strand).

[0124] In some embodiments, the disclosed methods further comprise processing optical signals (e.g., fluorescence signals) detected in images (e.g., fluorescence images) acquired during the cyclic series of base-by-base sequencing reactions to detect the presence or absence of complementary single nucleotides of nucleotide-fluorophore complexes in the extended priming strand in each sequencing cycle at the locations of each of a plurality of template nucleic acid molecules (i.e., the locations corresponding to each of a plurality of target analyte molecules and/or their associated target-specific barcode sequences), thereby enabling inference of the nucleotide sequence of the plurality of template nucleic acid molecules (e.g., the plurality of target analyte molecules and/or associated target-specific barcode sequences).

[0125] In some embodiments, the cyclic series of base-by-base sequencing reactions comprises performing at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, or more than 50 cycles of the base-by-base sequencing reaction.

[0126] In some embodiments, each cycle of base-by-base sequencing further comprises a first wash step following the contacting step to remove unbound polymerase and nucleotide-fluorophore complexes. In some embodiments, the first wash step comprises, for example, use of the same buffer used for contacting the primed template nucleic acid with a polymerase and a composition comprising nucleotide-fluorophore complexes (but without the polymerase and composition). In some embodiments, the first wash buffer does not include KCl and/or includes little to no DMSO. In some embodiments, the first wash buffer is similar to those used for wash buffers as used in wash steps of a Western blot (e.g., a wash buffer added in a Western blot after binding a primary antibody but washing prior to incubation with a secondary antibody, such as PBST). PBST is a phosphate-buffered saline with a low-concentration of detergent, such as 0.05% to 0.1% Tween.

[0127] In some embodiments, each cycle of base-by-base sequencing further comprises removing the fluorophores from the core of the incorporated nucleotide-fluorophore complex following the detection step. In some embodiments, at least a subset of fluorophores of the plurality in each nucleotide-fluorophore complex are each attached to the core through a cleavable linker. In some embodiments, each cycle of base-by-base sequencing further comprises cleaving the linker between a fluorophore and the core of the incorporated nucleotide-fluorophore complex. Exemplary cleavable linkers and reaction conditions for cleaving such linkers are described elsewhere herein.

[0128] In some embodiments, each cycle of base-by-base sequencing further comprises photobleaching the plurality of fluorophores of the incorporated nucleotide-fluorophore complex following the detection step. In some embodiments, the plurality of fluorophores of the incorporated nucleotide-fluorophore complex is unable to fluoresce after photobleaching. Any suitable photobleaching methods can be implemented. In some embodiments, the sample is exposed to a light source until the signal emitted by the plurality of fluorophores is eliminated.

[0129] In some embodiments, each cycle of base-by-base sequencing further comprises cleaving the core from the nucleotide of the incorporated nucleotide-fluorophore complex, leaving an incorporated nucleotide following the detection step. In some embodiments, the single nucleotide in each nucleotide-fluorophore complex is attached to the core through a cleavable linker. In some embodiments, each cycle of base-by-base sequencing further comprises cleaving the linker between the core and the incorporated nucleotide. Exemplary cleavable linkers and reaction conditions for cleaving such linkers are described elsewhere herein.

[0130] In some embodiments, the nucleotide of each nucleotide-fluorophore complex comprises a reversible terminator moiety. In some embodiments, each cycle of base-by-base sequencing further comprises cleaving the reversible terminator moiety after detecting the incorporated nucleotide-fluorophore complex.

[0131] In some embodiments, the detection step comprises the use of an optical imaging technique (e.g., a fluorescence imaging technique) and real time or post-processing measurement of optical signals (e.g., fluorescence signals or the absence thereof) associated with the presence of a specific nucleotide-fluorophore complex at a plurality of locations corresponding to a plurality of target analytes distributed throughout the biological sample or tethered to specific locations on a substrate surface (e.g., a flow cell surface).

[0132] FIG. 1 provides a non-limiting example of a flowchart for a process 100 for sequencing a template nucleic acid molecule in accordance with one implementation of the methods described herein. In some embodiments, the sequencing steps depicted in FIG. 1 is performed as part of an in situ sequencing method or as part of a flow cell sequencing method. In process 100, in some embodiments, some steps are combined, the order of some steps are changed, and some steps are omitted. In some embodiments, additional steps are performed in combination with the steps shown in process 100. Accordingly, the steps illustrated (and described in greater detail below) for process 100 are exemplary by nature, and as such, should not be viewed as limiting.

[0133] At step 102 in FIG. 1, a priming strand bound to the template nucleic acid molecule is contacted with: (A) a polymerase and (B) a first composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex comprises: i) a single nucleotide attached to a core; and ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide.

[0134] In some embodiments, the template nucleic acid molecule comprises an endogenous nucleic acid molecule (e.g., a DNA molecule, an RNA molecule, or an mRNA molecule) that has been reverse transcribed, amplified, and/or extracted from a biological sample (e.g., a cell sample or tissue sample).

[0135] In some embodiments, the template nucleic acid molecule comprises a barcode sequence (e.g., a nucleic acid barcode sequence) associated with a target analyte of interest (e.g., using the barcoding methods described elsewhere herein) that has been reverse transcribed, amplified, and/or extracted from a biological sample (e.g., a cell sample or tissue sample).

[0136] In some embodiments, the method further comprises hybridizing a circularizable probe to a target analyte (or to a labeling agent bound to the target analyte), ligating the circularizable probe to form a circularized probe, and performing rolling circle amplification of the circularized probe to generate the template nucleic acid molecule. In some embodiments, for example, the circularizable probe is a padlock probe sequence.

[0137] In some embodiments, the template nucleic acid molecule to be sequenced is attached to a solid support, e.g., a sequencing flow cell.

[0138] In some embodiments, the template nucleic acid molecule is sequenced in situ in a cell sample or tissue sample. In some embodiments, the cell sample comprises a layer of cells deposited on a surface.

[0139] In some embodiments, the method further comprises providing: i) one or more reagents comprising the polymerase and the first composition comprising nucleotide-fluorophore complexes

(each comprising a single nucleotide attached to a core and a plurality of fluorophores attached to the core), and ii) the priming strand bound to the template nucleic acid molecule.

[0140] In some embodiments, the polymerase comprises, e.g., Taq polymerase, Therminator™ DNA polymerase, a Klenow fragment of DNA polymerase I, or any combination thereof. In some embodiments, the polymerase is not labeled with a detectable label.

[0141] At step **104** in FIG. **1**, the method further comprises incorporating one of the nucleotide-fluorophore complexes of the first composition of nucleotide-fluorophore complexes into the priming strand to form an extended priming strand.

[0142] In some embodiments, the single nucleotide of each nucleotide-fluorophore complex is attached to the core using a linker, e.g., linear polyethylene glycol (PEG) linker where the polyethylene glycol (PEG) linker comprises from 2 to 24 ethylene glycol units. In some embodiments, the linker comprises a cleavable linker, e.g., a photocleavable linker, a Pd-cleavable linker, a phosphine-cleavable linker, or a disulfide linker.

[0143] In some embodiments, at least a subset of fluorophores in each nucleotide-fluorophore complex is attached to the core using a linker, e.g., a linear polyethylene glycol (PEG) linker where the polyethylene glycol (PEG) linker comprises from 2 to 24 ethylene glycol units. In some embodiments, the linker comprises a cleavable linker, e.g., a photocleavable linker, a Pd-cleavable linker, a phosphine-cleavable linker, or a disulfide linker. In some embodiments, the linker of the nucleotide and the linker of the fluorophores are of the same type. In some embodiments, the linker of the nucleotide and the linker of the fluorophores are different.

[0144] In some embodiments, the core of each of the nucleotide-fluorophore complexes comprises an avidin or streptavidin core, and wherein the nucleotide in each complex is bound to the core via a biotinylated linker (e.g., a biotinylated linear polyethylene glycol (PEG) linker). In some embodiments, at least a subset of the fluorophores in each nucleotide-fluorophore complex are bound to the core via biotin, optionally wherein at least a subset of the fluorophores in each nucleotide-fluorophore complex are bound to the core via a biotinylated linker (e.g., a biotinylated linear polyethylene glycol (PEG) linker).

[0145] In some embodiments, the core of each of the nucleotide-fluorophore complexes comprises a branched polymer core, and wherein the nucleotide and the plurality of fluorophores in each nucleotide-fluorophore complex are attached to the branched polymer core.

[0146] At step **106** in FIG. **1**, the method comprises detecting the incorporated nucleotide-fluorophore complex. In some embodiments, the method further comprises identifying a complementary nucleotide in the template nucleic acid molecule.

[0147] In some embodiments, detecting a presence of the incorporated nucleotide-fluorophore complex comprises detecting a signal associated with the plurality of fluorophores of the nucleotide-fluorophore complex. In some embodiments, identifying a complementary nucleotide in the template nucleic acid molecule comprises detecting an absence of a signal (e.g., wherein an unlabeled nucleotide of the composition is incorporated into the priming strand to form an extended priming strand). In some embodiments, the detection step is performed, e.g., using a fluorescence imaging technique as described elsewhere herein.

[0148] In some embodiments, a composition of nucleotide-fluorophore complexes comprises a mixture with more than one nucleobase type. In some embodiments, provided herein are sets of nucleotide-fluorophore complexes, wherein each set shares a same nucleobase (e.g., selected from A, T, U, C, and/or G) that differs from other sets. In some embodiments, a composition comprises 4 sets of nucleotide-fluorophore complexes, where each set comprises a same nucleobase (e.g., selected from A, T, U, C, and/or G) that differs from the other sets, and wherein, for each set, the plurality of fluorophores is a plurality of a same fluorophore that differs from the other sets.

[0149] In some embodiments, a mixture comprises three sets of nucleotide-fluorophore complexes, each set comprising a same nucleobase (e.g., selected from A, T, U, C, and/or G) that differs from the other sets, and wherein, for each set, the plurality of fluorophores is a plurality of a same

fluorophore that differs from the other sets, and wherein the mixture further comprises a set of unlabeled nucleotides (e.g., not conjugated to a fluorophore) comprising a same nucleobase (e.g., selected from A, T, U, C, and/or G) that differs from the nucleobase of the three sets of nucleotide-fluorophore complexes.

[0150] In some embodiments, a mixture includes two sets of nucleotide-fluorophore complexes, each set having nucleotides of a same nucleobase that differs from the other set. In some embodiments, a method disclosed herein includes use of a first composition including a mixture of two sets, the two sets having two of four nucleobase types (e.g., (i) A, (ii) T or U, (iii), G, and (iv) C), and the method further includes use of an additional composition, including a mixture of two additional sets, the two additional sets having a different two of the four nucleobase types. In some embodiments, the two sets of the first composition have two different fluorophores (e.g., red and green), and the two sets of the additional composition have two different fluorophores (e.g., red and green).

[0151] In some embodiments, the nucleotide of each nucleotide-fluorophore complex in a set is selected from A, T, U, C, and G. In some embodiments, the nucleotide of each nucleotide-fluorophore complex in a set is selected from A, T, C, and G.

[0152] In some embodiments, the nucleotide in a set of unlabeled nucleotides is selected from A, T, U, C, and G. In some embodiments, the nucleotide in a set of unlabeled nucleotides is selected from A, T, C, and G.

[0153] In some embodiments, the method (or process) depicted in FIG. 1 further comprises, after the detection step, removing the fluorophores from the core of the incorporated nucleotide-fluorophore complex.

[0154] In some embodiments, removing the fluorophores from the core involves photobleaching the fluorophores. In some embodiments, the plurality of fluorophores of the incorporated nucleotide-fluorophore complex is unable to fluoresce after photobleaching. Any suitable photobleaching methods can be implemented. In some embodiments, the sample is exposed to a light source until the signal emitted by the plurality of fluorophores is eliminated.

[0155] In some embodiments, at least a subset of the fluorophores of each nucleotide-fluorophore complex is attached to the core via a cleavable linker, wherein removing the fluorophores from the core involves a cleavage reaction. In some embodiments, performing the cleavage reaction comprises contacting the complex with a reagent capable of cleaving the cleavable linker. For example, in some embodiments, the cleavable linker comprises a disulfide linker, and the cleavage reagent comprises a disulfide reducing agent, optionally wherein the disulfide reducing agent comprises Dithiothreitol (DTT). In some embodiments, the cleavable linker comprises a photocleavable linker (e.g., cleavable using UV light), a peroxide-cleavable arylboronic acid linker, a palladium (Pd)-cleavable linker (e.g., a bi-functionalized propargyl carbamate linker cleavable using a palladium complex), a phosphine-cleavable linker (e.g., a phosphine-cleavable linker comprising a disulfide moiety that can be cleaved using water-soluble phosphines or phosphine (scc, e.g., U.S. Pat. No. 10,487,102, which is incorporated herein by reference in its entirety)), etc.

[0156] In some embodiments, the method (or process) depicted in FIG. 1 further comprises, after the detecting step, cleaving the core from the nucleotide of the incorporated nucleotide-fluorophore complex, leaving an incorporated nucleotide. In some embodiments, cleaving the core from the nucleotide of the incorporated nucleotide-fluorophore complex is performed in addition to removing the fluorophores from the core. In some embodiments, removing the fluorophores from the core and cleaving the core from the nucleotide are performed sequentially (in either order). In some embodiments, removing the fluorophores from the core and cleaving the core from the nucleotide are performed at the same time.

[0157] In some embodiments, the single nucleotide of each nucleotide-fluorophore complex is attached to the core via a cleavable linker, and performing the cleavage reaction comprises contacting the complex with a reagent capable of cleaving the cleavable linker. For example, in

some embodiments, the cleavable linker comprises a disulfide linker, and the cleavage reagent comprises a disulfide reducing agent, optionally wherein the disulfide reducing agent comprises Dithiothreitol (DTT). In some embodiments, the cleavable linker comprises a photocleavable linker (e.g., cleavable using UV light), a peroxide-cleavable arylboronic acid linker, a palladium (Pd)-cleavable linker (e.g., a bi-functionalized propargyl carbamate linker cleavable using a palladium complex), a phosphine-cleavable linker (e.g., a phosphine-cleavable linker comprising a disulfide moiety that can be cleaved using water-soluble phosphines or phosphine (see, e.g., U.S. Pat. No. 10,487,102, which is incorporated herein by reference in its entirety)), etc.

[0158] In some embodiments, a nucleotide of a nucleotide-fluorophore complex or an unlabeled nucleotide comprises a reversible terminator moiety. In some embodiments, the method (or process) depicted in FIG. 1 further comprises cleaving the reversible terminator moiety after detecting the incorporated nucleotide-fluorophore complex or unlabeled nucleotide. In some embodiments, cleaving the reversible terminator moiety is performed sequentially with other steps in the method (e.g., removing the fluorophores from the core and/or cleaving the core from the nucleotide) in any order. In some embodiments, cleaving the reversible terminator moiety is performed at the same time as other steps in the method (e.g., removing the fluorophores from the core and/or cleaving the core from the nucleotide).

[0159] In some embodiments, the method (or process) depicted in FIG. 1 further comprises performing a first wash step to remove unbound polymerase and unbound nucleotide-fluorophore complexes prior to performing the detecting step, as described elsewhere herein.

[0160] In some embodiments, the method (or process) depicted in FIG. 1 further comprises repeating steps (a)-(c) for at least one additional cycle using at least one additional composition (e.g., a second composition, third composition, fourth composition, etc.) comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex of the additional composition comprises: i) a single nucleotide attached to a core; and ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide and thereby identify at least one additional complementary nucleotide in the template nucleic acid molecule. In some embodiments, the at least one additional cycle comprises at least 2, 5, 10, 20, 30, 40, or 50 additional cycles.

[0161] In some embodiments, the method (or process) depicted in FIG. 1 further comprises: prior to performing a first contacting step in (a), hybridizing a primer to a primer binding site in the template nucleic acid molecule.

[0162] In some embodiments, the first composition comprising nucleotide-fluorophore complexes and at least one additional composition (e.g., a second composition, third composition, fourth composition, etc.) comprising nucleotide-fluorophore complexes comprise the same sets of nucleotide-fluorophore complexes (e.g., nucleotide-fluorophore complexes comprising the same set of nucleobase). In some embodiments, the first composition comprising nucleotide-fluorophore complexes and at least one additional composition (e.g., a second composition, third composition, fourth composition, etc.) comprising nucleotide-fluorophore complexes comprise different sets of modified nucleotide molecules (e.g., nucleotide-fluorophore complexes comprising different sets of nucleobase).

[0163] In some embodiments, the first composition comprising nucleotide-fluorophore complexes and at least one additional composition comprising nucleotide-fluorophore complexes each comprise a set of unlabeled nucleotides.

[0164] Methods for processing the series of optical signals detected over the course of performing a cyclic series of base-by-base sequencing reactions to identify a nucleotide sequence are described elsewhere herein.

IV. Systems and Kits

[0165] In some aspects, provided herein are systems or kits for sequencing nucleic acid molecules, including systems or kits for sequencing and analysis of target nucleic acids in a biological sample

according to any of the methods described herein.

[0166] In some aspects, provided herein is a system or kit comprising any of the compositions comprising nucleotide-fluorophore complexes described herein. In some embodiments, the system or kit further comprises any of the primers described herein. In some embodiments, the system or kit further comprises any of the polymerases described herein. In some embodiments, the system or kit further comprises any of the biological samples described herein (e.g., a tissue section).

[0167] In some aspects, provided herein is a system or kit for performing in situ sequencing comprising a composition comprising nucleotide-fluorophore complexes as described herein, and one or more further components for performing the in situ sequencing reaction. In some embodiments, the one or more further components include a polymerase, a primer, a support for a tissue or cell sample, or any combination thereof. In some embodiments, the system or kit further comprises any of the circular probes and/or circularizable probes or probe sets disclosed herein. In some embodiments, the system or kit comprises a polymerase for rolling circle amplification.

[0168] In some aspects, provided herein is a system or kit for flow cell sequencing comprising a composition comprising nucleotide-fluorophore complexes as described herein, and one or more further components for performing the flow cell sequencing reaction. In some embodiments, the one or more further components include a polymerase, a primer, a flow cell, primers, adapters for sequencing library preparation, or any combination thereof.

[0169] In some aspects, provided herein is a system or kit for sequencing a template nucleic acid molecule, comprising: a composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex comprises: i) a single nucleotide attached to a core; and ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide. In some embodiments, the composition comprising nucleotide-fluorophore complexes comprises at least three sets of nucleotide-fluorophore complexes, wherein each set comprises a same nucleobase that differs from the other sets, and wherein, for each set, the plurality of fluorophores is a plurality of same fluorophores that differ from the other sets. In some embodiments, the composition comprises four sets of nucleotide-fluorophore complexes. In any of the embodiments herein, the composition further comprises unlabeled nucleotides, wherein the unlabeled nucleotides each comprise a same nucleobase that is different from the nucleobase of the nucleotide-fluorophore complexes in the sets.

[0170] In some embodiments, the system or kit comprises an additional composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex comprises: i) a single nucleotide attached to a core; and ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide. In some embodiments, the first composition and the additional composition comprise a same set of nucleotide-fluorophore complexes. In some embodiments, the first composition and the additional composition comprise different sets of nucleotide-fluorophore complexes. In some embodiments, the first composition and the additional composition each comprise nucleotide-fluorophore complexes, wherein the nucleotide of each nucleotide-fluorophore complex comprises a reversible terminator moiety. In some embodiments, the first composition and the additional composition each comprise a set of unlabeled nucleotides (e.g., not labeled with a detectable label). In some embodiments, the nucleotide of each nucleotide-fluorophore complex in a set of the first composition and in a set of the additional composition is selected from A, T, C, and G. In some embodiments, the nucleotide of each nucleotide-fluorophore complex in a set of the first composition and in a set of the additional composition is selected from A, T, U, C, and G. In some embodiments, the system or kit comprises at least 2, at least 5, at least 10, at least 25, at least 50, at least 75, at least 100, at least 300, at least 1,000 additional compositions comprising nucleotide-fluorophore complexes as described herein.

[0171] In some aspects, provided herein is a system or kit for sequencing a template nucleic acid molecule, comprising: a composition comprising nucleotide-fluorophore complexes, wherein each

nucleotide-fluorophore complex comprises: i) a single nucleotide attached to a core; and ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide. In some embodiments, the core of each of the nucleotide-fluorophore complexes comprises an avidin or streptavidin core, and wherein the nucleotide in each complex is bound to the core via a biotinylated linker. In some embodiments, at least a subset of the fluorophores in each nucleotide-fluorophore complex are bound to the core via biotin, optionally wherein at least a subset of the fluorophores in each nucleotide-fluorophore complex are bound to the core via a biotinylated linker. In some embodiments, the biotinylated linker is a biotinylated linear polyethylene glycol (PEG) linker.

[0172] In some aspects, provided herein is a system or kit for sequencing a template nucleic acid molecule, comprising: a composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex comprises: i) a single nucleotide attached to a core; and ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide. In some embodiments, the core of each of the nucleotide-fluorophore complexes comprises a branched polymer core, and wherein the nucleotide and the plurality of fluorophores in each nucleotide-fluorophore complex are attached to the branched polymer core.

[0173] In some embodiments, the systems or kits contain reagents and/or consumables required for performing one or more steps of the provided methods. In some embodiments, the systems or kits contain reagents for fixing, embedding, and/or permeabilizing the biological sample. In some embodiments, the systems or kits contain reagents, such as enzymes and buffers for ligation and/or amplification, such as ligases and/or polymerases. In some embodiments, the system or kit also comprises any of the reagents described herein, e.g., wash buffer and ligation buffer. In some embodiments, the systems or kits contain reagents for detection and/or sequencing, such as barcode detection probes or detectable labels. In some embodiments, the systems or kits optionally contain other components, for example nucleic acid primers.

[0174] The various components of the kit provided herein may be present in separate containers or certain compatible components may be pre-combined into a single container. In some embodiments, the kits further contain instructions for using the components of the kit to practice the provided methods. In some embodiments, each composition comprising nucleotide-fluorophore complexes is provided in separate containers. In some embodiments, sets of nucleotide-fluorophore complexes (as described elsewhere) are provided together in a single container, such as a tube. In some embodiments, each nucleotide-fluorophore complex is provided in separate containers. In some embodiments, a first combination of nucleotide-fluorophore complexes is provided together in a first container, and a second combination of nucleotide-fluorophore complexes is provided in a second container.

V. Biological Samples & Sample Preparation

[0175] A sample disclosed herein can be or derived from any biological sample. The biological sample can include any number of macromolecules, for example, cellular macromolecules and organelles (e.g., mitochondria and nuclei). The biological sample can include nucleic acids (such as DNA or RNA), proteins/polypeptides, carbohydrates, and/or lipids. The biological sample can be obtained as a tissue sample, such as a tissue section, biopsy, a core biopsy, a cell pellet, a cell block, a needle aspirate, or fine needle aspirate. The sample can be a fluid sample, such as a blood sample, urine sample, or saliva sample. The sample can be a skin sample, a colon sample, a cheek swab, a histology sample, a histopathology sample, a plasma or serum sample, a tumor sample, living cells, cultured cells, a clinical sample such as, for example, whole blood or blood-derived products, blood cells, or cultured tissues or cells, including cell suspensions. In some embodiments, the biological sample comprises cells which are deposited on a surface.

[0176] Biological samples can be derived from a homogeneous culture or population of the subjects or organisms mentioned herein or alternatively from a collection of several different

organisms. Biological samples can include one or more diseased cells. A diseased cell can have altered metabolic properties, gene expression, protein expression, and/or morphologic features. Examples of diseases include inflammatory disorders, metabolic disorders, nervous system disorders, and cancer. Cancer cells can be derived from solid tumors, hematological malignancies, cell lines, or obtained as circulating tumor cells. Biological samples can also include fetal cells and immune cells.

[0177] In some instances, the biological sample is provided on a substrate. In some instances, a substrate herein is any support that is insoluble in aqueous liquid and which allows for positioning of biological samples, analytes, features, and/or reagents (e.g., probes) on the support. In some instances, a biological sample is attached to a substrate. Attachment of the biological sample can be irreversible or reversible, depending upon the nature of the sample and subsequent steps in the analytical method. In certain instances, the sample is attached to the substrate reversibly by applying a suitable polymer coating to the substrate, and contacting the sample to the polymer coating. The sample can then be detached from the substrate, e.g., using an organic solvent that at least partially dissolves the polymer coating. Hydrogels are examples of polymers that are suitable for this purpose. In some instances, the substrate is coated or functionalized with one or more substances to facilitate attachment of the sample to the substrate. Suitable substances that can be used to coat or functionalize the substrate include, but are not limited to, lectins, poly-lysine, antibodies, and polysaccharides.

[0178] A variety of steps can be performed to prepare or process a biological sample for and/or during an assay. Except where indicated otherwise, the preparative or processing steps described below can generally be combined in any manner and in any order to appropriately prepare or process a particular sample for and/or analysis.

Preparation

[0179] A biological sample can be harvested from a subject (e.g., via surgical biopsy, whole subject sectioning) or grown in vitro on a growth substrate or culture dish as a population of cells, and prepared for analysis as a tissue slice or tissue section. Grown samples may be sufficiently thin for analysis without further processing steps. Alternatively, grown samples, and samples obtained via biopsy or sectioning, can be prepared as thin tissue sections using a mechanical cutting apparatus such as a vibrating blade microtome. As another alternative, in some embodiments, a thin tissue section is prepared by applying a touch imprint of a biological sample to a suitable substrate material.

[0180] The thickness of the tissue section can be a fraction of (e.g., less than 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, or 0.1) the maximum cross-sectional dimension of a cell. However, tissue sections having a thickness that is larger than the maximum cross-section cell dimension can also be used. For example, cryostat sections can be used, which can be, e.g., 10-20 μm thick. More generally, the thickness of a tissue section typically depends on the method used to prepare the section and the physical characteristics of the tissue, and therefore sections having a wide variety of different thicknesses can be prepared and used. For example, the thickness of the tissue section can be at least 0.1, 0.2, 0.3, 0.4, 0.5, 0.7, 1.0, 1.5, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 20, 30, 40, or 50 μm . Thicker sections can also be used if desired or convenient, e.g., at least 70, 80, 90, or 100 μm or more. Typically, the thickness of a tissue section is between 1-100 μm , 1-50 μm , 1-30 μm , 1-25 μm , 1-20 μm , 1-15 μm , 1-10 μm , 2-8 μm , 3-7 μm , or 4-6 μm , but as mentioned above, sections with thicknesses larger or smaller than these ranges can also be analysed.

[0181] Multiple sections can also be obtained from a single biological sample. For example, multiple tissue sections can be obtained from a surgical biopsy sample by performing serial sectioning of the biopsy sample using a sectioning blade. Spatial information among the serial sections can be preserved in this manner, and the sections can be analysed successively to obtain three-dimensional information about the biological sample.

[0182] In some instances, the biological sample (e.g., a tissue section as described above) is

prepared by deep freezing at a temperature suitable to maintain or preserve the integrity (e.g., the physical characteristics) of the tissue structure. The frozen tissue sample can be sectioned, e.g., thinly sliced, onto a substrate surface using any number of suitable methods. For example, a tissue sample can be prepared using a chilled microtome (e.g., a cryostat) set at a temperature suitable to maintain both the structural integrity of the tissue sample and the chemical properties of the nucleic acids in the sample. Such a temperature can be, e.g., less than -15°C ., less than -20°C ., or less than -25°C .

[0183] In some instances, the biological sample is prepared using formalin-fixation and paraffin-embedding (FFPE), which are established methods. In some instances, cell suspensions and other non-tissue samples are prepared using formalin-fixation and paraffin-embedding. Following fixation of the sample and embedding in a paraffin or resin block, the sample can be sectioned as described above. Prior to analysis, the paraffin-embedding material can be removed from the tissue section (e.g., deparaffinization) by incubating the tissue section in an appropriate solvent (e.g., xylene) followed by a rinse (e.g., 99.5% ethanol for 2 minutes, 96% ethanol for 2 minutes, and 70% ethanol for 2 minutes).

[0184] As an alternative to formalin fixation described above, a biological sample can be fixed in any of a variety of other fixatives to preserve the biological structure of the sample prior to analysis. For example, a sample can be fixed via immersion in ethanol, methanol, acetone, paraformaldehyde (PFA)-Triton, and combinations thereof.

[0185] In some instances, the methods provided herein comprise one or more post-fixing (also referred to as post-fixation) steps. In some instances, one or more post-fixing step is performed after contacting a sample with a polynucleotide disclosed herein, e.g., one or more probes such as a circular or padlock probe. In some instances, one or more post-fixing step is performed after a hybridization complex comprising a probe and a target is formed in a sample. In some instances, one or more post-fixing step is performed prior to a ligation reaction disclosed herein.

[0186] In some instances, a method disclosed herein comprises de-crosslinking the reversibly cross-linked biological sample. The de-crosslinking does not need to be complete. In some instances, only a portion of crosslinked molecules in the reversibly cross-linked biological sample are de-crosslinked and allowed to migrate.

[0187] In some instances, a biological sample is permeabilized to facilitate transfer of species (such as probes) into the sample. If a sample is not permeabilized sufficiently, the transfer of species (such as probes) into the sample may be too low to enable adequate analysis. Conversely, if the tissue sample is too permeable, the relative spatial relationship of the analytes within the tissue sample can be lost. Hence, a balance between permeabilizing the tissue sample enough to obtain good signal intensity while still maintaining the spatial resolution of the analyte distribution in the sample is desirable.

[0188] In general, a biological sample can be permeabilized by exposing the sample to one or more permeabilizing agents. Suitable agents for this purpose include, but are not limited to, organic solvents (e.g., acetone, ethanol, and methanol), cross-linking agents (e.g., paraformaldehyde), detergents (e.g., saponin, Triton X-100™ or Tween-20™), and enzymes (e.g., trypsin, proteases). In some instances, the biological sample is incubated with a cellular permeabilizing agent to facilitate permeabilization of the sample. Additional methods for sample permeabilization are described, for example, in Jamur et al., *Method Mol. Biol.* 588:63-66, 2010, the entire contents of which are incorporated herein by reference. Any suitable method for sample permeabilization can generally be used in connection with the samples described herein.

[0189] In some instances, the biological sample is permeabilized by any suitable methods. For example, one or more lysis reagents can be added to the sample. Examples of suitable lysis agents include, but are not limited to, bioactive reagents such as lysis enzymes that are used for lysis of different cell types, e.g., gram positive or negative bacteria, plants, yeast, mammalian, such as lysozymes, achromopeptidase, lysostaphin, labiase, kitalase, lyticase, and a variety of other

commercially available lysis enzymes. Other lysis agents can additionally or alternatively be added to the biological sample to facilitate permeabilization. For example, surfactant-based lysis solutions can be used to lyse sample cells. Lysis solutions can include ionic surfactants such as, for example, sarcosyl and sodium dodecyl sulfate (SDS). More generally, chemical lysis agents can include, without limitation, organic solvents, chelating agents, detergents, surfactants, and chaotropic agents.

[0190] Additional reagents can be added to a biological sample to perform various functions prior to analysis of the sample. In some instances, DNase and RNase inactivating agents or inhibitors such as proteinase K, and/or chelating agents such as EDTA, are added to the sample. For example, a method disclosed herein comprises a step for increasing accessibility of a nucleic acid for binding, e.g., a denaturation step to open up DNA in a cell for hybridization by a probe. For example, proteinase K treatment may be used to free up DNA with proteins bound thereto.

Embedding

[0191] In some instances, the biological sample is embedded in a matrix (e.g., a hydrogel matrix). Embedding the sample in this manner typically involves contacting the biological sample with a hydrogel such that the biological sample becomes surrounded by the hydrogel. For example, the sample can be embedded by contacting the sample with a suitable polymer material, and activating the polymer material to form a hydrogel. In some instances, the hydrogel is formed such that the hydrogel is internalized within the biological sample. Biological samples can include analytes (e.g., protein, RNA, and/or DNA) embedded in a 3D matrix. In some instances, amplicons (e.g., rolling circle amplification products) derived from or associated with analytes (e.g., protein, RNA, and/or DNA) can be embedded in a 3D matrix. In some instances, a 3D matrix comprises a network of natural molecules and/or synthetic molecules that are chemically and/or enzymatically linked, e.g., by crosslinking. In some instances, a 3D matrix comprises a synthetic polymer. In some instances, a 3D matrix comprises a hydrogel.

[0192] In some embodiments, a biological sample is embedded in any of a variety of other embedding materials to provide structural substrate to the sample prior to sectioning and other handling steps. In some cases, the embedding material is removed e.g., prior to analysis of tissue sections obtained from the sample. Suitable embedding materials include, but are not limited to, waxes, resins (e.g., methacrylate resins), epoxies, and agar.

[0193] In some instances, the biological sample is embedded in a matrix (e.g., a hydrogel matrix). Embedding the sample in this manner typically involves contacting the biological sample with a hydrogel such that the biological sample becomes surrounded by the hydrogel. For example, the sample can be embedded by contacting the sample with a suitable polymer material, and activating the polymer material to form a hydrogel. In some instances, the hydrogel is formed such that the hydrogel is internalized within the biological sample.

[0194] In some instances, the biological sample is immobilized in the hydrogel via cross-linking of the polymer material that forms the hydrogel. Cross-linking can be performed chemically and/or photochemically, or alternatively by any other suitable hydrogel-formation method.

[0195] In some instances, the biological sample is reversibly cross-linked prior to or during an in situ assay. In some embodiments, the analytes, polynucleotides and/or amplification product (e.g., amplicon) of an analyte or a probe bound thereto are anchored to a polymer matrix. For example, the polymer matrix can be a hydrogel. In some instances, one or more of the polynucleotide probe(s) and/or amplification product (e.g., amplicon) thereof are modified to contain functional groups that are used as an anchoring site to attach the polynucleotide probes and/or amplification product to a polymer matrix. In some instances, a modified probe comprising oligo dT is used to bind to mRNA molecules of interest, followed by reversible or irreversible crosslinking of the mRNA molecules.

[0196] In some instances, the biological sample is immobilized in a hydrogel via cross-linking of the polymer material that forms the hydrogel. Cross-linking can be performed chemically and/or

photochemically, or alternatively by any other suitable hydrogel-formation method. A hydrogel may include a macromolecular polymer gel including a network. Within the network, some polymer chains can optionally be cross-linked, although cross-linking does not always occur. [0197] In some instances, a hydrogel includes hydrogel subunits, such as, but not limited to, acrylamide, bis-acrylamide, polyacrylamide and derivatives thereof, poly(ethylene glycol) and derivatives thereof (e.g. PEG-acrylate (PEG-DA), PEG-RGD), gelatin-methacryloyl (GelMA), methacrylated hyaluronic acid (MeHA), polyaliphatic polyurethanes, polyether polyurethanes, polyester polyurethanes, polyethylene copolymers, polyamides, polyvinyl alcohols, polypropylene glycol, polytetramethylene oxide, polyvinyl pyrrolidone, polyacrylamide, poly(hydroxyethyl acrylate), and poly(hydroxyethyl methacrylate), collagen, hyaluronic acid, chitosan, dextran, agarose, gelatin, alginate, protein polymers, methylcellulose, and the like, and combinations thereof.

[0198] In some instances, a hydrogel includes a hybrid material, e.g., the hydrogel material includes elements of both synthetic and natural polymers. Examples of suitable hydrogels are described, for example, in U.S. Pat. Nos. 6,391,937, 9,512,422, and 9,889,422, and in U.S. Patent Application Publication Nos. 2017/0253918, 2018/0052081 and 2010/0055733, the entire contents of each of which are incorporated herein by reference.

[0199] The composition and application of the hydrogel-matrix to a biological sample typically depends on the nature and preparation of the biological sample (e.g., sectioned, non-sectioned, type of fixation). As one example, where the biological sample is a tissue section, the hydrogel-matrix can include a monomer solution and an ammonium persulfate (APS) initiator/tetramethylethylenediamine (TEMED) accelerator solution. As another example, where the biological sample consists of cells (e.g., cultured cells or cells disassociated from a tissue sample), the cells can be incubated with the monomer solution and APS/TEMED solutions. For cells, hydrogel-matrix gels are formed in compartments, including but not limited to devices used to culture, maintain, or transport the cells. For example, hydrogel-matrices can be formed with monomer solution plus APS/TEMED added to the compartment to a depth ranging from about 0.1 μm to about 2 mm.

[0200] Additional methods and aspects of hydrogel embedding of biological samples are described for example in Chen et al., *Science* 347 (6221): 543-548, 2015, the entire contents of which are incorporated herein by reference.

[0201] In some instances, the hydrogel forms the substrate. In some embodiments, the substrate includes a hydrogel and one or more second materials. In some embodiments, the hydrogel is placed on top of one or more second materials. For example, the hydrogel can be pre-formed and then placed on top of, underneath, or in any other configuration with one or more second materials. In some instances, hydrogel formation occurs after contacting one or more second materials during formation of the substrate. Hydrogel formation can also occur within a structure (e.g., wells, ridges, projections, and/or markings) located on a substrate.

[0202] In some instances, hydrogel formation on a substrate occurs before, contemporaneously with, or after probes are provided to the sample. For example, hydrogel formation can be performed on the substrate already containing the probes.

[0203] In some instances, hydrogel formation occurs within a biological sample. In some embodiments, a biological sample (e.g., tissue section) is embedded in a hydrogel. In some instances, hydrogel subunits are infused into the biological sample, and polymerization of the hydrogel is initiated by an external or internal stimulus.

[0204] In instances in which a hydrogel is formed within a biological sample, functionalization chemistry is used. In some instances, functionalization chemistry includes hydrogel-tissue chemistry (HTC). Any hydrogel-tissue backbone (e.g., synthetic or native) suitable for HTC can be used for anchoring biological macromolecules and modulating functionalization. Non-limiting examples of methods using HTC backbone variants include CLARITY, PACT, ExM, SWITCH and

cPACT. In some instances, hydrogel formation within a biological sample is permanent. For example, biological macromolecules can permanently adhere to the hydrogel allowing multiple rounds of interrogation. In some instances, hydrogel formation within a biological sample is reversible. In some instances, HTC reagents are added to the hydrogel before, contemporaneously with, and/or after polymerization. In some instances, a cell labeling agent is added to the hydrogel before, contemporaneously with, and/or after polymerization. In some instances, a cell-penetrating agent is added to the hydrogel before, contemporaneously with, and/or after polymerization. [0205] In some instances, additional reagents are added to the hydrogel subunits before, contemporaneously with, and/or after polymerization. For example, additional reagents can include but are not limited to oligonucleotides (e.g., probes), endonucleases to fragment DNA, fragmentation buffer for DNA, DNA polymerase enzymes, dNTPs used to amplify the nucleic acid and to attach the barcode to the amplified fragments. Other enzymes can be used, including without limitation, RNA polymerase, ligase, proteinase K, and DNase. Additional reagents can also include reverse transcriptase enzymes, including enzymes with terminal transferase activity, primers, and oligonucleotides. In some instances, optical labels are added to the hydrogel subunits before, contemporaneously with, and/or after polymerization.

[0206] Hydrogels embedded within biological samples can be cleared using any suitable method. For example, electrophoretic tissue clearing methods can be used to remove biological macromolecules from the hydrogel-embedded sample. In some instances, a hydrogel-embedded sample is stored before or after clearing of hydrogel, in a medium (e.g., a mounting medium, methylcellulose, or other semi-solid mediums).

[0207] In some instances, a biological sample embedded in a matrix (e.g., a hydrogel) is isometrically expanded. Isometric expansion methods that can be used include hydration, a preparative step in expansion microscopy, as described in, e.g., Chen et al., *Science* 347 (6221): 543-548, 2015 and U.S. Pat. No. 10,059,990, which are herein incorporated by reference in their entireties. Isometric expansion of the sample can increase the spatial resolution of the subsequent analysis of the sample. The increased resolution in spatial profiling can be determined by comparison of an isometrically expanded sample with a sample that has not been isometrically expanded. In some instances, a biological sample is isometrically expanded to a size at least 2×, 2.1×, 2.2×, 2.3×, 2.4×, 2.5×, 2.6×, 2.7×, 2.8×, 2.9×, 3×, 3.1×, 3.2×, 3.3×, 3.4×, 3.5×, 3.6×, 3.7×, 3.8×, 3.9×, 4×, 4.1×, 4.2×, 4.3×, 4.4×, 4.5×, 4.6×, 4.7×, 4.8×, or 4.9× its non-expanded size. In some instances, the sample is isometrically expanded to at least 2× and less than 20× of its non-expanded size.

Staining and Immunohistochemistry (IHC)

[0208] To facilitate visualization, biological samples can be stained using a wide variety of stains and staining techniques. In some instances, for example, a sample can be stained using any number of stains and/or immunohistochemical reagents. One or more staining steps may be performed to prepare or process a biological sample for an assay described herein or may be performed during and/or after an assay. In some instances, the sample is contacted with one or more nucleic acid stains, membrane stains (e.g., cellular or nuclear membrane), cytological stains, or combinations thereof. In some examples, the stain is specific to proteins, phospholipids, DNA (e.g., dsDNA, SSDNA), RNA, an organelle or compartment of the cell. The sample may be contacted with one or more labeled antibodies (e.g., a primary antibody specific for the analyte of interest and a labeled secondary antibody specific for the primary antibody). In some instances, cells in the sample are segmented using one or more images taken of the stained sample.

[0209] In some instances, the stain is performed using a lipophilic dye. In some examples, the staining is performed with a lipophilic carbocyanine or aminostyryl dye, or analogs thereof (e.g., DiI, DiO, DiR, DiD). Other cell membrane stains may include FM and RH dyes or immunohistochemical reagents specific for cell membrane proteins. In some examples, the stain includes but is not limited to, acridine orange, acid fuchsin, Bismarck brown, carmine, coomassie

blue, cresyl violet, DAPI, cosin, ethidium bromide, acid fuchsin, haematoxylin, Hoechst stains, iodine, methyl green, methylene blue, neutral red, Nile blue, Nile red, osmium tetroxide, ruthenium red, propidium iodide, rhodamine (e.g., rhodamine B), or safranin, or derivatives thereof. In some instances, the sample is stained with haematoxylin and cosin (H&E).

[0210] The sample can be stained using hematoxylin and cosin (H&E) staining techniques, using Papanicolaou staining techniques, Masson's trichrome staining techniques, silver staining techniques, Sudan staining techniques, and/or using Periodic Acid Schiff (PAS) staining techniques. PAS staining is typically performed after formalin or acetone fixation. In some instances, the sample is stained using Romanowsky stain, including Wright's stain, Jenner's stain, Can-Grunwald stain, Leishman stain, and Giemsa stain.

[0211] In some instances, biological samples are destained. Any suitable methods of destaining or discoloring a biological sample may be utilized and generally depend on the nature of the stain(s) applied to the sample. For example, in some instances, one or more immunofluorescent stains are applied to the sample via antibody coupling. Such stains can be removed using techniques such as cleavage of disulfide linkages via treatment with a reducing agent and detergent washing, chaotropic salt treatment, treatment with antigen retrieval solution, and treatment with an acidic glycine buffer. Methods for multiplexed staining and destaining are described, for example, in Bolognesi et al., *J. Histochem. Cytochem.* 2017; 65 (8): 431-444, Lin et al., *Nat Commun.* **2015; 6:8390**, Pirici et al., *J. Histochem. Cytochem.* 2009; 57:567-75, and Glass et al., *J. Histochem. Cytochem.* 2009; 57:899-905, the entire contents of each of which are incorporated herein by reference.

VI. Analytes

[0212] A biological sample may comprise one or a plurality of analytes of interest. Methods for performing multiplexed assays to analyze two or more different analytes in a single biological sample are provided.

[0213] The methods and compositions disclosed herein can be used to detect and analyze a wide variety of different analytes. In some embodiments, an analyte includes any biological substance, structure, moiety, or component to be analyzed. In some embodiments, a target disclosed herein similarly includes any analyte of interest. In some examples, a target or analyte is directly or indirectly detected.

[0214] Analytes can be derived from a specific type of cell and/or a specific sub-cellular region. For example, analytes can be derived from cytosol, from cell nuclei, from mitochondria, from microsomes, and more generally, from any other compartment, organelle, or portion of a cell. Permeabilizing agents that specifically target certain cell compartments and organelles can be used to selectively release analytes from cells for analysis, and/or allow access of one or more reagents (e.g., probes for analyte detection) to the analytes in the cell or cell compartment or organelle.

[0215] The analyte may include any biomolecule or chemical compound, including a macromolecule such as a protein or peptide, a lipid or a nucleic acid molecule, or a small molecule, including organic or inorganic molecules. The analyte may be a cell or a microorganism, including a virus, or a fragment or product thereof. An analyte can be any substance or entity for which a specific binding partner (e.g. an affinity binding partner) can be developed. Such a specific binding partner may be a nucleic acid probe (for a nucleic acid analyte) and may lead directly to the generation of a RCA template (e.g. a padlock or other circularizable probe). Alternatively, the specific binding partner may be coupled to a nucleic acid, which may be detected using an RCA strategy, e.g. in an assay which uses or generates a circular nucleic acid molecule which can be the RCA template.

[0216] Analytes of particular interest may include nucleic acid molecules, such as DNA (e.g. genomic DNA, mitochondrial DNA, plastid DNA, viral DNA, etc.) and RNA (e.g. mRNA, microRNA, rRNA, snRNA, viral RNA, etc.), and synthetic and/or modified nucleic acid molecules, (e.g. including nucleic acid domains comprising or consisting of synthetic or modified nucleotides

such as LNA, PNA, morpholino, etc.), proteinaceous molecules such as peptides, polypeptides, proteins or prions or any molecule which includes a protein or polypeptide component, etc., or fragments thereof, or a lipid or carbohydrate molecule, or any molecule which comprise a lipid or carbohydrate component. The analyte may be a single molecule or a complex that contains two or more molecular subunits, e.g. including but not limited to protein-DNA complexes, which may or may not be covalently bound to one another, and which may be the same or different. Thus in addition to cells or microorganisms, such a complex analyte may also be a protein complex or protein interaction. Such a complex or interaction may thus be a homo- or hetero-multimer. Aggregates of molecules, e.g. proteins may also be target analytes, for example aggregates of the same protein or different proteins. The analyte may also be a complex between proteins or peptides and nucleic acid molecules such as DNA or RNA, e.g. interactions between proteins and nucleic acids, e.g. regulatory factors, such as transcription factors, and DNA or RNA.

Endogenous Analytes

[0217] In some embodiments, an analyte herein is endogenous to a biological sample and can include nucleic acid analytes and non-nucleic acid analytes. Methods and compositions disclosed herein can be used to analyze nucleic acid analytes (e.g., using a nucleic acid probe or probe set that directly or indirectly hybridizes to a nucleic acid analyte) and/or non-nucleic acid analytes (e.g., using a labeling agent that comprises a reporter oligonucleotide and binds directly or indirectly to a non-nucleic acid analyte) in any suitable combination.

Labelling Agents

[0218] In some instances, provided herein are methods and compositions for analyzing endogenous analytes (e.g., RNA, ssDNA, cell surface or intracellular proteins, and/or metabolites) in a sample using one or more labeling agents. In some instances, an analyte labeling agent includes an agent that interacts with an analyte (e.g., an endogenous analyte in a sample). In some instances, the labeling agents comprise a reporter oligonucleotide that is indicative of the analyte or portion thereof interacting with the labeling agent. For example, the reporter oligonucleotide may comprise a barcode sequence that permits identification of the labeling agent. In some cases, the sample contacted by the labeling agent is further contacted with a probe (e.g., a single-stranded probe sequence), that hybridizes to a reporter oligonucleotide of the labeling agent, in order to identify the analyte associated with the labeling agent. In some instances, the analyte labeling agent comprises an analyte binding moiety and a labeling agent barcode domain comprising one or more barcode sequences, e.g., a barcode sequence that corresponds to the analyte binding moiety and/or the analyte. An analyte binding moiety barcode includes to a barcode that is associated with or otherwise identifies the analyte binding moiety. In some instances, by identifying an analyte binding moiety by identifying its associated analyte binding moiety barcode, the analyte to which the analyte binding moiety binds is also identified. An analyte binding moiety barcode can be a nucleic acid sequence of a given length and/or sequence that is associated with the analyte binding moiety. An analyte binding moiety barcode can generally include any of the variety of aspects of barcodes described herein.

[0219] In some instances, the method comprises one or more post-fixing (also referred to as post-fixation) steps after contacting the sample with one or more labeling agents.

[0220] In the methods and systems described herein, one or more labeling agents capable of binding to or otherwise coupling to one or more features may be used to characterize analytes, cells and/or cell features. In some instances, cell features include cell surface features. Analytes may include, but are not limited to, a protein, a receptor, an antigen, a surface protein, a transmembrane protein, a cluster of differentiation protein, a protein channel, a protein pump, a carrier protein, a phospholipid, a glycoprotein, a glycolipid, a cell-cell interaction protein complex, an antigen-presenting complex, a major histocompatibility complex, an engineered T-cell receptor, a T-cell receptor, a B-cell receptor, a chimeric antigen receptor, a gap junction, an adherens junction, or any combination thereof. In some instances, cell features include intracellular analytes, such as

proteins, protein modifications (e.g., phosphorylation status or other post-translational modifications), nuclear proteins, nuclear membrane proteins, or any combination thereof.

[0221] In some instances, an analyte binding moiety includes any molecule or moiety capable of binding to an analyte (e.g., a biological analyte, e.g., a macromolecular constituent). A labeling agent may include, but is not limited to, a protein, a peptide, an antibody (or an epitope binding fragment thereof), a lipophilic moiety (such as cholesterol), a cell surface receptor binding molecule, a receptor ligand, a small molecule, a bi-specific antibody, a bi-specific T-cell engager, a T-cell receptor engager, a B-cell receptor engager, a pro-body, an aptamer, a monobody, an affimer, a darpin, and a protein scaffold, or any combination thereof. The labeling agents can include (e.g., are attached to) a reporter oligonucleotide that is indicative of the cell surface feature to which the binding group binds. For example, the reporter oligonucleotide may comprise a barcode sequence that permits identification of the labeling agent. For example, a labeling agent that is specific to one type of cell feature (e.g., a first cell surface feature) may have coupled thereto a first reporter oligonucleotide, while a labeling agent that is specific to a different cell feature (e.g., a second cell surface feature) may have a different reporter oligonucleotide coupled thereto. For a description of non-limiting examples of labeling agents, reporter oligonucleotides, and methods of use, see, e.g., U.S. Pat. No. 10,550,429; U.S. Pat. Pub. 20190177800; and U.S. Pat. Pub. 20190367969, which are each incorporated by reference herein in their entirety.

[0222] In some instances, an analyte binding moiety includes one or more antibodies or epitope-binding fragments thereof. The antibodies or epitope-binding fragments including the analyte binding moiety can specifically bind to a target analyte. In some instances, the analyte is a protein (e.g., a protein on a surface of the biological sample (e.g., a cell) or an intracellular protein). In some instances, a plurality of analyte labeling agents comprising a plurality of analyte binding moieties bind a plurality of analytes present in a biological sample. In some instances, the plurality of analytes includes a single species of analyte (e.g., a single species of polypeptide). In some instances in which the plurality of analytes includes a single species of analyte, the analyte binding moieties of the plurality of analyte labeling agents are the same. In some instances in which the plurality of analytes includes a single species of analyte, the analyte binding moieties of the plurality of analyte labeling agents are the different (e.g., members of the plurality of analyte labeling agents can have two or more species of analyte binding moieties, wherein each of the two or more species of analyte binding moieties binds a single species of analyte, e.g., at different binding sites). In some instances, the plurality of analytes includes multiple different species of analyte (e.g., multiple different species of polypeptides).

[0223] In other instances, e.g., to facilitate sample multiplexing, a labeling agent that is specific to a particular cell feature has a first plurality of the labeling agent (e.g., an antibody or lipophilic moiety) coupled to a first reporter oligonucleotide and a second plurality of the labeling agent coupled to a second reporter oligonucleotide.

[0224] In some embodiments, these reporter oligonucleotides comprise nucleic acid barcode sequences that permit identification of the labeling agent which the reporter oligonucleotide is coupled to. The selection of oligonucleotides as the reporter may provide advantages of being able to generate significant diversity in terms of sequence, while also being readily attachable to most biomolecules, e.g., antibodies, etc., as well as being readily detected, e.g., using the in situ detection techniques described herein.

[0225] Attachment (coupling) of the reporter oligonucleotides to the labeling agents may be achieved through any of a variety of direct or indirect, covalent or non-covalent associations or attachments. For example, oligonucleotides may be covalently attached to a portion of a labeling agent (such a protein, e.g., an antibody or antibody fragment) using chemical conjugation techniques (e.g., Lightning-Link® antibody labeling kits available from Innova Biosciences), as well as other non-covalent attachment mechanisms, e.g., using biotinylated antibodies and oligonucleotides (or beads that include one or more biotinylated linker, coupled to

oligonucleotides) with an avidin or streptavidin linker. Antibody and oligonucleotide biotinylation techniques are available. See, e.g., Fang, et al., "Fluoride-Cleavable Biotinylation Phosphoramidite for 5'-end-Labeling and Affinity Purification of Synthetic Oligonucleotides," *Nucleic Acids Res.* Jan. 15, 2003; 31 (2): 708-715, which is entirely incorporated herein by reference for all purposes. Likewise, protein and peptide biotinylation techniques have been developed and are readily available. See, e.g., U.S. Pat. No. 6,265,552, which is entirely incorporated herein by reference for all purposes. Furthermore, click reaction chemistry may be used to couple reporter oligonucleotides to labeling agents.

[0226] Commercially available kits, such as those from Thunderlink and Abcam, and techniques common in the art may be used to couple reporter oligonucleotides to labeling agents as appropriate. In another example, a labeling agent is indirectly (e.g., via hybridization) coupled to a reporter oligonucleotide comprising a barcode sequence that identifies the label agent. For instance, the labeling agent may be directly coupled (e.g., covalently bound) to a hybridization oligonucleotide that comprises a sequence that hybridizes with a sequence of the reporter oligonucleotide. Hybridization of the hybridization oligonucleotide to the reporter oligonucleotide couples the labeling agent to the reporter oligonucleotide. In some instances, the reporter oligonucleotides are releasable from the labeling agent, such as upon application of a stimulus. For example, the reporter oligonucleotide may be attached to the labeling agent through a labile bond (e.g., chemically labile, photolabile, thermally labile, etc.) as generally described for releasing molecules from supports elsewhere herein.

[0227] In some cases, the labeling agent comprises a reporter oligonucleotide and a label. A label can be fluorophore, a radioisotope, a molecule capable of a colorimetric reaction, a magnetic particle, or any other suitable molecule or compound capable of detection. The label can be conjugated to a labeling agent (or reporter oligonucleotide) either directly or indirectly (e.g., the label can be conjugated to a molecule that can bind to the labeling agent or reporter oligonucleotide). In some cases, a label is conjugated to a first oligonucleotide that is complementary (e.g., hybridizes) to a sequence of the reporter oligonucleotide.

[0228] In some instances, multiple different species of analytes (e.g., polypeptides) from the biological sample are subsequently associated with the one or more physical properties of the biological sample. For example, the multiple different species of analytes can be associated with locations of the analytes in the biological sample. Such information (e.g., proteomic information when the analyte binding moiety (ies) recognizes a polypeptide(s)) can be used in association with other spatial information (e.g., genetic information from the biological sample, such as DNA sequence information, transcriptome information (e.g., sequences of transcripts), or both). For example, a cell surface protein of a cell can be associated with one or more physical properties of the cell (e.g., a shape, size, activity, or a type of the cell). The one or more physical properties can be characterized by imaging the cell. The cell can be bound by an analyte labeling agent comprising an analyte binding moiety that binds to the cell surface protein and an analyte binding moiety barcode that identifies that analyte binding moiety. Results of protein analysis in a sample (e.g., a tissue sample or a cell) can be associated with DNA and/or RNA analysis in the sample.

Detectable Labels

[0229] In some instances, one or more hybridization probes or one or more nucleotides (or analogs thereof) are labeled with distinguishing and/or detectable tags or labels. The tags may be distinguishable by means of their differences in fluorescence, Raman spectrum, charge, mass, refractive index, luminescence, length, or any other measurable property. The tag may be attached to one or more different positions on the nucleotide, so long as the fidelity of binding to the polymerase-nucleic acid complex is sufficiently maintained to enable identification of the complementary base on the template nucleic acid correctly. In some instances, the tag is attached to the nucleobase of the nucleotide. Alternatively, a tag is attached to the gamma phosphate position of the nucleotide.

[0230] Detectable labels can be suitable for small scale detection and/or suitable for high-throughput screening. As such, suitable detectable labels include, but are not limited to, radioisotopes, fluorophores, chemiluminescent compounds, bioluminescent compounds, and dyes. The detectable label can be qualitatively detected (e.g., optically or spectrally), or it can be quantified. Qualitative detection generally includes a detection method in which the existence or presence of the detectable label is confirmed, whereas quantifiable detection generally includes a detection method having a quantifiable (e.g., numerically reportable) value such as an intensity, duration, polarization, and/or other properties. In some instances, the detectable label is bound to another moiety, for example, a nucleotide or nucleotide analog, and can include a fluorescent, a colorimetric, or a chemiluminescent label.

[0231] In some instances, a detectable label is attached to another moiety, for example, a nucleotide or nucleotide analog. In some instances, one or more nucleotides are labeled with a cleavable detectable tag or label. For example, the non-terminating fluorescently labeled nucleotides can include a DBCO-nucleotide conjugated to fluorescent compound with a disulfide linker. In some instances, a non-terminating fluorescently labeled nucleotide is incorporated into the strand without termination, and after imaging, the linker can be cleaved to remove fluorescent label. In some instances, a DBCO-nucleotide (e.g., 5-DBCO-PEG4-UTP) undergoes a click reaction with the cleavable linker conjugated to a fluorescent label (e.g., cleavable linker-ATTO647N), and a disulfide group can be cleaved by tris(2-carboxyethyl) phosphine (TCEP) reduction together with 3'-O-azidomethyl-dNTP.

[0232] In some instances, the detectable label is a fluorophore. For example, the fluorophore can be from a group that includes: 7-AAD (7-Aminoactinomycin D), Acridine Orange (+DNA), Acridine Orange (+RNA), Alexa Fluor® 350, Alexa Fluor® 430, Alexa Fluor® 488, Alexa Fluor® 532, Alexa Fluor® 546, Alexa Fluor® 555, Alexa Fluor® 568, Alexa Fluor® 594, Alexa Fluor® 633, Alexa Fluor® 647, Alexa Fluor® 660, Alexa Fluor® 680, Alexa Fluor® 700, Alexa Fluor® 750, Allophycocyanin (APC), AMCA/AMCA-X, 7-Aminoactinomycin D (7-AAD), 7-Amino-4-methylcoumarin, 6-Aminoquinoline, Aniline Blue, ANS, APC-Cy7, ATTO-TAG™ CBQCA, ATTO-TAG™ FQ, Auramine O-Feulgen, BCECF (high pH), BFP (Blue Fluorescent Protein), BFP/GFP FRET, BOBO™-1/BO-PRO™-1, BOBO™-3/BO-PRO™-3, BODIPY® FL, BODIPY® TMR, BODIPY® TR-X, BODIPY® 530/550, BODIPY® 558/568, BODIPY® 564/570, BODIPY® 581/591, BODIPY® 630/650-X, BODIPY® 650-665-X, BTC, Calcein, Calcein Blue, Calcium Crimson™, Calcium Green-1™, Calcium Orange™, Calcofluor® White, 5-Carboxyfluorescein (5-FAM), 5-Carboxynaphthofluorescein, 6-Carboxyrhodamine 6G, 5-Carboxytetramethylrhodamine (5-TAMRA), Carboxy-X-rhodamine (5-ROX), Cascade Blue®, Cascade Yellow™, CCF2 (GeneBLAzer™), CFP (Cyan Fluorescent Protein), CFP/YFP FRET, Chromomycin A3, CI-NERF (low pH), CPM, 6-CR 6G, CTC Formazan, Cy2®, Cy3®, Cy3.5®, Cy5®, Cy5.5®, Cy7®, Cychrome (PE-Cy5), Dansylamine, Dansyl cadaverine, Dansylchloride, DAPI, Dapoxyl, DCFH, DHR, DIA (4-Di-16-ASP), DiD (DilC18 (5)), DIDS, Dil (DilC18 (3)), DiO (DiOC18 (3)), DiR (DilC18 (7)), Di-4 ANEPPS, Di-8 ANEPPS, DM-NERF (4.5-6.5 pH), DsRed (Red Fluorescent Protein), EBFP, ECFP, EGFP, ELF®-97 alcohol, Eosin, Erythrosin, Ethidium bromide, Ethidium homodimer-1 (EthD-1), Europium (III) Chloride, 5-FAM (5-Carboxyfluorescein), Fast Blue, Fluorescein-dT phosphoramidite, FITC, Fluo-3, Fluo-4, FluorX®, Fluoro-Gold™ (high pH), Fluoro-Gold™ (low pH), Fluoro-Jade, FM® 1-43, Fura-2 (high calcium), Fura-2/BCECF, Fura Red™ (high calcium), Fura Red™/Fluo-3, GeneBLAzer™ (CCF2), GFP Red Shifted (rsGFP), GFP Wild Type, GFP/BFP FRET, GFP/DsRed FRET, Hoechst 33342 & 33258, 7-Hydroxy-4-methylcoumarin (pH 9), 1,5 IAEDANS, Indo-1 (high calcium), Indo-1 (low calcium), Indodicarbocyanine, Indotricarbocyanine, JC-1, 6-JOE, JOJO™-1/JO-PRO™-1, LDS 751 (+DNA), LDS 751 (+RNA), LOLO™-1/LO-PRO™-1, Lucifer Yellow, LysoSensor™ Blue (pH 5), LysoSensor™ Green (pH 5), LysoSensor™ Yellow/Blue (pH 4.2), LysoTracker® Green, LysoTracker® Red, Lyso Tracker® Yellow, Mag-Fura-2, Mag-Indo-1, Magnesium Green™,

Marina Blue®, 4-Methylumbelliferone, Mithramycin, MitoTracker® Green, MitoTracker® Orange, MitoTracker® Red, NBD (amine), Nile Red, Oregon Green® 488, Oregon Green® 500, Oregon Green® 514, Pacific Blue, PBF1, PE (R-phycoerythrin), PE-Cy5, PE-Cy7, PE-Texas Red, PerCP (Peridinin chlorophyll protein), PerCP-Cy5.5 (TruRed), PharRed (APC-Cy7), C-phycoerythrin, R-phycoerythrin, R-phycoerythrin (PE), PI (Propidium Iodide), PKH26, PKH67, POPO™-1/PO-PRO™-1, POPO™-3/PO-PRO™-3, Propidium Iodide (PI), PyMPO, Pyrene, Pyronin Y, Quantam Red (PE-Cy5), Quinacrine Mustard, R670 (PE-Cy5), Red 613 (PE-Texas Red), Red Fluorescent Protein (DsRed), Resorufin, RH 414, Rhod-2, Rhodamine B, Rhodamine Green™, Rhodamine Red™, Rhodamine Phalloidin, Rhodamine 110, Rhodamine 123, 5-ROX (carboxy-X-rhodamine), S65A, S65C, S65L, S65T, SBFI, SITS, SNAFL®-1 (high pH), SNAFL®-2, SNARF®-1 (high pH), SNARFR-1 (low pH), Sodium Green™, SpectrumAqua®, SpectrumGreen® #1, SpectrumGreen® #2, SpectrumOrange®, SpectrumRed®, SYTOR 11, SYTOR 13, SYTOR 17, SYTOR 45, SYTOX® Blue, SYTOX® Green, SYTOX® Orange, 5-TAMRA (5-Carboxytetramethylrhodamine), Tetramethylrhodamine (TRITC), Texas Red®/Texas Red®-X, Texas Red®-X (NHS Ester), Thiadicarbocyanine, Thiazole Orange, TOTO®-1/TO-PRO®-1, TOTOR-3/TO-PRO®-3, TO-PRO®-5, Tri-color (PE-Cy5), TRITC (Tetramethylrhodamine), TruRed (PerCP-Cy5.5), WW 781, X-Rhodamine (XRITC), Y66F, Y66H, Y66W, YFP (Yellow Fluorescent Protein), YOYO®-1/YO-PRO®-1, YOYO®-3/YO-PRO®-3, 6-FAM (Fluorescein), 6-FAM (NHS Ester), 6-FAM (Azide), HEX, TAMRA (NHS Ester), Yakima Yellow, MAX, TET, TEX615, ATTO 488, ATTO 532, ATTO 542, ATTO 550, ATTO 565, ATTO Rho101, ATTO 590, ATTO 633, ATTO 647N, TYE 563, TYE 665, TYE 705, 5' IRDye® 700, 5' IRDye® 800, 5' IRDye® 800CW (NHS Ester), WellRED D4 Dye, WellRED D3 Dye, WellRED D2 Dye, Lightcycler® 640 (NHS Ester), and Dy 750 (NHS Ester).

[0233] The detectable label can be directly detectable by itself (e.g., radioisotope labels or fluorescent labels) or, in the case of an enzymatic label, can be indirectly detectable, e.g., by catalyzing chemical alterations of a substrate compound or composition, which substrate compound or composition is directly detectable. The label can emit a signal or alter a signal delivered to the label so that the presence or absence of the label can be detected. In some cases, coupling is via a linker, which may be cleavable, such as photo-cleavable (e.g., cleavable under ultra-violet light), chemically-cleavable (e.g., via a reducing agent, such as dithiothreitol (DTT), tris(2-carboxyethyl) phosphine (TCEP)) or enzymatically cleavable (e.g., via an esterase, lipase, peptidase, or protease).
Generation of Products

[0234] In some instances, provided herein are methods and compositions for analyzing one or more products of an endogenous analyte and/or a labeling agent in a biological sample. In some instances, an endogenous analyte (e.g., a viral or cellular DNA or RNA) or a product (e.g., a hybridization product, a ligation product, an extension product (e.g., by a DNA or RNA polymerase), a replication product, a transcription/reverse transcription product, and/or an amplification product such as a rolling circle amplification (RCA) product) thereof is analyzed. In some instances, a labeling agent that directly or indirectly binds to an analyte in the biological sample is analyzed. In some instances, a product (e.g., a hybridization product, a ligation product, an extension product (e.g., by a DNA or RNA polymerase), a replication product, a transcription/reverse transcription product, and/or an amplification product such as a rolling circle amplification (RCA) product) of a labeling agent that directly or indirectly binds to an analyte in the biological sample is analyzed.

(a) Hybridization

[0235] In some instances, a hybridization product comprising the pairing of substantially complementary or complementary nucleic acid sequences within two different molecules is analyzed. For example, hybridization of an endogenous analyte or the labeling agent (e.g., reporter oligonucleotide attached thereto) with another endogenous molecule or another labeling agent or a probe can be analyzed. Pairing can be achieved by any process in which a nucleic acid sequence

joins with a substantially or fully complementary sequence through base pairing to form a hybridization complex. For purposes of hybridization, two nucleic acid sequences are “substantially complementary” if at least 60% (e.g., at least 70%, at least 80%, or at least 90%) of their individual bases are complementary to one another.

[0236] Various probes and probe sets can be hybridized to an endogenous analyte and/or a labeling agent and each probe may comprise one or more barcode sequences. Non-limiting examples of barcoded probes or probe sets may be based on a padlock probe, a gapped padlock probe, a SNAIL (Splint Nucleotide Assisted Intramolecular Ligation) probe set, a PLAYR (Proximity Ligation Assay for RNA) probe set, a PLISH (Proximity Ligation in situ Hybridization) probe set, and RNA-templated ligation probes. The specific probe or probe set design can vary.

(b) Ligation

[0237] In some instances, a ligation product of an endogenous analyte and/or a labeling agent is analyzed. In some instances, the ligation product is formed between two or more endogenous analytes. In some instances, the ligation product is formed between two or more labeling agents. In some instances, the ligation product is an intramolecular ligation of an endogenous analyte. In some instances, the ligation product is an intramolecular ligation product or an intermolecular ligation product, for example, the ligation product is generated by the circularization of a circularizable probe or probe set upon hybridization to a target sequence. The target sequence can be comprised in an endogenous analyte (e.g., nucleic acid such as a genomic DNA or mRNA) or a product thereof (e.g., cDNA from a cellular mRNA transcript), or in a labeling agent (e.g., the reporter oligonucleotide) or a product thereof.

[0238] In some instances, provided herein is a probe or probe set capable of DNA-templated ligation, such as from a cDNA molecule. Sec, e.g., U.S. Pat. No. 8,551,710, which is hereby incorporated by reference in its entirety. In some instances, provided herein is a probe or probe set capable of RNA-templated ligation. Sec, e.g., U.S. Pat. Pub. 2020/0224244 which is hereby incorporated by reference in its entirety. In some instances, the probe set is a SNAIL probe set. See, e.g., U.S. Pat. Pub. 20190055594, which is hereby incorporated by reference in its entirety. In some instances, provided herein is a multiplexed proximity ligation assay. Sec, e.g., U.S. Pat. Pub. 20140194311 which is hereby incorporated by reference in its entirety. In some instances, provided herein is a probe or probe set capable of proximity ligation, for instance a proximity ligation assay for RNA (e.g., PLAYR) probe set. Sec, e.g., U.S. Pat. Pub. 20160108458, which is hereby incorporated by reference in its entirety. In some instances, a circular probe is indirectly hybridized to the target nucleic acid. In some instances, the circular construct is formed from a probe set capable of proximity ligation, for instance a proximity ligation in situ hybridization (PLISH) probe set. Sec, e.g., U.S. Pat. Pub. 2020/0224243 which is hereby incorporated by reference in its entirety.

[0239] In some instances, the ligation involves chemical ligation (e.g., click chemistry ligation). In some instances, the chemical ligation involves template dependent ligation. In some instances, the chemical ligation involves template independent ligation. In some instances, the click reaction is a template-independent reaction (see, e.g., Xiong and Seela (2011), J. Org. Chem. 76 (14): 5584-5597, incorporated by reference herein in its entirety). In some instances, the click reaction is a template-dependent reaction or template-directed reaction. In some instances, the template-dependent reaction is sensitive to base pair mismatches such that reaction rate is significantly higher for matched versus unmatched templates. In some instances, the click reaction is a nucleophilic addition template-dependent reaction. In some instances, the click reaction is a cyclopropane-tetrazine template-dependent reaction.

[0240] In some instances, the ligation involves enzymatic ligation. In some instances, the enzymatic ligation involves use of a ligase. In some embodiments, the ligase used herein comprises an enzyme that is commonly used to join polynucleotides together or to join the ends of a single polynucleotide. An RNA ligase, a DNA ligase, or another variety of ligase can be used to ligate two

nucleotide sequences together. Ligases comprise ATP-dependent double-strand polynucleotide ligases, NAD-i-dependent double-strand DNA or RNA ligases and single-strand polynucleotide ligases, for example any of the ligases described in EC 6.5.1.1 (ATP-dependent ligases), EC 6.5.1.2 (NAD⁺-dependent ligases), EC 6.5.1.3 (RNA ligases). Specific examples of ligases comprise bacterial ligases such as *E. coli* DNA ligase, Tth DNA ligase, *Thermococcus* sp. (strain 9° N) DNA ligase (9° NTM DNA ligase, New England Biolabs), Taq DNA ligase, AmpligaseTM (Epicentre Biotechnologies) and phage ligases such as T3 DNA ligase, T4 DNA ligase and T7 DNA ligase and mutants thereof. In some instances, the ligase is a T4 RNA ligase. In some instances, the ligase is a splintR ligase. In some instances, the ligase is a single stranded DNA ligase. In some instances, the ligase is a T4 DNA ligase. In some instances, the ligase is a ligase that has an DNA-splinted DNA ligase activity. In some instances, the ligase is a ligase that has an RNA-splinted DNA ligase activity.

[0241] In some instances, the ligation herein is a direct ligation. In some instances, the ligation herein is an indirect ligation. “Direct ligation” means that the ends of the polynucleotides hybridize immediately adjacently to one another to form a substrate for a ligase enzyme resulting in their ligation to each other (intramolecular ligation). Alternatively, “indirect” means that the ends of the polynucleotides hybridize non-adjacently to one another, i.e., separated by one or more intervening nucleotides or “gaps”. In some instances, said ends are not ligated directly to each other, but instead occurs either via the intermediacy of one or more intervening (so-called “gap” or “gap-filling” (oligo) nucleotides) or by the extension of 3’ end of a probe to “fill” the “gap” corresponding to said intervening nucleotides (intermolecular ligation). In some cases, the gap of one or more nucleotides between the hybridized ends of the polynucleotides is “filled” by one or more “gap” (oligo) nucleotide(s) which are complementary to a splint, padlock probe, or target nucleic acid. The gap may be a gap of 1 to 60 nucleotides or a gap of 1 to 40 nucleotides or a gap of 3 to 40 nucleotides. In specific instances, the gap is a gap of about 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 or more nucleotides, of any integer (or range of integers) of nucleotides in between the indicated values. In some instances, the gap between said terminal regions is filled by a gap oligonucleotide or by extending 3’ end of a polynucleotide. In some cases, ligation involves ligating the ends of the probe to at least one gap (oligo) nucleotide, such that the gap (oligo) nucleotide becomes incorporated into the resulting polynucleotide. In some instances, the ligation herein is preceded by gap filling. In other instances, the ligation herein does not require gap filling.

[0242] In some instances, ligation of the polynucleotides produces polynucleotides with melting temperature higher than that of unligated polynucleotides. Thus, In some embodiments, ligation stabilizes the hybridization complex containing the ligated polynucleotides prior to subsequent steps, comprising amplification and detection.

[0243] In some embodiments, a high fidelity ligase, such as a thermostable DNA ligase (e.g., a Taq DNA ligase), is used. Thermostable DNA ligases are active at elevated temperatures, allowing further discrimination by incubating the ligation at a temperature near the melting temperature (T_{sub.m}) of the DNA strands. This selectively reduces the concentration of annealed mismatched substrates (expected to have a slightly lower T_{sub.m} around the mismatch) over annealed fully base-paired substrates. Thus, high-fidelity ligation can be achieved through a combination of the intrinsic selectivity of the ligase active site and balanced conditions to reduce the incidence of annealed mismatched dsDNA.

[0244] In some instances, the ligation herein is a proximity ligation of ligating two (or more) nucleic acid sequences that are in proximity with each other, e.g., through enzymatic means (e.g., a ligase). In some instances, proximity ligation includes a “gap-filling” step that involves incorporation of one or more nucleic acids by a polymerase, based on the nucleic acid sequence of a template nucleic acid molecule, spanning a distance between the two nucleic acid molecules of interest (see, e.g., U.S. Pat. No. 7,264,929, the entire contents of which are incorporated herein by reference). A wide variety of different methods can be used for proximity ligating nucleic acid

molecules, including (but not limited to) “sticky-end” and “blunt-end” ligations. Additionally, single-stranded ligation can be used to perform proximity ligation on a single-stranded nucleic acid molecule. Sticky-end proximity ligations involve the hybridization of complementary single-stranded sequences between the two nucleic acid molecules to be joined, prior to the ligation event itself. Blunt-end proximity ligations generally do not include hybridization of complementary regions from each nucleic acid molecule because both nucleic acid molecules lack a single-stranded overhang at the site of ligation.

(c) Primer Extension and Amplification

[0245] In some instances, a primer extension product of an analyte, a labeling agent, a probe or probe set bound to the analyte (e.g., a circularizable probe bound to genomic DNA, mRNA, or cDNA), or a probe or probe set bound to the labeling agent (e.g., a circularizable probe bound to one or more reporter oligonucleotides from the same or different labeling agents) is analyzed.

[0246] A primer is generally a single-stranded nucleic acid sequence having a 3' end that can be used as a substrate for a nucleic acid polymerase in a nucleic acid extension reaction. RNA primers are formed of RNA nucleotides, and are used in RNA synthesis, while DNA primers are formed of DNA nucleotides and used in DNA synthesis. Primers can also include both RNA nucleotides and DNA nucleotides (e.g., in a random or designed pattern). Primers can also include other natural or synthetic nucleotides described herein that can have additional functionality. In some examples, DNA primers are used to prime RNA synthesis and vice versa (e.g., RNA primers can be used to prime DNA synthesis). Primers can vary in length. For example, primers can be about 6 bases to about 120 bases. For example, primers can include up to about 25 bases. A primer, may in some cases, refer to a primer binding sequence. A primer extension reaction generally refers to any method where two nucleic acid sequences become linked (e.g., hybridized) by an overlap of their respective terminal complementary nucleic acid sequences (e.g., 3' termini). Such linking can be followed by nucleic acid extension (e.g., an enzymatic extension) of one, or both termini using the other nucleic acid sequence as a template for extension. Enzymatic extension can be performed by an enzyme including, but not limited to, a polymerase and/or a reverse transcriptase.

[0247] In some instances, a product of an endogenous analyte and/or a labeling agent is an amplification product of one or more polynucleotides, for instance, a circular probe or circularizable probe or probe set. In some instances, the amplifying is achieved by performing rolling circle amplification (RCA). In other instances, a primer that hybridizes to the circular probe or circularized probe is added and used as such for amplification. In some instances, the RCA comprises a linear RCA, a branched RCA, a dendritic RCA, or any combination thereof.

[0248] In some instances, the amplification is performed at a temperature between or between about 20° C. and about 60° C. In some instances, the amplification is performed at a temperature between or between about 30° C. and about 40° C. In some embodiments, the amplification step, such as the rolling circle amplification (RCA) is performed at a temperature between at or about 25° C. and at or about 50° C., such as at or about 25° C., 27° C., 29° C., 31° C., 33° C., 35° C., 37° C., 39° C., 41° C., 43° C., 45° C., 47° C., or 49° C.

[0249] In some instances, upon addition of a DNA polymerase in the presence of appropriate dNTP precursors and other cofactors, a primer is elongated to produce multiple copies of the circular template. This amplification step can utilize isothermal amplification or non-isothermal amplification. In some instances, after the formation of the hybridization complex and association of the amplification probe, the hybridization complex is rolling-circle amplified to generate a cDNA nanoball (i.e., amplicon) containing multiple copies of the cDNA. Techniques for rolling circle amplification (RCA) include linear RCA, a branched RCA, a dendritic RCA, or any combination thereof. (See, e.g., Baner et al, *Nucleic Acids Research*, 26:5073-5078, 1998; Lizardi et al, *Nature Genetics* 19:226, 1998; Mohsen et al., *Acc Chem Res.* 2016 Nov. 15; 49 (11): 2540-2550; Schweitzer et al. *Proc. Natl Acad. Sci. USA* 97:101 13-1 19, 2000; Faruqi et al, *BMC Genomics* 2:4, 2000; Nallur et al, *Nucl. Acids Res.* 29: el 18, 2001; Dean et al. *Genome Res.* 1

1:1095-1099, 2001; Schweitzer et al, *Nature Biotech.* 20:359-365, 2002; U.S. Pat. Nos. 6,054,274, 6,291,187, 6,323,009, 6,344,329 and 6,368,801, each of which is herein incorporated by reference in its entirety). Non-limiting examples of polymerases for use in RCA comprise DNA polymerase such phi29 (q29) polymerase, Klenow fragment, *Bacillus stearothermophilus* DNA polymerase (BST), T4 DNA polymerase, T7 DNA polymerase, or DNA polymerase I. In some embodiments, DNA polymerases that have been engineered or mutated to have desirable characteristics are employed. In some instances, the polymerase is phi29 DNA polymerase.

[0250] In some embodiments, during the amplification step, modified nucleotides are added to the reaction to incorporate the modified nucleotides in the amplification product (e.g., nanoball). Non-limiting examples of the modified nucleotides comprise amine-modified nucleotides. In some aspects of the methods, for example, for anchoring or cross-linking of the generated amplification product (e.g., nanoball) to a scaffold, to cellular structures and/or to other amplification products (e.g., other nanoballs). In some embodiments, the amplification products comprises a modified nucleotide, such as an amine-modified nucleotide. In some instances, the amine-modified nucleotide comprises an acrylic acid N-hydroxysuccinimide moiety modification. Examples of other amine-modified nucleotides comprise, but are not limited to, a 5-Aminoallyl-dUTP moiety modification, a 5-Propargylamino-dCTP moiety modification, a N6-6-Aminohexyl-dATP moiety modification, or a 7-Deaza-7-Propargylamino-dATP moiety modification.

[0251] In some embodiments, the polynucleotides and/or amplification product (e.g., amplicon) are anchored to a polymer matrix. For example, the polymer matrix can be a hydrogel. In some instances, one or more of the polynucleotide probe(s) are modified to contain functional groups that can be used as an anchoring site to attach the polynucleotide probes and/or amplification product to a polymer matrix. Non-limiting examples of modification and polymer matrix that can be employed in accordance with the provided instances comprise those described in, for example, WO 2014/163886, WO 2017/079406, US 2016/0024555, US 2018/0251833 and US 2017/0219465, each of which is herein incorporated by reference in its entirety. In some examples, the scaffold also contains modifications or functional groups that react with or incorporate the modifications or functional groups of the probe set or amplification product. In some examples, the scaffold comprises oligonucleotides, polymers or chemical groups, to provide a matrix and/or support structures.

[0252] The amplification products may be immobilized within the matrix generally at the location of the nucleic acid being amplified, thereby creating a localized colony of amplicons. The amplification products may be immobilized within the matrix by steric factors. The amplification products may also be immobilized within the matrix by covalent or noncovalent bonding. In this manner, the amplification products may be considered to be attached to the matrix. By being immobilized to the matrix, such as by covalent bonding or cross-linking, the size and spatial relationship of the original amplicons is maintained. By being immobilized to the matrix, such as by covalent bonding or cross-linking, the amplification products are resistant to movement or unraveling under mechanical stress.

[0253] In some embodiments, the amplification products are copolymerized and/or covalently attached to the surrounding matrix thereby preserving their spatial relationship and any information inherent thereto. For example, if the amplification products are those generated from DNA or RNA within a cell embedded in the matrix, the amplification products can also be functionalized to form covalent attachment to the matrix preserving their spatial information within the cell thereby providing a subcellular localization distribution pattern. In some instances, the provided methods involve embedding the one or more polynucleotide probe sets and/or the amplification products in the presence of hydrogel subunits to form one or more hydrogel-embedded amplification products. In some instances, the hydrogel-tissue chemistry described comprises covalently attaching nucleic acids to in situ synthesized hydrogel for tissue clearing, enzyme diffusion, and multiple-cycle sequencing while an existing hydrogel-tissue chemistry method cannot. In some instances, to

enable amplification product embedding in the tissue-hydrogel setting, amine-modified nucleotides are comprised in the amplification step (e.g., RCA), functionalized with an acrylamide moiety using acrylic acid N-hydroxysuccinimide esters, and copolymerized with acrylamide monomers to form a hydrogel.

[0254] In some instances, the RCA template comprises the target analyte, or a part thereof, where the target analyte is a nucleic acid, or it is provided or generated as a proxy, or a marker, for the analyte. In some instances, different analytes are detected in situ in one or more cells using a RCA-based detection system, e.g., where the signal is provided by generating an RCA product from a circular RCA template which is provided or generated in the assay, and the RCA product is detected to detect the corresponding analyte. The RCA product may thus be regarded as a reporter which is detected to detect the target analyte. However, the RCA template may also be regarded as a reporter for the target analyte; the RCA product is generated based on the RCA template, and comprises complementary copies of the RCA template. The RCA template determines the signal which is detected, and is thus indicative of the target analyte. As will be described in more detail below, the RCA template may be a probe, or a part or component of a probe, or may be generated from a probe, or it may be a component of a detection assay (e.g., a reagent in a detection assay), which is used as a reporter for the assay, or a part of a reporter, or signal-generation system. The RCA template used to generate the RCP may thus be a circular (e.g. circularized) reporter nucleic acid molecule, namely from any RCA-based detection assay which uses or generates a circular nucleic acid molecule as a reporter for the assay. Since the RCA template generates the RCP reporter, it may be viewed as part of the reporter system for the assay.

[0255] In some instances, a product herein includes a molecule or a complex generated in a series of reactions, e.g., hybridization, ligation, extension, replication, transcription/reverse transcription, and/or amplification (e.g., rolling circle amplification), in any suitable combination.

VI. Fluorescence Detection and Imaging

Fluorescence Detection

[0256] Fluorescence detection in tissue samples can often be hindered by the presence of strong background fluorescence. "Autofluorescence" is the general term used to distinguish background fluorescence (that can arise from a variety of sources, including aldehyde fixation, extracellular matrix components, red blood cells, lipofuscin, and the like) from the desired immunofluorescence from the fluorescently labeled antibodies or probes. Tissue autofluorescence can lead to difficulties in distinguishing the signals due to fluorescent antibodies or probes from the general background. In some instances, a method disclosed herein utilizes one or more agents to reduce tissue autofluorescence, for example, Autofluorescence Eliminator (Sigma/EMD Millipore), TrueBlack Lipofuscin Autofluorescence Quencher (Biotium), MaxBlock Autofluorescence Reducing Reagent Kit (Max Vision Biosciences), and/or a very intense black dye (e.g., Sudan Black, or comparable dark chromophore).

[0257] Examples of fluorescent labels and nucleotides and/or polynucleotides conjugated to such fluorescent labels comprise those described elsewhere herein and those described in, for example, Hoagland, Handbook of Fluorescent Probes and Research Chemicals, Ninth Edition (Molecular Probes, Inc., Eugene, 2002); Keller and Manak, DNA Probes, 2nd Edition (Stockton Press, New York, 1993); Eckstein, editor, Oligonucleotides and Analogues: A Practical Approach (IRL Press, Oxford, 1991); and Wetmur, Critical Reviews in Biochemistry and Molecular Biology, 26:227-259 (1991), each of which is herein incorporated by reference in its entirety. In some instances, non-limiting examples of techniques and methods applicable to the provided embodiments comprise those described in, for example, U.S. Pat. Nos. 4,757,141, 5,151,507 and 5,091,519, each of which is herein incorporated by reference in its entirety. In some instances, one or more fluorescent dyes are used as labels for labeled target sequences, for example, as described in U.S. Pat. No. 5,188,934 (4,7-dichlorofluorescein dyes); U.S. Pat. No. 5,366,860 (spectrally resolvable rhodamine dyes); U.S. Pat. No. 5,847,162 (4,7-dichlororhodamine dyes); U.S. Pat. No. 4,318,846 (ether-substituted

fluorescein dyes); U.S. Pat. No. 5,800,996 (energy transfer dyes); U.S. Pat. No. 5,066,580 (xanthine dyes); and U.S. Pat. No. 5,688,648 (energy transfer dyes), each of which is herein incorporated by reference in its entirety. Labelling can also be carried out with quantum dots, as described in U.S. Pat. Nos. 6,322,901, 6,576,291, 6,423,551, 6,251,303, 6,319,426, 6,426,513, 6,444,143, 5,990,479, 6,207,392, US 2002/0045045 and US 2003/0017264, each of which is herein incorporated by reference in its entirety. In some instances, a fluorescent label comprises a signaling moiety that conveys information through the fluorescence absorption and/or emission properties of one or more molecules. Non-limiting examples of fluorescence properties comprise fluorescence intensity, fluorescence lifetime, emission spectrum characteristics and energy transfer.

Imaging

[0258] In some embodiments, the detection (comprising imaging) is carried out using any of a number of different types of microscopy, e.g., confocal microscopy, two-photon microscopy, light-field microscopy, intact tissue expansion microscopy, and/or CLARITY™-optimized light sheet microscopy (COLM).

[0259] In some instances, fluorescence microscopy is used for detection and imaging of the sample. In some embodiments, a fluorescence microscope is an optical microscope that uses fluorescence and phosphorescence instead of, or in addition to, reflection and absorption to study properties of organic or inorganic substances. In fluorescence microscopy, a sample is illuminated with light of a wavelength which excites fluorescence in the sample. The fluoresced light, which is usually at a longer wavelength than the illumination, is then imaged through a microscope objective. Two filters may be used in this technique; an illumination (or excitation) filter which ensures the illumination is near monochromatic and at the correct wavelength, and a second emission (or barrier) filter which ensures none of the excitation light source reaches the detector. Alternatively, these functions may both be accomplished by a single dichroic filter. The fluorescence microscope can be or comprise any microscope that uses fluorescence to generate an image, whether it is a more simple set up like an epifluorescence microscope, or a more complicated design such as a confocal microscope, which uses optical sectioning to achieve better z-axis resolution of the sample to be imaged.

[0260] In some instances, confocal microscopy is used for detection and imaging of the sample. Confocal microscopy uses point illumination and a pinhole in an optically conjugate plane in front of the detector to eliminate out-of-focus signal. As only light produced by fluorescence very close to the focal plane can be detected, the image's optical resolution, particularly in the sample depth direction, is much better than that of wide-field microscopes. However, as much of the light from sample fluorescence is blocked at the pinhole, this increased resolution is at the cost of decreased signal intensity-so long exposures are often required. As only one point in the sample is illuminated at a time, 2D or 3D imaging requires scanning over a regular raster (i.e., a rectangular pattern of parallel scanning lines) in the specimen. The achievable thickness of the focal plane is defined mostly by the wavelength of the used light divided by the numerical aperture of the objective lens, but also by the optical properties of the specimen. The thin optical sectioning possible makes these types of microscopes particularly good at 3D imaging and surface profiling of samples.

CLARITY™-optimized light sheet microscopy (COLM) provides an alternative microscopy for fast 3D imaging of large clarified samples. COLM interrogates large immune-stained tissues, permits increased speed of acquisition and results in a higher quality of generated data.

[0261] Other types of microscopy that can be employed comprise bright field microscopy, oblique illumination microscopy, dark field microscopy, phase contrast, differential interference contrast (DIC) microscopy, interference reflection microscopy (also known as reflected interference contrast, or RIC), single plane illumination microscopy (SPIM), super-resolution microscopy, laser microscopy, electron microscopy (EM), Transmission electron microscopy (TEM), Scanning electron microscopy (SEM), reflection electron microscopy (REM), Scanning transmission electron microscopy (STEM) and low-voltage electron microscopy (LVEM), scanning probe microscopy

(SPM), atomic force microscopy (AFM), ballistic electron emission microscopy (BEEM), chemical force microscopy (CFM), conductive atomic force microscopy (C-AFM), electrochemical scanning tunneling microscope (ECSTM), electrostatic force microscopy (EFM), fluidic force microscope (FluidFM), force modulation microscopy (FMM), feature-oriented scanning probe microscopy (FOSPM), kelvin probe force microscopy (KPFM), magnetic force microscopy (MFM), magnetic resonance force microscopy (MRFM), near-field scanning optical microscopy (NSOM) (or SNOM, scanning near-field optical microscopy, SNOM, Piezoresponse Force Microscopy (PFM), PSTM, photon scanning tunneling microscopy (PSTM), PTMS, photothermal microspectroscopy/microscopy (PTMS), SCM, scanning capacitance microscopy (SCM), SECM, scanning electrochemical microscopy (SECM), SGM, scanning gate microscopy (SGM), SHPM, scanning Hall probe microscopy (SHPM), SICM, scanning ion-conductance microscopy (SICM), SPSM spin polarized scanning tunneling microscopy (SPSM), SSRM, scanning spreading resistance microscopy (SSRM), SThM, scanning thermal microscopy (SThM), STM, scanning tunneling microscopy (STM), STP, scanning tunneling potentiometry (STP), SVM, scanning voltage microscopy (SVM), and synchrotron x-ray scanning tunneling microscopy (SXSTM), and intact tissue expansion microscopy (exM).

[0262] In some instances, a method herein comprises subjecting the sample to expansion microscopy methods and techniques. Expansion allows individual targets (e.g., mRNA or RNA transcripts) which are densely packed within a cell, to be resolved spatially in a high-throughput manner. Expansion microscopy techniques are known in the art and can be performed as described in US 2016/0116384 and Chen et al., *Science*, 347, 543 (2015), each of which is incorporated herein by reference in its entirety. In some instances, the method does not comprise subjecting the sample to expansion microscopy. In some instances, the method does not comprise dissociating a cell from the sample such as a tissue or the cellular microenvironment. In some instances, the method does not comprise lysing the sample or cells therein. In some instances, the method does not comprise embedding the sample or molecules from the sample in an exogenous matrix.

[0263] In some cases, analysis is performed on one or more images captured, and comprises processing the image(s) and/or quantifying signals observed. In some instances, images of signals from different fluorescent channels and/or nucleotide incorporation cycles are compared and analyzed. In some instances, images of signals (or absence thereof) at a particular location in a sample from different fluorescent channels and/or sequential incorporation cycles are aligned to analyze an analyte at the location. For instance, a particular location in a sample can be tracked and signal spots from sequential incorporation cycles can be analyzed to detect a target polynucleotide sequence (e.g., a barcode sequence or subsequence thereof) in an analyte at the location. The analysis may comprise processing information of one or more cell types, one or more types of analytes, a number or level of analyte, and/or a number or level of cells detected in a particular region of the sample. In some instances, the analysis comprises detecting a sequence e.g., a barcode sequence present in an amplification product at a location in the sample. In some instances, the number of signals detected in a unit area in the biological sample is quantified. In some instances, the signals detected at a corresponding position in the biological sample in a plurality of images taken at different z positions (e.g., in the depth direction) is quantified and analyzed.

VII. Barcode Sequences

[0264] In some instances, an analyte described herein is associated with one or more barcode(s), e.g., at least two, three, four, five, six, seven, eight, nine, ten, or more barcodes. Barcodes can be used to spatially-resolve molecular components found in biological samples, for example, within a cell or a tissue sample. A barcode can be attached to an analyte or to another moiety or structure (e.g., a target-specific antibody) in a reversible or irreversible manner. In some embodiments, a barcode comprises about 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, or more than 30 nucleotides.

[0265] In some instances, a barcode includes two or more sub-barcodes (or barcode segments) that

together function as a single barcode. For example, a polynucleotide barcode can include two or more polynucleotide sequences (e.g., sub-barcode) that are contiguous or that are separated by one or more non-barcode sequences. In some instances, a barcode comprises about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more than 10 sub-barcode (or barcode segments). In some instances, each sub-barcode (or barcode segment) comprises about 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, or more than 30 nucleotides. In some instances, each non-barcode sequence comprises about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, or more than 30 nucleotides.

[0266] In some instances, the one or more barcode(s) also provide a platform for targeting functionalities, such as oligonucleotides, oligonucleotide-antibody conjugates, oligonucleotide-streptavidin conjugates, modified oligonucleotides, affinity purification, detectable moieties, enzymes, enzymes for detection assays or other functionalities, and/or for detection and identification of the polynucleotide. In any of the preceding instances, the methods provided herein can include analyzing the barcodes by performing sequential hybridization and detection with a plurality of labelled probes (e.g., detection oligos) or by performing in situ sequencing.

[0267] In some instances, e.g., in a barcode sequencing method, barcode sequences are detected for identification of other molecules including nucleic acid molecules (DNA or RNA) that are longer than the barcode sequences themselves, as opposed to direct sequencing of the longer nucleic acid molecules. In some instances, an N-mer barcode sequence comprises up to 4^N unique sequences given a sequencing read of N bases, and a much shorter sequencing read may be required for molecular identification compared to non-barcoded sequencing methods such as direct sequencing. For example, 1024 molecular species may be identified using a 5-nucleotide barcode sequence ($4^{5.0}=1024$), whereas 8 nucleotide barcodes can be used to identify up to 65,536 molecular species, a number greater than the total number of distinct genes in the human genome. In some instances, the barcode sequences contained in the probes or RCPs are detected, rather than endogenous sequences, which can be an efficient read-out in terms of information per cycle of sequencing. Because the barcode sequences are pre-determined, they can also be designed to feature error detection and correction mechanisms, scc, e.g., U.S. Pat. Pub. 20190055594 and U.S. Pat. Pub 20210164039, each of which is herein incorporated by reference in its entirety.

VIII. Cyclic Array Sequencing in a Flow Cell

[0268] In some instances, a method of sequencing as provided herein includes “cyclic array sequencing” of amplified template nucleic acid molecules. Cyclic array flow cell sequencing methods generally involve performing multiple cycles of an enzymatic reaction on an array of spatially-separated oligonucleotide features (e.g., clonally-amplified colonies of template nucleic acid fragments tethered to a support surface, e.g., a flow cell surface). In some instances, the template nucleic acid is modified with known adapter sequence(s) comprising, e.g., amplification and/or sequencing primer binding sites, and then affixed to the support surface (e.g., the lumen surface(s) of a flow cell) in a random or patterned array by hybridization to surface-tethered complementary capture probes (complementary to adapter sequences) on the support surface, clonally amplified, and then probed using the aforementioned sequencing reaction as described herein. In some embodiments, the flow cell sequencing comprises massively parallel sequencing reaction, whereby each enzymatic reaction cycle is used to query only one base (the “interrogation” nucleobase) of the template nucleic acid fragment in each oligonucleotide feature, but thousands to billions of template nucleic acid molecules may be processed in parallel. Performing repeated cycles is then used to progressively identify the nucleic acid sequence of each template nucleic acid molecule based on patterns of detection of a signal or detection of an absence of a signal associated with binding of a nucleotide of a nucleotide-fluorophore complex to the template, as detected over the course of multiple reaction cycles. In some embodiments, detection is often based on the use of the plurality of fluorophores of the nucleotide-fluorophore complexes and fluorescence imaging of the array.

Nucleic Acid Extraction

[0269] Nucleic acid extraction from cells or other biological samples may be performed using any of a variety of techniques known to those of skill in the art. For example, a typical DNA extraction procedure may comprise: (i) collection of a cell or tissue sample from which DNA is to be extracted, (ii) disruption of cell membranes (i.e., cell lysis) to release DNA and other cytoplasmic components, (iii) treatment of the lysed sample with a concentrated salt solution to precipitate proteins, lipids, and RNA, followed by centrifugation to separate out the precipitated proteins, lipids, and RNA, and (iv) purification of DNA from the supernatant (e.g., using spin columns or paramagnetic beads) to remove detergents, proteins, salts, or other reagents used during the cell membrane lysis step. Non-limiting examples of methods for performing nucleic acid (e.g., DNA and RNA) extraction are described in, for example, Ali et al. (2017) “Current Nucleic Acid Extraction Methods and Their Implications to Point-of-Care Diagnostics”, *BioMed Research International* 2017:9306564, and Dairawan et al. (2020), “The Evolution of DNA Extraction Methods”, *Am J Biomed Sci & Res* 8 (1): 39-45, the entire contents of each of which are incorporated herein by reference.

[0270] A variety of suitable commercial nucleic acid extraction and purification kits are consistent with the disclosure herein. Examples include, but are not limited to, the QIAamp® kits (for isolation of genomic DNA from human samples) and DNAcasy kits (for isolation of genomic DNA from animal or plant samples) from Qiagen (Germantown, Md.), or the Maxwell® and ReliaPrep™ series of kits from Promega (Madison, Wis.).

Sequencing Library Preparation

[0271] Sequence library preparation may be performed using any of a variety of techniques known to those of skill in the art. Library preparation typically comprises performing the steps of, e.g., end repair, tailing, and ligation of adapter sequences to template nucleic acid fragments. Extracted nucleic acid molecules (e.g., DNA molecule), or fragments thereof, that are typically used as the input for sequencing library preparation often have overhangs containing single-stranded DNA (ssDNA overhangs), breaks in the phosphodiester backbone that exist on just one strand (nicks), and/or ssDNA regions internal to the duplex molecule (ssDNA gaps). End repair reactions (using, e.g., a combination of 3' exonuclease digestion to remove 3' overhangs and a strand displacing polymerase reaction using dNTPs to fill nicks and gaps) are used to correct these defects in order to maximize the yield for capturing and sequencing the extracted DNA, and result in the generation of blunt-ended, double-stranded DNA (dsDNA) molecules.

[0272] Tailing (e.g., A tailing) is an enzymatic method (using, e.g., a Taq DNA polymerase) for adding a non-templated nucleotide (e.g., an A nucleotide) to the 3' end of a blunt-ended, double-stranded DNA molecule that facilitates the ligation of the adapter sequences used for sequencing.

[0273] One or more adapter sequences may then be ligated to the ends of the end-repaired and tailed template nucleic acid molecules. The adapter sequences may comprise, for example, (i) capture sequences (e.g., the Illumina p5 and p7 adapter sequences) that allow the nucleic acid molecules of the library to bind to a flow cell surface comprising complementary capture probes, (ii) amplification primer binding sites for use in performing reverse transcription and/or for generating clonally-amplified clusters on a flow cell surface, (iii) sequencing primer binding sites (e.g., the Illumina Rd1 and Rd2 sequencing primer binding site sequences) used to initiate sequencing. In addition to amplification and/or sequencing primer binding sites, in some instances the adapters comprise a barcode sequence, e.g., a sample identification barcode sequence (such as the Illumina Index 1 and Index 2 sample identifier sequences).

[0274] Non-limiting examples of methods for performing sequencing library preparation are described in, for example, Head et al. (2014), “Library construction for next-generation sequencing: Overviews and challenges”, *BioTechniques* 56 (1): 61-77, and Hess et al. (2020), “Library preparation for next generation sequencing: A review of automation strategies”, *Biotechnology Advances* 41:107537, the entire contents of each of which are incorporated herein by

reference.

Nucleic Acid Amplification (Including Clonal Amplification)

[0275] In some instances, the disclosed methods for performing nucleic acid sequencing (e.g., in vitro and/or flow cell sequencing) comprise performing one or more steps (e.g., 1, 2, 3, 4, 5, or more than 5) steps of nucleic acid amplification. Amplification reactions with respect to in situ based sequencing methods as described herein are discussed previously. In some instances, for example, one or more steps of nucleic acid amplification are performed as part of sequencing library preparation and/or following sequencing library preparation. In some instances, one or more steps of nucleic acid amplification (e.g., using a solid-phase amplification technique such as bridge amplification) are performed after the template molecules of a sequencing library have been tethered to a support surface (e.g., a flow cell surface) to generate clonally-amplified colonies of the tethered template nucleic acid fragments.

[0276] In some instances, nucleic acid amplification is performed to amplify all of the nucleic acid molecules extracted from a biological sample (e.g., using a random primer sequence). In some instances, nucleic acid amplification is performed to amplify a selected subset of nucleic acid molecules extracted from a biological sample (e.g., using one or more primer sequences designed to hybridize to portions of the sequences for one or more target nucleic acid molecules of interest, or to sequences adjacent thereto).

[0277] In some instances, nucleic acid amplification is performed to amplify an entire sequencing library (e.g., using a primer sequence that hybridizes to a common amplification primer binding site in the sequencing library adapters). In some instances, nucleic acid amplification is performed to amplify selected portions of the sequencing library (e.g., using one or more primer sequences designed to hybridize to one or more amplification primer binding sites associated with one or more identifier sequences (or barcodes) included in the sequencing library adapters).

[0278] Nucleic acid amplification may be performed using any of a variety of nucleic acid amplification techniques known to those of skill in the art, including both thermal and/or isothermal nucleic acid amplification techniques. Examples of suitable thermal nucleic acid amplification techniques include, but are not limited to, polymerase chain reaction (PCR), multiplexed PCR, nested PCR, bridge PCR, reverse transcription PCR (RT-PCR). Examples of suitable isothermal nucleic acid amplification techniques include, but are not limited to, rolling circle amplification (RCA), nucleic acid sequence-based amplification (NASBA), loop-mediated isothermal amplification (LAMP), strand displacement amplification (SDA), helicase-dependent amplification (HDA), nicking enzyme amplification reaction (NEAR), and recombinase polymerase amplification (RPA). Examples of methods for performing nucleic acid amplification are described in, for example, Gill et al. (2008), "Nucleic Acid Isothermal Amplification Technologies-A Review", *Nucleosides, Nucleotides, and Nucleic Acids* 27:224-243, Fakruddin et al. (2013), "Nucleic acid amplification: Alternative method of polymerase chain reaction", *J Pharm Bioallied Sci.* 5 (4): 245-252, and U.S. Pat. No. 8,143,008, the entire contents of each of which are incorporated herein by reference.

Nucleotides

[0279] In some instances, the disclosed methods for performing nucleic acid sequencing (e.g., in situ and/or flow cell sequencing) comprise the use of modified versions (e.g., comprising a first functional group or a second function group, as described elsewhere herein) of any of a variety of naturally-occurring nucleotides and/or functional analogs thereof (e.g., nucleotide analogs capable of hybridizing to a nucleic acid sequence in a sequence-specific/correctly base-paired manner) to probe a template nucleic acid sequence. Naturally-occurring nucleotides include deoxyribonucleotides (found in DNA) that comprise a deoxyribose sugar moiety, and ribonucleotides (found in RNA) that comprise a ribose sugar moiety. Naturally-occurring deoxyribonucleotides comprise a nucleobase (or "base") selected from the group consisting of adenine (A), thymine (T), cytosine (C), or guanine (G). Naturally-occurring ribonucleotides

comprise a nucleobase selected from the group consisting of uracil (U), adenine (A), cytosine (C), or guanine (G). In some instances, the nucleotides are terminated (e.g., reversibly terminated). In some instances, the nucleotides are conjugated to a detectable label, e.g., a fluorophore. In some instances, the nucleotides are conjugated to other moieties, e.g., reactive functional groups.

Primers (or Primer Sequences)

[0280] In some instances, the disclosed methods for performing nucleic acid sequencing (e.g., in situ and/or flow cell sequencing) comprise the use of primer sequences that are complementary to, e.g., a subsequence (or primer binding site) that is part of an endogenous nucleic acid target sequence or a sequence (or primer binding site) that is located at or near a barcode (identifier) sequence associated with a target analyte. In some instances, a primer sequence is designed to hybridize to a primer binding site associated with a single target analyte sequence and/or an associated target-specific barcode sequence. In some instances, a primer sequence is designed to hybridize to a sequence (or primer binding site) that is associated with a plurality of target analyte sequences and/or associated target-specific barcode sequences (e.g., at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 150, 200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900, 950, 1000, or more than 1000 target analyte sequences and/or associated target-specific barcode sequences).

Polymerases

[0281] In some instances, the disclosed methods for performing nucleic acid sequencing (e.g., in situ and/or flow cell sequencing) comprise performing one or more steps of nucleic acid amplification or replication using one or more polymerases. Examples of polymerases that may be used for amplification include, but are not limited to, DNA polymerases (e.g., Taq DNA polymerase), RNA polymerases, and/or reverse transcriptases.

[0282] As noted elsewhere herein, non-limiting examples of polymerases for use in rolling circle amplification (RCA) comprise DNA polymerases such as phi29 ((29) polymerase, Klenow fragment, *Bacillus stearothermophilus* DNA polymerase (BST), T4 DNA polymerase, T7 DNA polymerase, or DNA polymerase I. In some embodiments, DNA polymerases that have been engineered or mutated to have desirable characteristics are employed. In some embodiments, the polymerase is phi29 DNA polymerase.

Base-Calling

[0283] As noted elsewhere herein, the disclosed methods for performing nucleic acid sequencing (e.g., in situ and/or flow cell sequencing) may comprise inferring the sequence of a template nucleic acid molecule from a series of optical signals (e.g., fluorescence signals) detected in images acquired during a repetitive series of sequencing reaction cycles in a process referred to as “base-calling”. The interplay of sequencing chemistry, opto-fluidics hardware, optical sensors, and signal processing software utilized in sequencing platforms affects the types of errors made during sequencing (see, e.g., Lederberger et al. (2011), “Base-calling for next-generation sequencing platforms”, *Brief Bioinform.* 12 (5): 489-497). The characterization of errors associated with the sequencing process and implementation of chemistry-, imaging-, and/or signal processing software-based methods for minimizing sequence errors are thus important for maximizing the accuracy of sequencing results.

[0284] In four-color sequencing-by-synthesis methods, for example, a set of four images—one image for each of four detection channels corresponding to the emission wavelengths for four fluorophores used to label the reversibly terminated nucleotides—are acquired in each sequencing cycle. Processing of the images to detect fluorescence intensity signals produces an intensity quadruple for the location of each sequencing colony on a flow cell surface (or the location of each target analyte, or amplified representation thereof (e.g., an RCP) in the case of in situ sequencing), where each value represents the intensity of the fluorescence signal for the detection channels corresponding to A, C, G and T. Ideally, the channel in which the maximum intensity occurs would be the base that is “called” for a given RCP or sequencing colony (or target analyte) in a given

cycle. However, the chemical processes involved in sequencing are imperfect, leading to errors in base-calling (scc, e.g., Cacho, et al. (2016), "A Comparison of Base-calling Algorithms for Illumina Sequencing Technology", Briefings in Bioinformatics 17 (5): 786-795). In some sequencing-by-synthesis (SBS) platforms, for example, sources of error may include phasing (or lagging; e.g., where the primed template nucleic acid molecules at one or more locations fail to incorporate the next base due to variation in polymerase reaction kinetics), pre-phasing (or leading; e.g., where more than one nucleotide is incorporated in a single cycle due to, e.g., impurities in the reversibly terminated nucleotides), signal decay (due to, e.g., photobleaching and/or loss of template nucleic acid during the sequencing process), and cross-talk (e.g., when two or more fluorophore emission spectra overlap, which may cause a positive correlation between signal intensities measured in the corresponding detection channels).

[0285] A variety of statistical approaches have been developed to correct for, or minimize, such errors and generate more accurate base-calls. Examples include, but are not limited to, AYB (Goldman Group, European Molecular Biology Laboratory-European Bioinformatics Institute, Cambridgeshire, UK), and Bustard (Illumina, Inc., San Diego, CA).

Sequence Reads

[0286] The output of the base-calling process applied to optical signals detected in a series of images of a biological sample or flow cell surface acquired during a cycling sequencing process consists of a plurality of sequence reads, e.g., the nucleotide sequences determined for all or a portion of a template nucleic acid molecule (e.g., an endogenous nucleic acid analyte or a barcode sequence associated with a target analyte).

[0287] In some instances, the sequence reads generated using the disclosed methods for in situ and/or flow cell sequencing comprise sequence reads of at least about 10, 20, 30, 40, 50, 60, 70, 80, 90, 100 or more nucleotides or base pairs of the template nucleic acid sequences. In some instances, the sequence reads generated using the disclosed methods comprise sequence reads of at least about 100, 120, 140, 160, 180, 200, 220, 240, 260, 280, 300, 320, 340, 360, 380, 400, or more than 400 nucleotides or base pairs of the template nucleic acid sequences.

[0288] In some instances, the disclosed methods for in situ or flow cell sequencing generate at least about 100, 200, 300, 400, 500, 600, 700, 800, 900, 1,000, or more sequencing reads per run. In some instances, the disclosed method generates at least about 1,000, 1,500, 2,000, 2,500, 3,000, 3,500, 4,000, 4,500, 5,000, 5,500, 6,000, 6,500, 7,000, 7,500, 8,000, 8,500, 9,000, 9,500, 10,000, 20,000, 30,000, 40,000, 50,000, 60,000, 70,000, 80,000, 90,000, 100,000, 200,000, 300,000, 400,000, 500,000, 600,000, 700,000, 800,000, 900,000, 10×10^6 , 5×10^6 , 10×10^7 , or more than 10×10^7 sequencing reads per run.

Sequence Assembly & Alignment

[0289] In some instances, the disclosed methods for in situ and/or flow cell sequencing comprise assembly of longer template nucleic acid sequences, e.g., genome fragments or whole genomes, from a plurality of relatively short sequence reads. Sequence assembly may be performed by identifying the overlapping sequences from multiple short sequence reads to assemble longer, contiguous sections of sequence.

[0290] In some instances, the disclosed methods for in situ and/or flow cell sequencing comprise identifying a code word corresponding to a sequence read or an assembled sequence, where the code word is one of a plurality of code words in a codebook that includes assignment of each of the plurality of code words to a target analyte of interest. The sequence read or assembled sequence may thus be used to identify a specific target analyte (based on the corresponding code word) in, e.g., a multiplexed in situ detection or sequencing assay.

[0291] In some instances, the disclosed methods for in situ and/or flow cell sequencing comprise alignment of sequence reads and/or assembled sequences to a known reference sequence or consensus sequence (e.g., the GRCh38 human reference genome (Genome Reference Consortium)) from the same or a similar organism. Alignment to a reference sequence or consensus sequence

may be used to identify gaps, errors, or variants in the assembled sequence. Any of a variety of bioinformatics software programs known to those of skill in the art may be used to assemble longer sequences from relatively short sequence reads. Examples include, but are not limited to, DBG2OLC (see, e.g., Ye et al. (2016), “DBG2OLC: Efficient Assembly of Large Genomes Using Long Erroneous Reads of the Third Generation Sequencing Technologies”, *Scientific Reports* 6:31900), SPAdes (see, e.g., Bankevich et al. (2012), “SPAdes: A New Genome Assembly Algorithm and Its Applications to Single-Cell Sequencing”, *J. Computational Biol.* 19 (5): 455-477), SparseAssembler (see, e.g., Ye et al. (2012), “Exploiting Sparseness in de novo Genome Assembly”, *BMC Bioinformatics* 13 (Suppl 6): S1), Fermi (see, e.g., Li (2012), “Exploring Single-Sample SNP and INDEL Calling with Whole-Genome de novo Assembly”, *Bioinformatics* 28 (14): 1838-1844), and String Graph Assembler (SGA) (see, e.g., Simpson et al. (2012), “Efficient de novo Assembly of Large Genomes Using Compressed Data Structures”, *Genome Res.* 22:549-556), each of which are herein incorporated by reference in its entirety.

IX. Opto-Fluidic Instrument

[0292] In some instances, the sequencing methods described herein (e.g., in situ sequence or flow cell sequencing) include using instruments having integrated optics and fluidics modules (“opto-fluidic instruments” or “opto-fluidic systems”) for detecting target molecules (e.g., nucleic acids, proteins, antibodies, etc.) in biological samples (e.g., one or more cells or a tissue sample) as described herein.

[0293] In an opto-fluidic instrument, the fluidics module is configured to deliver one or more reagents (e.g., compositions comprising nucleotide-fluorophore complexes, unlabeled nucleotides, primers, detectable-labeled probes and/or non-labeled probes, polymerases and/or other enzymes, deprotection reagents, buffers, etc.) to the biological sample (e.g., to a sample cartridge within which the biological sample is contained) or to a flow cell (e.g., within which nucleic acid molecules extracted from the biological sample have been tethered) and/or to remove spent reagents therefrom. In some instances, one or more sample preparation steps (e.g., fixing, embedding, sample clearing, and/or nucleic acid extraction (in the case that nucleic acid molecules are to be extracted and sequenced in a flow cell)) are performed prior to the sample being placed on the instrument. In some instances, the fluidics module is configured to deliver one or more further reagents (e.g., primary probe(s) such as circular probe(s) or circularizable probe(s) or probe set(s)) and/or to remove non-specifically hybridized probe(s). In some instances, the fluidics module is configured to deliver one or more detectably labeled probes and optionally intermediate probes to detect the target analytes, or amplified representatives thereof (e.g., RCP(s)) in the biological sample. In some instances, the fluidics module is configured to deliver one or more composition (e.g., composition of nucleotide-fluorophore complexes and/or unlabeled nucleotides, as well as primers, polymerases, deprotection reagents, etc.) to sequence, e.g., native nucleic acid sequences, barcode sequences associated with target analytes, or amplified copies thereof (e.g., barcode sequences included in RCP(s)) in the biological sample. In some instances, the fluidics module is configured to deliver one or more compositions (e.g., compositions of nucleotide-fluorophore complexes and/or unlabeled nucleotides, as well as primers, polymerases, deprotection reagents, etc.) to a flow cell to sequence, e.g., native nucleic acid sequences, barcode sequences, or amplified copies thereof extracted from the biological sample.

[0294] Additionally, the optics module is configured to illuminate the biological sample (or flow cell) with light having one or more spectral emission curves (over a range of wavelengths) and subsequently capture one or more images of emitted light signals from the biological sample (or flow cell) during one or more decoding (e.g., probing or sequencing) cycles. In various instances, the captured images are processed in real time and/or at a later time to determine the presence of the one or more target molecules in the biological sample, as well as two-dimensional and/or three-dimensional position information associated with each detected target molecule within the biological sample. In various instances, the captured images of a flow cell surface are processed in

real time and/or at a later time to determine the sequence of the one or more nucleic acid sequences (e.g., barcode sequences associated with one or more target molecules) that have been extracted from a biological sample. In some embodiment, the optics module further comprises an autofocus mechanism configured to maintain focus at a specified sample plane (e.g., a plane that is perpendicular to the optical axis of an objective lens of the optics module).

[0295] Additionally, the opto-fluidics instrument includes a sample module configured to receive (and, optionally, secure) one or more biological samples (e.g., biological samples contained within one or more sample cartridges), or to receive (and, optionally, secure) one or more flow cells. In some instances, the sample module includes an X-Y stage configured to move the biological sample (or flow cell) along an X-Y plane (e.g., perpendicular to the optical axis of an objective lens of the optics module).

[0296] In various instances, the opto-fluidic instrument is configured to analyze one or more target molecules (e.g., one or more target RNAs) in their naturally occurring place (i.e., in situ) within the biological sample. In some instances, the opto-fluidic instrument is configured to analyze one or more target RNAs in relative spatial locations within the biological sample. For example, an opto-fluidic instrument may be an in-situ analysis system used to analyze a biological sample and detect target molecules including, but not limited to, DNA, RNA, proteins, antibodies, and/or the like. In some instances, the in situ analysis system is used to detect one or more target RNAs using target-primed rolling circle amplification (RCA) according to the methods disclosed herein.

[0297] In various instances, the opto-fluidic instrument is configured to perform in situ target molecule detection via base-by-base sequencing (e.g., by sequencing an identifier sequence such as a barcode sequence associated with a target molecule) and/or any imaging or target molecule detection technique. That is, for example, an opto-fluidic instrument may include a fluidics module that includes fluids needed for establishing the experimental conditions required for the probing or sequencing of target molecules (or associated barcode sequences) in the sample. Further, such an opto-fluidic instrument may also include a sample module configured to receive the sample, and an optics module including an imaging system for illuminating (e.g., exciting one or more fluorescent probes within the sample) and/or imaging light signals received from the probed sample. The in-situ analysis system may also include other ancillary modules configured to facilitate the operation of the opto-fluidic instrument, such as, but not limited to, cooling systems, motion calibration systems, etc.

[0298] FIG. 2 shows an example workflow of analysis of a biological sample **210** (e.g., cell or tissue sample) using an opto-fluidic instrument or system **200**, according to various instances. In various instances, the sample **210** is a biological sample (e.g., a tissue) that includes molecules such as DNA, RNA, proteins, antibodies, etc. For example, the sample **210** can be a sectioned tissue that is treated to access the RNA thereof for probe (e.g., circularizable probe) hybridization and sequencing (e.g., using a sequencing primer that hybridizes to RCPs to sequence barcode sequences in the RCPs) described elsewhere herein.

[0299] In various instances, the sample **210** is placed in the opto-fluidic instrument or system **200** for analysis and detection of the molecules in the sample **210**. In various instances, the opto-fluidic instrument or system **200** is a system configured to facilitate the experimental conditions conducive for the detection of the target molecules. For example, the opto-fluidic instrument or system **200** can include a fluidics module **230**, an optics module **240**, a sample module **250**, and an ancillary module **260**, and these modules may be operated by a system controller **220** to create the experimental conditions for hybridization probe-based detection and/or base-by-base sequencing of nucleic acid molecules in the sample **210**, as well as to facilitate the imaging of the sample (e.g., by an imaging system of the optics module **240**). In various instances, the various modules of the opto-fluidic instrument or system **200** are separate components in communication with each other, or at least some of them are integrated together.

[0300] In various instances, the sample module **250** is configured to receive the sample **210** into the

opto-fluidic instrument or system **200**. For instance, the sample module **260** may include a sample interface module (SIM) that is configured to receive a sample device (e.g., cassette) onto which the sample **210** can be deposited. That is, the sample **210** may be placed in the opto-fluidic instrument or system **200** by depositing the sample **210** (e.g., the sectioned tissue) on a sample device that is then inserted into the SIM of the sample module **250**. In some instances, the sample module **250** includes an X-Y stage onto which the SIM is mounted. The X-Y stage may be configured to move the SIM mounted thereon (e.g., and as such the sample device containing the sample **210** inserted therein) in perpendicular directions along the two-dimensional (2D) plane of the opto-fluidic instrument or system **200**.

[0301] The experimental conditions that are conducive for the detection of the molecules in the sample **210** may depend on the target molecule detection technique that is employed by the opto-fluidic instrument or system **200**. For example, in various instances, the opto-fluidic instrument or system **200** can be a system that is configured to detect molecules (e.g., by detecting hybridization probes that hybridize to nucleic molecules (e.g., barcode sequences) and/or by nucleotides incorporated into extending sequencing primers using an identifier sequence as a template) in the sample **210**.

[0302] In various instances, the fluidics module **230** includes one or more components that may be used for storing the reagents, as well as for transporting said reagents to and from the sample device containing the sample **210**. For example, the fluidics module **230** may include reservoirs configured to store the reagents, as well as a waste container configured for collecting the reagents (e.g., and other waste) after use by the opto-fluidic instrument or system **200** to analyze and detect the molecules of the sample **210**. Further, the fluidics module **230** may also include pumps, tubes, pipettes, etc., that are configured to facilitate the transport of the reagent to the sample device (e.g., and as such the sample **210**). For instance, the fluidics module **230** may include pumps (“reagent pumps”) that are configured to pump washing/stripping reagents to the sample device for use in washing/stripping the sample **210** (e.g., as well as other washing functions such as washing an objective lens of the imaging system of the optics module **240**).

[0303] In various instances, the ancillary module **260** is a cooling system of the opto-fluidic instrument or system **200**, and the cooling system includes a network of coolant-carrying tubes that are configured to transport coolants to various modules of the opto-fluidic instrument or system **200** for regulating the temperatures thereof. In such cases, the fluidics module **230** may include coolant reservoirs for storing the coolants and pumps (e.g., “coolant pumps”) for generating a pressure differential, thereby forcing the coolants to flow from the reservoirs to the various modules of the opto-fluidic instrument or system **200** via the coolant-carrying tubes. In some instances, the fluidics module **230** includes returning coolant reservoirs that are configured to receive and store returning coolants, e.g., heated coolants flowing back into the returning coolant reservoirs after absorbing heat discharged by the various modules of the opto-fluidic instrument or system **200**. In such cases, the fluidics module **230** may also include cooling fans that are configured to force air (e.g., cool and/or ambient air) into the returning coolant reservoirs to cool the heated coolants stored therein. In some instance, the fluidics module **230** includes cooling fans that are configured to force air directly into a component of the opto-fluidic instrument or system **200** so as to cool said component. For example, the fluidics module **230** may include cooling fans that are configured to direct cool or ambient air into the system controller **220** to cool the same.

[0304] As discussed above, the opto-fluidic instrument or system **200** may include an optics module **240** which include the various optical components of the opto-fluidic instrument or system **200**, such as but not limited to a camera, an illumination module (e.g., LEDs), an objective lens, and/or the like. The optics module **240** may include a fluorescence imaging system that is configured to image the fluorescence emitted by the detectably labeled nucleotides are incorporated in extending sequencing primers in the sample **210** after the detectable labels are excited by light from the illumination module of the optics module **240**.

[0305] In some instances, the optics module **240** also includes an optical frame onto which the camera, the illumination module, and/or the X-Y stage of the sample module **250** may be mounted.

[0306] In various instances, the system controller **220** is configured to control the operations of the opto-fluidic instrument or system **200** (e.g., and the operations of one or more modules thereof). The system controller **220** may take various forms, including a processor, a single computer (or computer system), or multiple computers in communication with each other. In various instances, the system controller **220** is communicatively coupled with data storage, set of input devices, display system, or a combination thereof. In some cases, some or all of these components are considered to be part of or otherwise integrated with the system controller **220**, are separate components in communication with each other, or are integrated together. In other examples, the system controller **220** is in communication with a cloud computing platform.

[0307] In various instances, the opto-fluidic instrument or system **200** analyzes the sample **210** and generates the output **270** that includes indications of the presence of the target molecules in the sample **210**. For instance, with respect to instances discussed above where the opto-fluidic instrument or system **200** employs a sequencing technique for detecting molecules, the opto-fluidic instrument or system **200** may cause the sample **210** to undergo successive sequencing cycles, where during the same sequencing cycle the sample is imaged to detect signals associated with nucleotide binding and/or incorporation events at some locations in the sample **210**, as well as to detect an absence of signals at other locations in the sample. In such cases, the output **270** may include a series of optical signals (e.g., a code word) specific to each identifier sequence (e.g., a barcode sequence), which allow the identification of the target molecules.

[0308] FIG. **3** illustrates an example of a computing device or system in accordance with one or more examples of the disclosure. Device **300** can be a host computer connected to a network. Device **300** can be a client computer or a server. As shown in FIG. **3**, device **300** can be any suitable type of microprocessor-based device, such as a personal computer, workstation, server, or handheld computing device (portable electronic device), such as a phone or tablet. The device can include, for example, one or more of processor **310**, input device **320**, output device **330**, memory/storage **340**, and communication device **360**. Input device **320** and output device **330** can generally correspond to those described above, and they can either be connectable or integrated with the computer.

[0309] Input device **320** can be any suitable device that provides input, such as a touch screen, keyboard or keypad, mouse, or voice-recognition device. Output device **330** can be any suitable device that provides output, such as a touch screen, haptics device, or speaker.

[0310] Storage **340** can be any suitable device that provides storage, such as an electrical, magnetic, or optical memory including a RAM, cache, hard drive, or removable storage disk. Communication device **360** can include any suitable device capable of transmitting and receiving signals over a network, such as a network interface chip or device. The components of the computer can be connected in any suitable manner, such as via a physical bus **370** or wirelessly.

[0311] Software **350**, which can be stored in memory/storage **340** and executed by processor **310**, can include, for example, the programming that embodies the functionality of the present disclosure (e.g., as embodied in the methods and systems described above). Software **350** can also be stored and/or transported within any non-transitory computer-readable storage medium for use by or in connection with an instruction execution system, apparatus, or device, such as those described above, that can fetch instructions associated with the software from the instruction execution system, apparatus, or device and execute the instructions. In the context of this disclosure, a computer-readable storage medium can be any medium, such as storage **340**, that can contain or store programming for use by or in connection with an instruction execution system, apparatus, or device.

[0312] Software **350** can also be propagated within any transport medium for use by or in connection with an instruction execution system, apparatus, or device, such as those described

above, that can fetch instructions associated with the software from the instruction execution system, apparatus, or device and execute the instructions. In the context of this disclosure, a transport medium can be any medium that can communicate, propagate, or transport programming for use by or in connection with an instruction execution system, apparatus, or device. The transport readable medium can include, but is not limited to, an electronic, magnetic, optical, electromagnetic, or infrared wired or wireless propagation medium.

[0313] Device **300** may be connected to a network, which can be any suitable type of interconnected communication system. The network can implement any suitable communications protocol and can be secured by any suitable security protocol. The network can comprise network links of any suitable arrangement that can implement the transmission and reception of network signals, such as wireless network connections, T1 or T3 lines, cable networks, DSL, or telephone lines.

[0314] Device **300** can implement any operating system suitable for operating on the network. Software **350** can be written in any suitable programming language, such as C, C++, Java, or Python. In various implementations, application software embodying the functionality of the present disclosure is deployed in different configurations, such as in a client/server arrangement or through a web browser as a web-based application or web service, for example.

X. Terminology

[0315] Unless defined otherwise, all terms of art, notations and other technical and scientific terms or terminology used herein are intended to have the same meaning as is commonly understood by one of ordinary skill in the art to which the claimed subject matter pertains. In some cases, terms with commonly understood meanings are defined herein for clarity and/or for ready reference, and the inclusion of such definitions herein should not necessarily be construed to represent a substantial difference over what is generally understood in the art.

[0316] The terms “polynucleotide,” and “nucleic acid molecule,” used interchangeably herein, refer to polymeric forms of nucleotides of any length, either ribonucleotides or deoxyribonucleotides. Thus, this term comprises, but is not limited to, single-, double-, or multi-stranded DNA or RNA, genomic DNA, cDNA, DNA-RNA hybrids, or a polymer comprising purine and pyrimidine bases or other natural, chemically or biochemically modified, non-natural, or derivatized nucleotide bases. The backbone of the polynucleotide can comprise sugars and phosphate groups (as may typically be found in RNA or DNA), or modified or substituted sugar or phosphate groups.

[0317] A “primer” used herein can be an oligonucleotide, either natural or synthetic, that is capable, upon forming a duplex with a polynucleotide template, of acting as a point of initiation of nucleic acid synthesis and being extended from its 3' end along the template so that an extended duplex is formed. The sequence of nucleotides added during the extension process is determined by the sequence of the template polynucleotide. Primers usually are extended by a DNA polymerase.

[0318] “Ligation” may refer to the formation of a covalent bond or linkage between the termini of two or more nucleic acids, e.g., oligonucleotides and/or polynucleotides, in a template-driven reaction. The nature of the bond or linkage may vary widely and the ligation may be carried out enzymatically or chemically. As used herein, ligations are usually carried out enzymatically to form a phosphodiester linkage between a 5' carbon terminal nucleotide of one oligonucleotide with a 3' carbon of another nucleotide.

[0319] The term “about” as used herein refers to the usual error range for the respective value readily known to the skilled person in this technical field. Reference to “about” a value or parameter herein comprises (and describes) embodiments that are directed to that value or parameter per se.

[0320] As used herein, the singular forms “a,” “an,” and “the” comprise plural referents unless the context clearly dictates otherwise. For example, “a” or “an” means “at least one” or “one or more.”

[0321] Throughout this disclosure, various aspects of the claimed subject matter are presented in a range format. It should be understood that the description in range format is merely for

convenience and brevity and should not be construed as an inflexible limitation on the scope of the claimed subject matter. Accordingly, the description of a range should be considered to have specifically disclosed all the possible sub-ranges as well as individual numerical values within that range. For example, where a range of values is provided, it is understood that each intervening value, between the upper and lower limit of that range and any other stated or intervening value in that stated range is encompassed within the claimed subject matter. The upper and lower limits of these smaller ranges may independently be comprised in the smaller ranges, and are also encompassed within the claimed subject matter, subject to any specifically excluded limit in the stated range. Where the stated range comprises one or both of the limits, ranges excluding either or both of those comprised limits are also comprised in the claimed subject matter. This applies regardless of the breadth of the range.

[0322] Use of ordinal terms such as “first”, “second”, “third”, etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements. Similarly, use of a), b), etc., or i), ii), etc. does not by itself connote any priority, precedence, or order of steps in the claims. Similarly, the use of these terms in the specification does not by itself connote any required priority, precedence, or order.

[0323] The disclosures of all publications, patents, and patent applications referred to herein are each hereby incorporated by reference in their entireties. To the extent that any reference incorporated by reference conflicts with the instant disclosure, the instant disclosure shall control.

EXAMPLES

[0324] The following examples are included for illustrative purposes only and are not intended to limit the scope of the present disclosure.

Example 1: Sequencing Using Nucleotide-Fluorophore Complexes In Situ

[0325] This example provides a workflow for using a composition comprising nucleotide-fluorophore complexes to sequence a template nucleic acid molecule (e.g., DNA or RNA) in a tissue section. Use of nucleotide-fluorophore complexes for sequencing may provide certain advantages such as increased signal intensity, signal-to-noise ratios, and sensitivity.

[0326] A tissue sample is obtained and cryosectioned onto a glass slide for processing. The tissue is fixed by incubating in 3.7% paraformaldehyde (PFA). In other cases, a FFPE sample can be de-paraffinized and processed to be used. One or more washes is performed and the tissue is then permeabilized. To prepare for probe hybridization, a wash buffer is added to the tissue sample. The washed tissue sample is then contacted with a circularizable probe comprising a template nucleic acid molecule and a barcode sequence. The barcode sequence identifies a target analyte within the tissue sample. The circularizable probe is allowed to hybridize to the target analyte. The tissue sample is then contacted with a ligation reaction mix including ligase, and the circularizable probe is ligated to form a circular template for rolling circle amplification (RCA). The tissue sample is then incubated with an RCA mixture containing a Phi29 DNA polymerase and dNTP for RCA of the circularized probes. From this amplification, the RCA product (e.g., RCP) comprises the template nucleic acid molecule and a barcode sequence.

[0327] A non-limiting example of the disclosed sequencing methods is illustrated schematically in FIG. 4. In the case of in situ sequencing, the tissue sample is washed and then contacted with a primer. The primer is allowed to hybridize to the template nucleic acid molecule in the RCP (step 1 in FIG. 4, where the upper strand is the primer and the lower strand is the template nucleic acid molecule). The tissue sample is washed and then contacted (in one or more steps) with a polymerase and a first composition comprising nucleotide-fluorophore complexes. Each nucleotide-fluorophore complex comprises, e.g., (i) a single reversibly terminated nucleotide attached to an avidin core through a biotinylated linker, wherein the single nucleotide is attached to

the biotinylated linker through a disulfide bond and (ii) a plurality of fluorophores, wherein each fluorophore is attached to the avidin core through a biotinylated linker. The ratio of the fluorophores to the nucleotide in each of the nucleotide-fluorophore complexes of the first composition is at or at least about 3:1. The first composition comprising nucleotide-fluorophore complexes comprises, e.g., a set of four nucleotide-fluorophore complexes, where each set of the four nucleotide-fluorophore complexes comprises a different nucleobase and a different type of fluorophore. An extension reaction is performed to incorporate one of the nucleotide-fluorophore complexes of the first composition into the priming strand to form an extended priming strand (step 2 in FIG. 4). In an example, the nucleotide-fluorophore complex incorporated into the extended priming strand identifies a complementary nucleotide in the template nucleic acid molecule. The tissue sample is then washed to remove polymerase and unbound nucleotide-fluorophore complexes of the first composition.

[0328] Fluorescence imaging is used to detect a signal associated with the presence of the nucleotide-fluorophore complex of the first composition incorporated into the priming strand and to identify a complementary nucleotide in the template nucleic acid molecule. Images for each of a plurality of detection channels configured to detect signals arising from labels (e.g., fluorescent dyes) conjugated to nucleotide-fluorophore complexes present in the priming strand are acquired in each cycle of a multicycle sequencing run.

[0329] The tissue sample is contacted with a buffer comprising dithiothreitol (DTT) and a deprotection reagent. The reducing agent is allowed to disrupt the disulfide bond between the single nucleotide and the biotinylated linker to release the avidin core from the extended priming strand. The primer bound to the template nucleic acid molecule is deprotected (step 3 in FIG. 4). The tissue sample is washed to remove the released avidin core. After washing the tissue sample, the entire cycle as described above is repeated with additional compositions of nucleotide-fluorophore complexes until the barcode is sequenced.

Example 2: Sequencing Using Nucleotide-Fluorophore Complexes in a Flow Cell

[0330] This example provides a workflow for using a composition comprising nucleotide-fluorophore complexes to sequence a template nucleic acid molecule (e.g., DNA or RNA) immobilized on a flow cell surface. Use of nucleotide-fluorophore complexes for sequencing may provide certain advantages such as increased signal intensity, signal-to-noise ratios, and sensitivity.

[0331] Again, the sequencing process is illustrated schematically in FIG. 4. The template nucleic acid molecules to be sequenced are attached to the surface of a flow cell at discrete locations (e.g., where each location comprises multiple clonally-amplified copies of a single template nucleic acid molecule created using bridge amplification or a similar technique). A primer is introduced to the flow cell. The primer is allowed to hybridize to an immobilized template nucleic acid molecule (step 1 in FIG. 4, where the upper strand is the primer and the lower strand is the template nucleic acid molecule). The flow cell is washed and then contacted (in one or more steps) with a polymerase and a first composition comprising nucleotide-fluorophore complexes. Each nucleotide-fluorophore complex comprises, e.g., (i) a single reversibly terminated nucleotide attached to an avidin core through a biotinylated linker, wherein the single nucleotide is attached to the biotinylated linker through a disulfide bond and (ii) a plurality of fluorophores, wherein each fluorophore is attached to the avidin core through a biotinylated linker. The ratio of the fluorophores to the nucleotide in each of the nucleotide-fluorophore complexes of the first composition is at or at least about 3:1. The first composition comprising nucleotide-fluorophore complexes comprises, e.g., a set of four nucleotide-fluorophore complexes, where each set of the four nucleotide-fluorophore complexes comprises a different nucleobase and a different type of fluorophore. An extension reaction is performed to incorporate one of the nucleotide-fluorophore complexes of the first composition into the priming strand to form an extended priming strand (step 2 in FIG. 4). The flow cell is washed to remove polymerase and unbound nucleotide-fluorophore complexes of the first composition.

[0332] Fluorescence imaging is used to detect a signal associated with the presence of the nucleotide-fluorophore complex of the first composition incorporated into the priming strand and to identify a complementary nucleotide in the template nucleic acid molecule. Images for each of a plurality of detection channels configured to detect signals arising from labels (e.g., fluorescent dyes) conjugated to nucleotide-fluorophore complexes present in the priming strand are acquired in each cycle of a multicycle sequencing run.

[0333] The flow cell is contacted with a buffer comprising dithiothreitol (DTT) and a deprotection reagent. The reducing agent is allowed to disrupt the disulfide bond between the single nucleotide and the biotinylated linker to release the avidin core from the extended priming strand. The primer bound to the template nucleic acid molecule is deprotected (step 3 in FIG. 4). The flow cell is washed to remove the released the released avidin core. After washing the flow cell, the entire cycle as described above is repeated with additional compositions of nucleotide-fluorophore complexes until the template nucleic acid molecule is sequenced.

[0334] The present disclosure is not intended to be limited in scope to the particular disclosed embodiments, which are provided, for example, to illustrate various aspects of the present disclosure. Various modifications to the compositions and methods described will become apparent from the description and teachings herein. Such variations may be practiced without departing from the true scope and spirit of the disclosure and are intended to fall within the scope of the present disclosure.

Claims

1. A method for sequencing a template nucleic acid molecule, comprising: (a) contacting a priming strand bound to the template nucleic acid molecule with: (A) a polymerase; and (B) a composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex comprises: (i) a single nucleotide attached to a core; and (ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide; (b) incorporating one of the nucleotide-fluorophore complexes of the composition of nucleotide-fluorophore complexes into the priming strand to form an extended priming strand; and (c) detecting the incorporated nucleotide-fluorophore complex.
2. The method of claim 1, wherein the ratio of the fluorophores to the nucleotide in each of the nucleotide-fluorophore complexes of the first composition is at least 3:1.
3. The method of claim 1, further comprising removing the fluorophores from the incorporated nucleotide-fluorophore complex.
4. The method of claim 1, further comprising, after the detecting step, cleaving the core from the nucleotide of the incorporated nucleotide-fluorophore complex, leaving an incorporated nucleotide.
5. The method of claim 1, wherein the core of each of the nucleotide-fluorophore complexes comprises an avidin or streptavidin core, and wherein the nucleotide in each complex is bound to the core via a biotinylated linker.
6. The method of claim 5, wherein at least a subset of the fluorophores in each nucleotide-fluorophore complex are bound to the core via biotin.
7. The method of claim 5, wherein the biotinylated linker is a biotinylated linear polyethylene glycol (PEG) linker.
8. The method of claim 1, wherein the core of each of the nucleotide-fluorophore complexes comprises a branched polymer core, and wherein the nucleotide and the plurality of fluorophores in each nucleotide-fluorophore complex are attached to the branched polymer core.
9. The method of claim 1, wherein the nucleotide comprises a reversible terminator moiety, and wherein the method further comprises cleaving the reversible terminator moiety after detecting the incorporated nucleotide-fluorophore complex.
10. The method of claim 1, further comprising: (d) contacting a priming strand bound to the

template nucleic acid molecule with: (A) an additional polymerase; and (B) an additional composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex of the additional composition comprises: (i) a single nucleotide attached to a core; and (ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide; (e) incorporating one of the nucleotide-fluorophore complexes of the additional composition of nucleotide-fluorophore complexes into the extended priming strand; and (f) detecting the incorporated nucleotide-fluorophore complex of the additional composition of nucleotide-fluorophore complexes.

11. The method of claim 10, comprising repeating steps (d) through (f) e) and d) for at least one additional cycle using at least one additional composition comprising nucleotide-fluorophore complexes.

12. The method of claim 11, wherein the at least one additional cycle comprises at least 2 additional cycles.

13. The method of claim 1, wherein the nucleotide-fluorophore complexes of the composition comprises four sets of nucleotide-fluorophore complexes, each set comprising a same nucleobase that differs from the other sets, and wherein, for each set, the plurality of fluorophores is a plurality of a same fluorophore that differs from the other sets.

14-18. (canceled)

19. The method of claim 1, wherein the template nucleic acid molecule comprises a target analyte nucleic acid molecule.

20. The method of claim 1, wherein the template nucleic acid molecule comprises a barcode sequence associated with a target analyte.

21. The method of claim 20, further comprising hybridizing a circularizable probe or probe set to the target analyte or to a labeling agent bound to the target analyte and ligating the circularizable probe or probe set to form a circularized probe, wherein the method further comprises performing rolling circle amplification of the circularized probe to generate the template nucleic acid molecule.

22-23. (canceled)

24. The method of claim 1, wherein the template nucleic acid molecule to be sequenced is attached to a solid support.

25. The method of claim 24, wherein the solid support comprises a sequencing flow cell.

26. The method of claim 1, wherein the template nucleic acid molecule is sequenced in situ in a cell sample or tissue sample.

27. (canceled)

28. A kit for sequencing a template nucleic acid molecule comprising: a composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex comprises: (i) a single nucleotide attached to a core; and (ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide.

29. The kit of claim 28, wherein the composition of nucleotide-fluorophore complexes comprises at least three sets of nucleotide-fluorophore complexes, wherein each set comprises a same nucleobase that differs from the other sets, and wherein, for each set, the plurality of fluorophores is a plurality of same fluorophores that differ from the other sets.

30. (canceled)

31. The kit of claim 29, wherein the composition further comprises unlabeled nucleotides, wherein the unlabeled nucleotides each comprise a same nucleobase that is different from the nucleobase of the nucleotide-fluorophore complexes in the sets.

32. The kit of claim 28, wherein the core of each of the nucleotide-fluorophore complexes comprises an avidin or streptavidin core, and wherein the nucleotide in each complex is bound to the core via a biotinylated linker.

33. The kit of claim 32, wherein at least a subset of the fluorophores in each nucleotide-fluorophore complex are bound to the core via biotin.

34. (canceled)

35. The kit of claim 28, wherein the core of each of the nucleotide-fluorophore complexes comprises a branched polymer core, and wherein the nucleotide and the plurality of fluorophores in each nucleotide-fluorophore complex are attached to the branched polymer core.

36. A composition comprising: a composition comprising nucleotide-fluorophore complexes, wherein each nucleotide-fluorophore complex comprises: (i) a single nucleotide attached to a core; and (ii) a plurality of fluorophores attached to the core; wherein each nucleotide-fluorophore complex comprises no more than one nucleotide.

37. The composition of claim 36, wherein the nucleotide comprises a reversible terminator moiety.

38. The composition of claim 36, wherein the composition of nucleotide-fluorophore complexes comprises at least three sets of nucleotide-fluorophore complexes, wherein each set comprises a same nucleobase that differs from the other sets, and wherein, for each set, the plurality of fluorophores is a plurality of same fluorophores that differ from the other sets.

39-40. (canceled)

41. The composition of claim 36, wherein the core of each of the nucleotide-fluorophore complexes comprises an avidin or streptavidin core, and wherein the nucleotide in each complex is bound to the core via a biotinylated linker.

42. The composition of claim 41, wherein at least a subset of the fluorophores in each nucleotide-fluorophore complex are bound to the core via biotin.

43-44. (canceled)

45. The method of claim 1, further comprising, after step (c), identifying a complementary nucleotide in the template nucleic acid based on the detecting.
